

A Multi-Task NLP Framework for Barangay-Level Validation of Crowdsourced Flood Reports

Harley Love N. Cuba
Franz Sedrick F. Embuscado
Jan-Aceryl N. Futalan

ATENEO DE DAVAO UNIVERSITY
SCHOOL OF ARTS AND SCIENCES
DAVAO CITY

April 2026

Acknowledgement

Writing this thesis paper was a massive challenge, and we could not have made it past the starting line without the proper guidance of the people who helped pave the way.

To our thesis professors, Dr. Oniel B. Victoriano and Ma'am Grace Tacadao: first and foremost, thank you, Dr. Victoriano, for trusting us with our topic, and thank you, Ma'am Tacadao, for helping us understand the challenges of training an AI model. Thank you both for being supportive and for giving us the constructive critiques that we needed.

To our adviser, Sir Antonio Bulao II, thank you for your guidance. Your comments and critiques on our paper have been a great help, as they allowed us to break through the parts where we were stuck. Your dedication to academic excellence constantly pushed us to refine our paper and maintain a high standard.

To the distinguished members of our panel, thank you for your evaluation, constructive critiques, and invaluable recommendations that significantly elevated the quality of this manuscript. We are also thankful for your suggestion to reduce the number of models in this study.

We are deeply grateful to the faculty, staff, and administration of the Computer Science Department of Ateneo de Davao University for providing the intellectual environment, resources, and academic support necessary to carry out this research. Without their help, most of this paper would not exist.

Above all, we express our deepest gratitude to Almighty God for granting us wisdom, resilience, health, and spiritual strength throughout the arduous journey of conceptualizing, engineering, and completing this research.

Finally, our deepest love and gratitude go to our families and loved ones, whose endless care, sacrifices, patience, and unwavering belief in our capabilities sustained us through sleepless nights and challenging technical hurdles. To our friends and colleagues, thank you for the camaraderie, encouragement, and shared perseverance.

Table of Contents

Acknowledgement **i**

Abstract **viii**

1 INTRODUCTION **1**
1.1 Background of the Study 1
1.2 Problem Statement 2
1.3 Objective of the Study 2
1.4 Significance of the Study 3
1.5 Scope and Limitation of the Study 4

2 REVIEW OF RELATED WORKS **5**
2.1 Social Media and Crowdsourcing as Social Sensors in Disaster Informatics . 5
2.1.1 The Human Sensor Paradigm: VGI versus Physical Telemetry . . . 5
2.1.2 Global and Regional Precedents in Crowdsourced Urban Flood Sensing 6
2.1.3 Data Integrity Bottlenecks and the Imperative for Computational Filtering 6
2.2 Text Classification Baselines and Evolution in Crisis Informatics 7
2.2.1 First-Generation Crisis NLP: Lexical Engineering and Statistical Classifiers 7
2.2.2 Second-Generation Crisis NLP: Distributed Word Embedding and Recurrent Architectures 8
2.2.3 Third-Generation Crisis NLP: Pre-trained Contextual Transformers and Cross-Lingal Encoders 9
2.2.4 Architectural Synthesis and Methodological Justification of Benchmark Baselines 9
2.3 Multilingual NLP and Low-Resource Code-Switching (Tagalog and Cebuano) 10
2.3.1 Sociolinguistic Dynamics of Crisis Communication: Taglish, Bislish, and Informal Orthography 10
2.3.2 Subword Tokenization Mechanics: WordPiece versus SentencePiece under Dialectal Mixing 11
2.3.3 Cross-Lingual Representation and Low-Resource Transfer in Transformer Backbones 11

2.3.4 Linguistic Justification for Davao City's Tri-Lingual Disaster Environment . . . 12
2.4 Location Extraction, named Entity Recognition (NER), and Geoparsing . . 13
2.4.1 Geotagging Sparsity and the Geoparsing Imperative in Crisis Informatics . . . 13
2.4.2 Toponym Recognition: Gazetteers versus Statistical Sequence Labeling . . . 14
2.4.3 Deep Sequential NER: Deconstructing the BERT-BiLSTM-CRF Paradigm . . . 14
2.4.4 Disjoint Pipeline Latency, Cascading Errors, and the Shift to Unified Token Heads . . . 15
2.5 Multi-Task Learning: Architecture and Multi-Objective Loss Balancing . . 15
2.5.1 Hard Parameter Sharing and Inductive Bias Transfer . . . 15
2.5.2 Homoscedastic Uncertainty Loss Balancing . . . 16
2.6 Data Quality, Fuzzy Deduplication, and Imbalance Regularization . . . 16
2.6.1 Preprocessing Noise and MinHash/LSH Deduplication . . . 16
2.6.2 SMOTE on Dense Embeddings and Transformer Regularization . . 17
2.7 Spatial-Temporal Clustering, Consensus, and Anomaly Filtering . . . 17
2.7.1 Spatial-Temporal Vector Formulation and Algorithm Selection (K-Means++ vs. ST-DBSCAN) . . . 17
2.7.2 Centroid Anomaly Filtering and Consensus Thresholding . . . 17
2.8 Algorithmic Bias Mitigation, Disaster Silencescapes, and Infrastructure Fragility . . . 18
2.8.1 Digital Divide Skews and Spatial Representation Biases . . . 18
2.8.2 Infrastructure Blackout Voids and Source Authority Override . . . 18
2.9 Web Scraping Ethics, Legal Precedents, and Regulatory Compliance . . . 18
2.9.1 International Jurisprudential Precedents on Automated Public Data Extraction . . . 18
2.9.2 Philippine Statutory Protection and Privacy-Preserving Compliance 19
2.10 Local Philippine Hydrological Dynamics and Social Media Reporting Behaviors . . . 19
2.10.1 Catchment Hydrography and Tri-Basin Inundation Dynamics of Davao City . . . 19
2.10.2 Social Sensing Behaviors and Physical Telemetry Gap Filling . . . 20
2.11 Statistical Evaluation Methodology and Inferential Significance Testing . . 20
2.11.1 Metric Selection Under Severe Class Imbalance . . . 20
2.11.2 Inferential Hypothesis Testing and Multi-Task Neural Evaluation . 21
2.12 Multi-Tier Theoretical Grounding of the SDSS Framework . . . 21
2.12.1 Computational Linguistics Layer and Cross-Lingual Representation 21
2.12.2 Integrated SDSS Theoretical Synthesis . . . 21
2.13 Theoretical Framework . . . 22
2.13.1 Linguistic Analysis Layer: Multi-Task XLM-RoBERTa . . . 22
2.13.2 Validation & Verification Layer: K-Means++ . . . 23
2.13.3 Theory-Grounded Validated Information . . . 23

3 Methodology 24

3.1 Research Design . . . 24
3.1.1 Case Study Scope, Geographic Boundaries, and Bias Controls . . . 24
3.1.2 Five-Stage Computational Architecture . . . 25
3.1.3 Comparative Benchmark and Statistical Framework . . . 28
3.2 Conceptual Framework . . . 28
3.2.1 Case Study Scope, Inherent Biases, and Sampling Gaps . . . 30
3.2.2 Multilingual Preprocessing and Multi-Task Analysis . . . 30
3.2.3 Multilingual Preprocessing and Scraping Noise Mitigation . . . 30
3.2.4 Multi-Task Model Regularization, Balancing, and Overfitting Control 30
3.2.5 Mathematical Assumptions and Algorithmic Bias Control . . . 31
3.2.6 Intelligence Synthesis and Output Phase . . . 31
3.3 Operational Framework . . . 31
3.3.1 Operational Framework of the AGOS Disaster Intelligence Platform 32
3.3.2 Citizen Mobile Application Workflow . . . 32
3.3.3 LGU Administrative Portal Workflow . . . 33
3.3.4 Feedback and Notifications Workflow . . . 34
3.3.5 Integration of the Operational Framework . . . 34
3.3.6 Real-Time Data Ingestion with Selenium Browser Automation . . . 35
3.3.7 Multi-Task NLP Engine (Relevance & Spatial Extraction) . . . 35
3.3.8 Ground Truth Annotation and Dataset Curation . . . 36
3.3.9 Multi-Task Model Regularization, Balancing, and Overfitting Control 37
3.3.10 Mathematical Assumptions and Algorithmic Bias Control . . . 37
3.3.11 Automated Verification and Consensus Building . . . 39
3.3.12 Alert Generation and LGU Dashboard (Output) . . . 39
3.4 Hardware and Software Environment . . . 40
3.4.1 Team Development and Benchmarking Machines . . . 40
3.5 Version Control and Reproducibility . . . 42
3.5.1 Davao Flood Scraper . . . 42
3.5.2 Davao Flood Cleaner . . . 43
3.5.3 Davao Flood Model XLM-RoBERTa . . . 43
3.5.4 AGOS Web App . . . 44
3.6 Baseline Selection . . . 44
3.6.1 Task A: Flood Classification . . . 46
3.6.2 Task B: Location Extraction (NER) . . . 46
3.6.3 Full-System Pipeline Baselines . . . 47
3.7 Evaluation Metrics . . . 47
3.8 Hyperparameter Tuning . . . 49
3.9 Statistical Model Comparison . . . 51
3.9.1 Hypothesis Testing and Significance Testing . . . 52

4 Results and Discussion 53
4.1 Overview of Experimental Setup . . . 53
4.1.1 Dataset partitioning and Stratification . . . 53
4.1.2 Execution Environment and Hardware Specifications . . . 54
4.1.3 Model Hyperparameters and Regularization Configuration . . . 55
4.2 Machine Learning Evaluation on Unseen Test Set . . . 56

4.2.1 Narrative Interpretation of Test Set Results . . . 58
4.3 Evaluation Across the Research Objectives and Graph Specifications . . . 59
4.3.1 Research Objective 1: Classification Performance (Task A) . . . 59
4.3.2 Research Objective 2: Preprocessing & Noise Reduction . . . 61
4.3.3 Research Objective 3: Location Entity Extraction & Spatial Consensus (Task B + K-Means++) . . . 63
4.3.4 Research Objective 4: System Performance Gain over Baselines . . . 65
4.3.5 Research Objective 5: Overall System Reliability & Real-Time Processing Latency . . . 67
4.4 Baseline Comparison and Statistical Validation . . . 69
4.4.1 5-Fold Cross-Validation Performance Comparison . . . 69
4.4.2 Inferential Hypothesis Testing . . . 71
4.5 Feature Importance and Component Ablation . . . 72
4.6 Error Analysis, Edge Cases and Discussion . . . 73
4.6.1 Qualitative Error Analysis and Edge Cases . . . 73
4.6.2 Integration with Existing Literature . . . 74
4.6.3 Practical Implications for the AGOS LGU Dashboard . . . 74

5 Conclusions and Recommendations 76
5.1 Conclusions . . . 76
5.1.1 Overarching Conclusion on Main Problem Statement & General Objective . . . 76
5.1.2 Conclusion on Specific Objective 1: Task A Flood Relevance Classification . . . 76
5.1.3 Conclusion on Specific Objective 2: Quantifying Preprocessing Noise Reduction . . . 77
5.1.4 Conclusion on Specific Objective 3: Task B Barangay Location Extraction & K-Means++ Validation . . . 77
5.1.5 Conclusion on Specific Objective 4: System Performance Gain (ΔF1) Over Baselines . . . 77
5.1.6 Conclusion on Specific Objective 5: Total System Reliability, Scalability, and Deployment Viability . . . 78
5.2 Future Work . . . 78
5.2.1 Recommendations for Future Researchers (Technical Extensions on the Model) . . . 78
5.2.2 Limitations . . . 80
5.2.3 Extensions . . . 81

Bibliography 85

List of Figures

1 Theoretical framework 22

2 Data Acquisition Phase 25
3 Data Processing Phase 26
4 Multi-Task Analysis Phase 26
5 Verification Phase 27
6 Output Phase 27
7 Conceptual Framework 29
8 Operational Framework 31

9 Task A Classification Metric Comparison & Precision-Recall Curve 60
10 Confusion Matrix of Multi-Task XLM-RoBERTa (N=664 61
11 Empirical data retention and noise mitigation efficiency. 63
12 Evaluation of Task B (Barangay Named Entity Recognition) 64
13 Geospatial consensus clustering and anomaly detection across Davao City Barangays. 65
14 Relative performance gain (Delta F1) in system reliability 67
15 Operational Reliability trade-off showing False Alarm Rate (FAR), Miss Rate (MR), and end-to-end processing latency across model architectures . 69

List of Tables

1 Hardware and Software Specifications 41
2 Baseline Selection Table 45
3 Hyperparameter Search Space and Optimal Configuration 50
4 Statistical Model Comparison Table 51

5 Team Development and Benchmarking Machines 54
6 Machine Learning Evaluation Metrics on Unseen Held-Out Test Set 57
7 Task A Classification Performance Comparison 59
8 Data Retention & Noise Filtering Funnel Across Pipeline Stages 62
9 Barangay Location Named Entity Recognition (Task B) Comparison . . . 63
10 System Performance Gain (Delta F1) Over Established Baselines 66
11 Operational System Reliability & Real-Time Latency Benchmarks 68
12 5-Fold Cross-Validation Performance Comparison 70
13 Component Ablation System Matrix 72

Abstract

Flooding remains a critical hazard in the Philippines, necessitating accurate, real-time, localized monitoring to support effective disaster response. While social media platforms serve as pervasive sensors for immediate updates, the reliability of crowdsourced data is frequently compromised by high levels of noise and misinformation. Existing research has explored machine learning for classification, yet many systems lack integrated verification mechanisms to distinguish credible reports from noise at the barangay level.

This study proposes a Multi-Task NLP framework that integrates supervised text classification and fine-grained location extraction using a single Multi-Task XLM-RoBERTa model. The framework further utilizes K-Means++ clustering to implement consensus-building logic and identify anomalous reports, ensuring that only high-confidence data is validated. By synthesizing these techniques into a unified pipeline, the system transforms raw crowdsourced data from platforms like Twitter (X), Facebook, and Reddit into reliable disaster intelligence. The system's performance is evaluated based on its ability to reduce noise and its overall accuracy, precision, recall, and F1-score in distinguishing credible flood reports to support localized decision-making for LGUs and emergency responders.

Keywords: Flooding, Natural Language Processing (NLP), Multi-Task Learning, Named Entity Recognition (NER), K-Means++ Clustering, Barangay-Level Monitoring

Chapter 1

INTRODUCTION

1.1 Background of the Study

Flooding is one of the most common natural hazards in the Philippines, and it continues to cause serious damage to infrastructure, communities, and public safety, especially during periods of heavy rainfall and typhoons. Urban areas are particularly at risk due to rapid development and insufficient drainage systems (Besarra et al., 2024). One of the main difficulties in managing floods is the lack of real-time, localized information. Traditional monitoring systems often depend on manual reporting, which can lead to delays in response. Because of this, people have increasingly turned to social media to share immediate updates. However, these posts are often unverified, which makes them unreliable for official use (Erokhin, 2024).

To address these challenges, different technologies and approaches have been developed to improve real-time awareness and disaster response. Crowdsourced reporting systems, for example, allow the public to share flood updates quickly. While useful, the data that can be collected is often inconsistent, incomplete, and sometimes affected by misinformation. Participation also tends to vary across different locations, which can lead to gaps in coverage (Erokhin, 2024; SMLD Review, 2026). At the same time, machine learning and deep learning methods have been applied to classify disaster-related content. However, these approaches still struggle with issues such as noisy text data, computational overhead, imbalanced datasets, and limited adaptability in real-world situations (SMLD Review, 2026).

More recently, advances in Natural Language Processing (NLP) have opened new possibilities for better analyzing unstructured social media data without relying on heavy, disjointed model pipelines. Transformer-based models like XLM-RoBERTa can handle multilingual text highly effectively, making them ideal for processing flood-related posts written in English, Tagalog, Cebuano, or a code-switched mix of the three (Zhang et al., 2026). By transitioning to a Multi-Task Learning (MTL) architecture, a single XLM-RoBERTa engine can be optimized to simultaneously classify a post's relevance and perform Named Entity Recognition (NER) to extract specific locations like barangays and streets, entirely eliminating the computational overhead of separate extraction models.

Furthermore, spatial validation techniques like K-Means++ clustering can group similar reports geographically, allowing for the identification of high-density consensus areas while filtering out isolated, anomalous reports that fall outside these cluster centroids (Hanny & Resch, 2024; Eze et al., 2023).

Despite these developments, many existing systems still treat classification, location extraction, and validation as isolated tasks, leading to fragmented analysis and heavy processing requirements. There is also a lack of studies focusing on cohesive, barangay-level flood monitoring using a streamlined approach. This gap highlights the need for "A Multi-Task NLP Framework for Barangay-Level Validation of Crowdsourced Flood Reports." By integrating XLM-RoBERTa as a unified engine and refining spatial validation through K-Means++, this study aims to synthesize textual relevance and spatial consensus into a focused Confidence Score. Furthermore, the framework incorporates a Source Authority Override to mitigate the "Cold Start Problem" during the early stages of a disaster. Ultimately, this research shifts away from an ensemble of disparate models toward a cohesive, localized intelligence system designed to improve the accuracy, reliability, and real-time usefulness of crowdsourced flood data.

1.2 Problem Statement

This study aims to develop an NLP-based flood monitoring system that integrates text classification, location extraction, and validation mechanism to improve reliability of crowdsourced flood reports. Specifically, the study seeks to answer the following questions:

1. How accurate is the XLM-RoBERTa model in classifying flood-related and non-flood-related textual data based on metrics such as accuracy, precision, recall, and F1-score?
2. To what extent can preprocessing techniques reduce text cleaning, tokenization, normalization, stopword removal, and duplicate filtering reduce noise and irrelevant information in crowdsourced flood reports before classification?
3. How effective is the integration of a single Multi-Task NLP model with consensus based clustering (K-Means++) in improving the overall reliability of identifying credible flood reports and filtering out spatial anomalies?
4. How much improvement in system performance can be achieved when the multi-task XLM-RoBERTa model is integrated with K-Means++ clustering, compared to a baseline classification only approach?
5. How reliable is the overall system in distinguishing credible and non credible flood reports based on evaluation metrics such as accuracy, precision, recall, and F1-score after integrating the selected models into a unified framework?

1.3 Objective of the Study

The general objective of this study is to develop a Multi-Task NLP-based flood monitoring system that integrates classification, fine-grained extraction, and consensus-based

validation to improve the reliability of crowdsourced flood reports at the barangay level.

Specifically, the study aims to:

1. Measure the performance of the XLM-RoBERTa model in identifying flood-related versus non-flood-related textual data using metrics such as accuracy, precision, recall, and F1-score.
2. Quantify the reduction of noise in crowdsourced reports achieved through preprocessing techniques, including cleaning, tokenization, normalization, and duplicate filtering.
3. Evaluate the effectiveness of the Multi-Task XLM-RoBERTa model for fine-grained location extraction and K-Means++ clustering for consensus-based validation to identify credible reports and filter out spatial anomalies.
4. Measure the increase in system performance (e.g., F1-score) when the Multi-Task XLM-RoBERTa model is integrated with K-Means++ clustering compared to a baseline classification-only approach.
5. Measure the system's total reliability in distinguishing between credible and non-credible flood reports through a unified framework applied to localized monitoring.

1.4 Significance of the Study

This study contributes to the fields of Disaster Informatics, Flood Monitoring Systems, and Natural Language Processing (NLP) by proposing a streamlined Multi-Task NLP-based framework that integrates multilingual classification and location extraction with consensus-based validation. Unlike existing approaches that focus primarily on classification, this research advances text-based disaster monitoring by incorporating automated verification mechanisms that transform raw crowdsourced data into reliable disaster intelligence. Technologically, the study strengthens the application of NLP in real-time social media analysis, specifically by demonstrating how a single Multi-Task model can handle noisy, unstructured, and code-switched datasets (English, Tagalog, and Cebuano) within localized barangay contexts.

For stakeholders, the study provides multiple practical benefits. Local Government Units (LGUs) and DRRMOs benefit from a decision-support system that provides accurate and verified flood information at the barangay level, enabling better planning, faster response times, and optimized resource allocation. Emergency Responders gain the ability to filter credible reports from high volumes of social media noise, allowing them to prioritize urgent situations based on algorithmic confidence scores and multi-user corroboration. Communities and Citizens benefit from access to validated flood updates, which enhances localized awareness, personal preparedness, and overall safety during extreme weather events. Furthermore, researchers and developers can use this framework as a reference for building more efficient and scalable disaster monitoring systems that utilize multi-task learning to reduce computational overhead.

In relation to the United Nations Sustainable Development Goals (SDGs), this

study supports SDG 11: Sustainable Cities and Communities by improving disaster resilience through reliable monitoring, and SDG 13: Climate Action by enhancing the use of technology in disaster preparedness. By enabling accurate, real-time, and localized flood information, the study contributes to building safer, data-driven, and more resilient communities.

1.5 Scope and Limitation of the Study

The scope of this research centers on the development and evaluation of a unified NLP-based flood monitoring system that processes and validates crowdsourced textual data from social media such as Facebook public posts/pages. The system follows a structured pipeline beginning with data acquisition through web scraping and keyword/hashtag filtering, followed by rigorous preprocessing including cleaning, normalization, and duplicate removal. To ensure robust and efficient analysis, the framework utilizes Multi-Task XLM-RoBERTa, a transformer-based model capable of handling multilingual text effectively. This single model performs dual functions: identifying flood-relevant reports written in English, Tagalog, and Cebuano and performing fine-grained Named Entity Recognition (NER) to extract localized locations at the barangay level. The system further performs validation through a module utilizing K-Means++ for consensus-building logic, which groups reports by aligning extracted spatial features with posting timestamps. The system's reliability and performance are evaluated using standard metrics such as accuracy, precision, recall, and F1-score, with the ultimate goal of providing validated disaster intelligence to Local Government Units (LGUs) and emergency responders within a barangay-level context.

Conversely, the study faces several limitations, most notably its exclusive reliance on textual data, excluding context that could be provided by images, videos, or IoT-based sensors. The accuracy of the system is highly dependent on the availability and quality of crowdsourced data, which is often subject to noise, misinformation, and uneven geographic participation. Although the standalone Isolation Forest model has been removed to streamline the pipeline, the system continues to identify "noise" and outliers through the K-Means++ module. Specifically, reports that do not meet the minimum density threshold to form a cluster or those that fall significantly far from established centroids are treated as spatial anomalies and flagged as low-confidence. Consequently, the system still faces a "cold start" challenge during the onset of a disaster, where early, single reports may be flagged as low-confidence due to a lack of corroborating data.

Chapter 2

REVIEW OF RELATED WORKS

2.1 Social Media and Crowdsourcing as Social Sensors in Disaster Informatics

2.1.1 The Human Sensor Paradigm: VGI versus Physical Telemetry

Traditional hydrometeorological monitoring relies heavily on physical telemetry, such as automated river stage gauges, rain gauges, and satellite remote sensing networks. While physical sensors provide calibrated quantitative measurements, they suffer from fundamental structural limitations during sudden urban inundations: sparse spatial density, prohibitive capital and maintenance costs, transmission delays, and catastrophic sensor washouts during extreme flood surges. In dense, tropical urban environments, river gauges are frequently situated kilometers apart along primary channels, leaving tributaries, localized street drainage channels, and low-lying residential basins completely unmonitored.

To address these spatial coverage voids, disaster informatics has increasingly embraced Volunteered Geographic Information (VGI) (Goodchild, 2007) and the Social Sensing Paradigm (Sakaki et al., 2010). VGI conceptualizes citizens as distributed, real-time "human sensors" who actively or passively transmit observations about their immediate environment (Goodchild, 2007). Rather than relying solely on static physical hardware, social sensing leverages ubiquitous mobile devices and social media platforms to capture micro-level situational awareness at zero sensor deployment cost (Sakaki et al., 2010; Helmrich et al., 2021). Citizens experiencing flash floods post immediate street-level reports containing granular observational details such as rising water depths relative to landmarks, road impassability, trapped vehicles, and immediate rescue requirements (Chen et al., 2022; Helmrich et al., 2021). Consequently, crowdsourced social sensing complements physical telemetry by transforming passive citizens into an active, hyper-local observation network capable of providing continuous situational updates during high-velocity urban disasters.

2.1.2 Global and Regional Precedents in Crowdsourced Urban Flood Sensing

The empirical utility of crowdsourced social sensing for urban flood management has been demonstrated across diverse geographic scales. In tropical Southeast Asian urban centers characterized by intense monsoonal cloudbursts and rapid hydrological response times, social media feeds have proven indispensable for rapid flood hazard mapping. In Jakarta, Indonesia an urban environment highly analogous to Philippine metropolitan centers in climate, rapid urbanization, and high social media penetration Eilander et al. (2016) harvested social media posts to construct near real-time inundation maps, demonstrating that citizen reports can track flood extent with high spatiotemporal fidelity. At a global scale, de Bruijn et al. (2019) aggregated multilingual social media streams to detect and track major flood events across national boundaries, proving that cross-lingual social text can serve as an effective proxy for global hazard monitoring.

Recent computational studies have further confirmed the operational value of crowdsourced disaster intelligence across developing urban regions. In Zhengzhou, China, Peng and Zhang (2024) applied spatiotemporal mining to microblogging feeds during extreme storm events, demonstrating that social media activity strongly correlates with localized inundation severity and offers fine-grained hazard assessment where physical sensors failed. Similarly, in Mumbai, India a coastal megacity sharing climate and infrastructure vulnerabilities with major Philippine urban centers Navalkar et al. (2025) deployed Natural Language Processing (NLP) frameworks on crowdsourced feeds to extract real-time flood reports and identify urban waterlogging hotspots. Furthermore, critical reviews by Roy et al. (2021) and Gopal et al. (2026) highlight that social media data, when systematically harvested and processed, yields actionable insights for municipal emergency response, evacuation planning, and post-disaster damage assessment.

2.1.3 Data Integrity Bottlenecks and the Imperative for Computational Filtering

Despite its high spatiotemporal resolution, unvalidated crowdsourced data cannot be directly consumed by Local Government Units (LGUs) or emergency dispatchers. Social media feeds suffer from severe data integrity bottlenecks (Gopal et al., 2026; Songchon et al., 2021). Utilizing social media for disaster response presents significant challenges, primarily because less than 5% of raw posts contain actionable emergency requests (Roy et al., 2021). This extreme signal-to-noise ratio is exacerbated by feeds heavily polluted with bot spam, political commentary, viral reposts, and commercial marketing that opportunistically leverages disaster-related search terms (Songchon et al., 2021).

Furthermore, informal social text frequently blends regional dialects and figurative idioms—such as the metaphorical "baha sa orders"—which severely increases false positive rates in simple keyword filters (Navalkar et al., 2025). Data extraction is also hindered by demographic skews that concentrate digital activity in affluent urban centers, while power and cellular outages in the most devastated areas create "disaster silencescapes" where vulnerable communities become digitally silent while less affected regions remain vocal (Chen et al., 2022; Gopal et al., 2026). Ultimately, these inherent data quality

and veracity deficits risk the rapid spread of rumors and unverified claims, threatening dispatch credibility and inducing responder fatigue (Songchon et al., 2021).

These integrity bottlenecks establish an imperative for automated, multi-stage computational filtering before crowdsourced reports reach operational dashboards (Navalkar et al., 2025; Songchon et al., 2021). Simple keyword scrapers or standalone text classifiers are fundamentally insufficient; a post mentioning a flood keyword does not guarantee a real emergency at that location, nor does it resolve fine-grained spatial boundaries.

To bridge raw social feeds with official municipal decision-support systems, a framework must execute concurrent multilingual semantic classification and fine-grained Named Entity Recognition (NER) to extract localized barangay names. Furthermore, extracted spatial entities must undergo spatial-temporal consensus verification (e.g., K-Means++ clustering over coordinate vectors $\mathbf{p}_i = [x_i, y_i, t_i]^T$) to ensure that alerts are backed by multi-user spatial density ($\theta \geq 3$ reports) while isolating spatial anomalies. Finally, to prevent discarding single-source critical alerts from official responders during the initial minutes of a flash flood, the architecture must integrate a Source Authority Override ($CS = 1.0$), bypassing spatial consensus for pre-verified handles. Without this integrated computational pipeline, crowdsourced social sensing remains an uncurated, high-risk data stream unsuitable for life-safety emergency dispatch.

2.2 Text Classification Baselines and Evolution in Crisis Informatics

2.2.1 First-Generation Crisis NLP: Lexical Engineering and Statistical Classifiers

Early research in crisis information relied on classification machine learning paradigms that coupled with hand crafted feature engineering with linear statistical classifiers. To help filter disaster-related social media posts during sudden environmental hazards, foundational systems helps leverage Term Frequency-Inverse Document Frequency (TF-IDF) metrics, n-grams bag-of-words representations, and domain-specific lexicons processed through Logistic Regression, Support Vector Machines (SVM), or Naive Bayes classifiers (Imran et al., 2015; Olteanu et al., 2014). The release of standardized disaster lexicons, such as CrisisLex (Olteanu et al., 2014) formalized keyword-based filtering by matching post tokens against curated dictionaries of disaster terminology.

Operationally, first-generation statistical classifiers offer distinct advantages: linear time computational complexity O(N), minimal memory footprints, high inference throughput, and transparent model interpretability via inspection of feature coefficients (Imran et al., 2015). However, when deployed on uncurated, crowdsourced disaster feeds, lexical engineering exhibits severe structural failure modes.

The limitations of traditional bag-of-words and n-gram representations stem primarily from their strict conditional independence assumption between tokens, which com-

pletely discards syntactic word order and sentence dependencies. This structural limitation makes linear classifiers incapable of processing negation syntax and local contextual modifiers; for example, in Philippine social media feeds, expressions like "walang baha" (no flood) or commercial real estate ads boasting "100% Flood-Free" contain the high-weight hazard token "baha," causing sparse TF-IDF models to invariably trigger severe false-positive spikes (Olteanu et al., 2014). Additionally, frequency-based features conflate literal disaster reports with lexical polysemy and idiomatic expressions, such as "baha sa orders" or "bumaha ng ayuda," leading to systematic misclassifications. Finally, exact token-matching algorithms exhibit extreme out-of-vocabulary (OOV) fragility, failing when confronted with panicked character repetitions like "baaaaaaahaaaaaa," colloquial contractions, regional orthographic variations, or informal social media slang (Imran et al., 2015).

2.2.2 Second-Generation Crisis NLP: Distributed Word Embedding and Recurrent Architectures

The transition from sparse lexical representations to dense, continuous vector spaces marked the second generation of disaster text classification. Distributed representation algorithms such as Word2Vec, Global Vectors for Word Representation (GloVe), and FastText—mapped words into low-dimensional vector spaces where semantic relationships were preserved through geometric proximity (Nguyen et al., 2017). In crisis informatics, these static embeddings were paired with deep neural architectures, primarily Convolutional Neural Networks (CNNs) for localized n-gram feature extraction and Bidirectional Long Short-Term Memory (BiLSTM) networks for sequential context modeling (Nguyen et al., 2017).

Although second-generation deep learning architectures mitigated the out-of-vocabulary limitations of rigid keywords matching, they introduced fundamental structural bottlenecks. The principal among these is their reliance on context-independent static lookups, which can be assigned a single vector to each token regardless of the surrounding sentence syntax; consequently, a term like “baha” maps to an identical point in a vector space whether it denotes a severe inundation like “pabaha sa matina”, a real estate claims such as “100% flood free”, or a metaphorical phrase like “baha ng tao or baha sa orders”. Additionally, recurrent network architectures like BiLSTMs process inputs sequentially, resulting in O(N) temporal recurrence latency during inference, remaining susceptible to vanishing gradients over long sequences, and failing to exploit parallel matrix processing across GPU hardware (Nguyen et al., 2017)., Finally, while while charactering n-gram model such as FastText improved handling of unseen terms, static embeddings still failed to dynamically adjust token representations based on a broader syntactic context, severely limiting their performance across morphologically rich regional dialects.

2.2.3 Third-Generation Crisis NLP: Pre-trained Contextual Transformers and Cross-Lingal Encoders

The emergence of the Transformer architecture (Vaswani et al., 2017) and self-attention mechanisms fundamentally altered crisis computing by replacing sequential recurrence with fully parallelizable bidirectional contextual encodings. Models such as Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) dynamically compute token representations based on full sentence context, capturing complex syntactic dependencies, negation, and semantic polysemy. The utility of transformer encoders in disaster management was further formalized through large-scale, human-annotated crisis benchmarks, such as HumAID and CrisisBench (Alam et al., 2021).

To process social streams in multilingual and low-resource regions, research expanded to multilingual pre-trained encoders, specifically comparing Multilingual BERT (mBERT) and XLM-RoBERTa (Conneau et al., 2020). The operational divergence between these architectures lies in their underlying pre-training corpora and subword tokenization mechanics. Fine-tuned on Wikipedia text across 104 languages using a 110,000-token WordPiece vocabulary, mBERT relies on a corpus dominated by formal prose; consequently, its WordPiece tokenizer severely fragments informal, agglutinative Tagalog and Cebuano affixes (e.g., "pabaha", "gibaha", "gipabakwit") into sub-optimal subword chunks, degrading semantic encoding in regional dialects (Conneau et al., 2020; Alam et al., 2021). In contrast, XLM-RoBERTa is pre-trained on 2.5 TB of filtered CommonCrawl web text across 100 languages using a expanded 250,000-token SentencePiece vocabulary (Conneau et al., 2020). Because SentencePiece operates directly on raw text streams without requiring language-specific whitespace tokenizers, XLM-RoBERTa better preserves the morphological integrity of regional Philippine code-switched dialects such as Taglish and Bislish (Conneau et al., 2020).

2.2.4 Architectural Synthesis and Methodological Justification of Benchmark Baselines

To eliminate "solution-first bias" and establish a rigorous empirical foundation, the baseline selection in Chapter 3 (Section 3.6.1) and performance evaluation in Chapter 4 (Section 4.3.1) directly reflect the historical evolution of crisis NLP. Following the comparative evaluation framework established by Abiola et al. (2025) for low-resource multilingual classification, this study implements a three-tiered baseline hierarchy to scientifically isolate and quantify the performance gains of the proposed architecture:

Tier 1: Non-Deep Learning Lexical Baseline (Logistic Regression with TF-IDF): Establishes the baseline for computational speed and linear separability (Alam et al., 2021; Abiola et al., 2025). In Chapter 4 (Table 4.2), Logistic Regression yields an inferior Macro F1-score of 0.7796. This empirical baseline confirms that frequency-based n-gram representations fail to process negation syntax ("walang baha") and commercial real estate noise, validating the necessity of deep neural representations.

Tier 2: Single-Task Multilingual Transformer Baseline (mBERT): Establishes the baseline for single-task deep learning without cross-task parameter sharing (Abiola et

al., 2025). In Table 4.2, mBERT achieves a Macro F1-score of 0.8691. Benchmarking against mBERT isolates the exact performance limitations imposed by Wikipedia-skewed pre-training and WordPiece subword fragmentation when processing informal Philippine vernaculars.

Tier 3: Single-Task vs. Multi-Task XLM-RoBERTa Comparison: Isolates the specific architectural contribution of XLM-RoBERTa's SentencePiece vocabulary (Single-Task F1 = 0.9147) and demonstrates the benefit of Multi-Task Learning (Proposed System Macro F1 = 0.9347, PR-AUC = 0.9628) (Abiola et al., 2025). Jointly fine-tuning the shared transformer backbone on Task A (Relevance Classification) and Task B (Barangay NER) acts as an inductive regularizer, forcing the shared self-attention layers to encode contextual hazard-location relationships.

2.3 Multilingual NLP and Low-Resource Code-Switching (Tagalog and Cebuano)

2.3.1 Sociolinguistic Dynamics of Crisis Communication: Taglish, Bislish, and Informal Orthography

During emergency situations in the Philippines, citizen communication on social media platforms deviates substantially from standard grammatical structures. Crisis text is characterized by fluid, multi-layered code-switching that operates across both intra-sentential (mixing dialects within a single clause) and inter-sentential (alternating languages across sentence boundaries) modes (Aguilar et al., 2020; Oco & Roxas, 2012). In major urban centers such as Davao City, social media feeds combine English (the primary language of official agency advisories), Tagalog/Filipino (the national administrative language), and Cebuano/Bisaya (the dominant regional language), giving rise to localized linguistic constructs known as Taglish and Bislish (Oco & Roxas, 2012). A single crowdsourced report frequently blends all three registers (e.g., "Grabe na ang baha sa Matina Crossing, please stay safe everyone, dili na maagian ang dalan"), presenting complex syntax that defeats monolingual natural language processing pipelines (Aguilar et al., 2020).

The difficulty of parsing Philippine crisis text is exacerbated by informal orthography and dialectal shifts (Oco & Roxas, 2012). Under extreme distress, citizens introduce panic character elongations (e.g., "baaaaaaahaaaaaa" or "bhaaaa"), non-standard phonetic contractions, and emoji sequences to convey urgency. Furthermore, regional negation shifts alter sentence polarity across dialectal boundaries. While Tagalog utilizes "walang baha" (no flood) to denote safety, Cebuano/Bisaya employs "walay baha" or "layo sa baha" (far from the flood) (Oco & Roxas, 2012). When unhandled, these negation markers cause keyword-based filters to misinterpret promotional real-estate listings (e.g., "House and Lot in Matina: 100% Flood-Free / Walay Baha") as active emergencies. Finally, colloquial expressions and figurative idioms introduce severe semantic ambiguity (Aguilar et al., 2020). Terms such as "lapas-tao" (literally "over-person height", denoting chest-deep water) are frequently repurposed in commercial text to describe a flood of customer orders, while the Cebuano verb "nibaha" ("flooded") is used metaphorically to

describe a flood of relief donations ("nibaha ang tabang"), creating complex false-positive edge cases for automated classifiers.

2.3.2 Subword Tokenization Mechanics: WordPiece versus SentencePiece under Dialectal Mixing

The foundational bottleneck in processing low-resource, code-switched crisis text lies in subword tokenization mechanics. Traditional pre-trained multilingual transformers, such as Multilingual BERT (mBERT), rely on the WordPiece tokenization algorithm operating over a 110,000-token vocabulary (Conneau et al., 2020; Kudo, 2018). Under the "curse of multilinguality," a fixed vocabulary space partitioned across 104 languages allocates minimal token budgets to regional, low-resource Malayo-Polynesian dialects (Conneau et al., 2020). Because mBERT's pre-training corpus was heavily skewed toward formal prose from Wikipedia, its WordPiece tokenizer lacks representation for regional agglutinative affixes. In Tagalog and Cebuano, grammatical relationships are expressed through complex affixation patterns attached to root words (e.g., the root "baha" transforms into "pabaha", "gibaha", "gipabakwit", or "mabaha") (Cruz & Cheng, 2020). When encountering these morphologically rich forms, WordPiece excessively fragments words into atomic, non-semantic subword or byte-level chunks (e.g., Gi + ##bah + ##a), destroying contextual boundaries and triggering severe Out-Of-Vocabulary (OOV) bottlenecks (Kudo, 2018)

In contrast, XLM-RoBERTa employs a 250,000-token SentencePiece vocabulary trained on 2.5 TB of filtered CommonCrawl web text across 100 languages (Conneau et al., 2020). SentencePiece treats the input text as a raw character stream without requiring language-specific whitespace pre-tokenizers, making it well suited for informal social media text (Kudo, 2018). Its expanded subword vocabulary covers agglutinative prefixes and suffixes native to Philippine languages, allowing the tokenizer to construct intact subword units (e.g., mapping "Gibaha" to a single coherent subword token _Gibaha) (Conneau et al., 2020; Kudo, 2018). By preserving the morphological integrity of regional code-switched text, SentencePiece tokenization prevents subword fragmentation, stabilizes self-attention weights, and provides a dense representation space for downstream classification and entity extraction.

2.3.3 Cross-Lingual Representation and Low-Resource Transfer in Transformer Backbones

Massively multilingual transformer encoders overcome low-resource data scarcity through cross-lingual representation learning (Cruz & Cheng, 2020; Zhang & Wang, 2026). By optimizing a Masked Language Modeling (MLM) objective over multi-gigabyte corpora, architectures like XLM-RoBERTa construct a shared, language-agnostic latent space where semantically equivalent concepts across different languages are mapped to proximate geometric vectors (Conneau et al., 2020; George et al., 2025). Through shared self-attention layers, syntactic structures and disaster concepts learned from high-resource languages (e.g., English crisis corpora) automatically transfer to low-resource regional dialects (e.g., Tagalog and Cebuano) without requiring parallel translation matrices (Cruz & Cheng,

2020; Zhang & Wang, 2026).

Semantic Equivalence Across Languages: It demonstrates how semantically identical flood descriptions across three different languages/dialects—English ("Severe Inundation"), Tagalog ("Matinding Baha"), and Cebuano ("Lapas-Tao na Baha")—map to proximate continuous vectors in the same shared geometric space without needing separate translation dictionaries or parallel corpora. Inductive Cross-Lingual Transfer: It illustrates how the shared self-attention layers of the transformer backbone allow high-resource language representations (English/Tagalog) to regularize and improve representations for low-resource regional dialects (Cebuano/Bisaya).

In multilingual disaster informatics, cross-lingual representation learning allows a single model to understand semantic concepts across different languages without requiring parallel bilingual translation datasets (Cruz & Cheng, 2020; Zhang & Wang, 2026). XLM-RoBERTa projects semantically equivalent disaster expressions across English ("Severe Inundation"), Tagalog ("Matinding Baha"), and Cebuano ("Lapas-Tao na Baha") into close geometric proximity within a shared continuous embedding space. Rather than parsing each regional dialect through disjoint, siloed tokenizers, the shared self-attention layers exploit inductive cross-lingual transfer. This enables the model to transfer contextual emergency knowledge learned from high-resource English and Tagalog corpora directly into lower-resource, colloquial Cebuano and Bislish crisis feeds common in Davao City (Zhang & Wang, 2026).

This cross-lingual alignment is vital for Philippine disaster informatics, where domain-specific disaster training datasets in regional dialects are small. Pre-trained multilingual embeddings enable inductive transfer, allowing the model to project localized Cebuano hazard expressions (e.g., "taas ang tubig", "gipabakwit") into the same semantic cluster as English emergency terms ("flash flood", "forced evacuation") (Cruz & Cheng, 2020; George et al., 2025). Furthermore, fine-tuning the shared encoder within a Multi-Task Learning (MTL) framework strengthens cross-task feature representations (Zhang & Wang, 2026). Simultaneously optimizing sequence classification (Task A) and token-level Named Entity Recognition (Task B) forces the shared self-attention heads to attend to spatial modifiers alongside hazard terms, reinforcing entity boundaries across code-switched inputs.

2.3.4 Linguistic Justification for Davao City's Tri-Lingual Disaster Environment

The sociolinguistic, tokenization, and cross-lingual transfer principles outlined above directly inform the methodology and empirical evaluation of this thesis. First, standard transformer fine-tuning protocols fix the maximum sequence length at 128 subword tokens, which leads to severe tail-end truncation due to the long intra-sentential code-switching in Davao City conversational social posts that combine situational narratives, landmark directions, and safety wishes (Conneau et al., 2020). Expanding the maximum sequence length to 256 subword tokens in Chapter 4 (Section 4.1.3) successfully accommodated 90.0% of raw conversational posts without truncation, preserving critical spatial entities located at the end of longer messages. Second, to prevent panicked character elongations

from inflating subword token counts and exceeding this 256-token limit, Chapter 3 (Section 3.3.3) implements a regex-based pre-tokenization cleaner that normalizes elongated sequences (such as mapping "baaaaaaahaaaaaa" back to dictionary forms like "baha") in Tagalog and Cebuano prior to tokenization, ensuring clean input representations for the SentencePiece tokenizer. Finally, the empirical limitations documented in Chapter 4 (Section 4.6.1) directly reflect residual linguistic edge cases, where despite XLM-RoBERTa's 250,000-token SentencePiece vocabulary, highly creative figurative expressions such as the commercial use of "lapas-tao" for customer orders or the metaphorical use of "nibaha" for charity donations accounted for 41.6% of the 24 false-positive errors on the test set, confirming that idiomatic language shifts remain an active boundary in computational disaster informatics.

2.4 Location Extraction, named Entity Recognition (NER), and Geoparsing

2.4.1 Geotagging Sparsity and the Geoparsing Imperative in Crisis Informatics

In crowdsourced disaster informatics, translating raw social media streams into actionable spatial intelligence requires converting unstructured text into validated geographic coordinates. This process is formally defined as geoparsing, a two-stage computational framework comprising toponym recognition (detecting and demarcating place-name entities within text) and toponym resolution (disambiguating candidate locations and mapping identified names to explicit spatial coordinates, such as latitude and longitude) (de Bruijn et al., 2018). While early location-based social network research assumed the availability of native device metadata, empirical studies consistently reveal a severe geotagging deficit: native Global Navigation Satellite System (GNSS) or GPS coordinates embedded in social media metadata headers are present in fewer than 1–2% of public disaster microblogs (de Bruijn et al., 2018; Middleton et al., 2014).

Geoparsing operates as a sequential two-stage pipeline: Stage 1 isolates content-mentioned spatial tokens via statistical Named Entity Recognition (Toponym Recognition), while Stage 2 grounds these entity spans into absolute geographic coordinates via gazetteer lookup and disambiguation (Toponym Resolution) (de Bruijn et al., 2018).

This geotagging sparsity stems from user privacy preferences, battery conservation during emergency power grid failures, and default platform configurations that strip location metadata (Middleton et al., 2014). Consequently, emergency management agencies cannot rely on native device coordinates to map hazardous events. Extracting spatial entities mentioned directly within the message body content-mentioned geoparsing is therefore an indispensable operational requirement for crowdsourced disaster response (de Bruijn et al., 2018; Middleton et al., 2014).

2.4.2 Toponym Recognition: Gazetteers versus Statistical Sequence Labeling

To extract place names from text, early disaster management systems relied on static gazetteers pre-compiled geographical databases such as GeoNames or OpenStreetMap and string-matching rules (Gelernter & Balaji, 2013). While gazetteer lookups execute in linear time, they exhibit fatal failure modes when applied to informal Philippine emergency streams (Gelernter & Balaji, 2013):

The processing of location data from informal social media text faces several complex challenges that cause exact dictionary-matching lookups to consistently fail. Primarily, informal abbreviations and orthographic noise, such as users shortening administrative terms to "brgy" or introducing erratic capitalization, create an immediate hurdle for standard tools. Crucially, critical toponymic information is often entirely missing from official records, with emergency reports frequently referencing non-administrative, colloquial landmarks like 'crossing', 'gym', 'overpass', or 'near chapel' that are absent from government gazetteers but vital for responder navigation (Gelernter & Balaji, 2013). This problem is further exacerbated by semantic ambiguity arising from name-entity homonymy and polysemy, where static lookups cannot resolve when a place name matches a personal name or a common noun. Finally, the nested hierarchical structure of Philippine spatial geography creates unique difficulties when unstructured text concatenates multiple layers, a task that static gazetteers struggle with but statistical Named Entity Recognition (NER) models handle by analyzing surrounding syntactic, morphological, and positional context to accurately label complex spatial expressions (Gelernter & Balaji, 2013).

2.4.3 Deep Sequential NER: Deconstructing the BERT-BiLSTM-CRF Paradigm

To overcome the rigid limits of gazetteers, deep learning architectures formulate location extraction as a token-level sequence labeling task (Lample et al., 2016; Wang et al., 2023). Among sequential architectures, the BERT-BiLSTM-CRF framework has long represented the state-of-the-art paradigm in disaster entity extraction (Li et al., 2025; Wang et al., 2023). This architecture employs a specialized three-layer stack: The architecture combines multiple specialized neural components to accurately identify spatial entities from raw text streams. First, a Contextual Transformer Encoder generates dense contextual subword embeddings directly from raw text input. These transformer representations are then passed through a Bidirectional LSTM Layer, which processes forward and backward hidden states to effectively model long-range sequential token dependencies. Finally, a Conditional Random Field (CRF) Layer computes an optimal global tag sequence across the sentence by modeling label transition probabilities between adjacent tags (Lample et al., 2016).

Token labeling operates under the standard Inside-Outside-Beginning (IOB / BIO) annotation scheme (Ramshaw & Marcus, 1995). In this scheme, B-LOC marks the initial token of a spatial entity, I-LOC denotes interior tokens of a multi-word entity, and O represents non-spatial tokens (Ramshaw & Marcus, 1995). The CRF layer mathematically penalizes syntactically illegal tag transitions such as an I-LOC tag directly

following an O tag without an intervening B-LOC tag thereby enforcing structural consistency across multi-word place names (e.g., Brgy [B-LOC], Matina [I-LOC], Crossing [I-LOC]) (Lample et al., 2016; Ramshaw & Marcus, 1995). Despite its high precision, the BERT-BiLSTM-CRF framework incurs substantial computational costs when deployed in real-time disaster monitoring environments.

2.4.4 Disjoint Pipeline Latency, Cascading Errors, and the Shift to Unified Token Heads

In production disaster monitoring systems, NER models are frequently deployed as downstream components within multi-stage disjoint pipelines, where an independent text classifier first filters relevant posts before passing them to a separate NER model (Li et al., 2025; Wang et al., 2023). Critical evaluation reveals two fundamental structural failure modes in these decoupled systems, as benchmarked in Chapter 4 (Section 4.3.4, Baseline Pipeline B): Cascading Error Propagation: Decoupled pipelines suffer from unidirectional error transfer. If the upstream relevance classifier produces a false negative misclassifying an emergency report containing vital location details as non-disaster-related, the post is immediately discarded. The downstream NER model is permanently starved of input, resulting in an unrecoverable loss of critical spatial intelligence. Redundant Computational Latency and VRAM Allocation: Running disjoint neural networks requires passing input text through multiple separate transformer encoders, doubling forward-pass matrix operations and GPU memory allocations. In Chapter 4 (Table 4.6), Baseline Pipeline B exhibits a processing latency of 384.2 ms per report, compared to 142.1 ms per report for the proposed unified framework. In fast-moving urban flash floods, a 2.7 latency penalty slows real-time stream ingestion and reduces system throughput.

To eliminate these architectural bottlenecks, recent crisis informatics research favors replacing heavy BiLSTM-CRF stacks with a direct token classification head attached to a shared multilingual transformer backbone (Li et al., 2025; Wang et al., 2023). Fine-tuning a shared Multi-Task XLM-RoBERTa model concurrently on Task A (Relevance Classification) and Task B (Barangay NER) enables inductive parameter sharing. The shared self-attention layers learn joint representations where sentence-level disaster intent informs token-level spatial boundaries, and vice versa. This design eliminates upstream error cascading, reduces VRAM usage, and slashes processing latency to 142.1 ms per report, providing the throughput required for real-time municipal emergency response dashboards.

2.5 Multi-Task Learning: Architecture and Multi-Objective Loss Balancing

2.5.1 Hard Parameter Sharing and Inductive Bias Transfer

In sudden-onset disaster monitoring, deploying separate, sequential deep learning models for relevance classification and location extraction doubles computational latency and induces upstream error propagation (Zhang & Wang, 2026). Multi-Task Learning (MTL)

addresses these architectural bottlenecks by utilizing hard parameter sharing across a single, shared transformer encoder backbone (such as XLM-RoBERTa), enabling inductive bias transfer between related NLP tasks (Abiola et al., 2025; Zhang & Wang, 2026). In this shared representation space, sequence-level disaster intent informs token-level spatial boundaries, and vice versa, which improves generalization performance across informal, code-switched social media streams (George et al., 2025; Zhang & Wang, 2026).

2.5.2 Homoscedastic Uncertainty Loss Balancing

A fundamental challenge in multi-task transformer fine-tuning is optimizing a joint multi-objective loss function where individual tasks exhibit varying loss scales and learning dynamics (Zhang & Wang, 2026). Naive manual weighting often causes the gradients of a dominant task (e.g., token-level NER) to overwhelm minority task heads (Abiola et al., 2025). To maintain dynamic gradient equilibrium, modern MTL frameworks implement homoscedastic uncertainty loss balancing—pioneered by Kendall et al. (2018) and continuously refined in transformer-based multi-task architectures (Zhang & Wang, 2026). By modeling task-dependent homoscedastic uncertainty as learnable noise variance parameters (σ_1, σ_2), the model dynamically scales classification loss ($\mathcal{L}_{\text{cls}}$) and sequence labeling loss ($\mathcal{L}_{\text{ner}}$):

$$\mathcal{L}_{\text{total}} = \frac{1}{2\sigma_1^2}\mathcal{L}_{\text{cls}} + \frac{1}{2\sigma_2^2}\mathcal{L}_{\text{ner}} + \log \sigma_1 + \log \sigma_2 \tag{2.1}$$

While Kendall et al. (2018) formulate dynamic homoscedastic uncertainty weighting in the literature, the empirical implementation in Chapters 3 and 4 of this work adopts a deterministic, grid-search-optimized equal weighting ($\alpha = 1.0, \beta = 1.0$) to ensure reproducible gradient stability within a 4 GB VRAM ceiling. This formulation regularizes learning, prevents task dominance, and ensures optimal multi-task convergence without requiring manual hyperparameter tuning over loss weights (Zhang & Wang, 2026).

2.6 Data Quality, Fuzzy Deduplication, and Imbalance Regularization

2.6.1 Preprocessing Noise and MinHash/LSH Deduplication

Raw social sensing streams harvested via web scrapers or streaming APIs contain extreme data noise, automated bot amplification, and high volumes of commercial noise (Erokhin & Komendantova, 2024; Sufi, 2022). Prior to model fine-tuning, deterministic fuzzy deduplication must eliminate high-volume commercial real-estate spam (e.g., ads containing terms like "flood-free") and viral reshares (Navalkar et al., 2025; Songchon et al., 2021). Because exact-string comparison fails on minor character variations, recent literature leverages MinHash Locality-Sensitive Hashing (LSH) (Ikotun et al., 2023). By converting character n-gram representations into fixed-size MinHash signatures and hashing candidate pairs into identical buckets, MinHash LSH enables sub-linear Jaccard

similarity filtering ($\geq 85\%$), stripping redundant noise prior to deep model ingestion (Ikotun et al., 2023).

2.6.2 SMOTE on Dense Embeddings and Transformer Regularization

Disaster feeds present extreme class imbalance, where urgent rescue requests comprise less than 5% of raw social volume compared to general chatter (Esparza et al., 2023; Roy et al., 2021). Traditional random oversampling leads to transformer overfitting, while under-sampling discards valuable training signals (Esparza et al., 2023). To balance minority urgency classes, recent crisis informatics models execute Synthetic Minority Over-sampling Technique (SMOTE) directly on dense multilingual transformer embeddings rather than raw tokens (Abiola et al., 2025). Combined with dropout regularization ($p = 0.1$), layer-wise decay, and early stopping, embedding-level SMOTE regularizes linear prediction heads and improves recall on critical flood reports (Abiola et al., 2025; Javed et al., 2026).

2.7 Spatial-Temporal Clustering, Consensus, and Anomaly Filtering

2.7.1 Spatial-Temporal Vector Formulation and Algorithm Selection (K-Means++ vs. ST-DBSCAN)

While linguistic NLP models evaluate individual post relevance, validating the scale and truthfulness of a physical flood requires spatial-temporal consensus building among independent observers (Hanny & Resch, 2024; Liu, 2025). Geoparsed location entities (x_i, y_i) and normalized post timestamps t_i are structured into continuous spatial-temporal coordinate vectors $\mathbf{p}_i = [x_i, y_i, t_i]^T$ to evaluate local event density (Binetti et al., 2025; Mehta et al., 2023). In spatial-temporal event detection, density-based algorithms like ST-DBSCAN are frequently referenced due to their ability to discover arbitrary spatial shapes (Peng & Zhang, 2024; Sathianarayanan et al., 2024). However, ST-DBSCAN requires sensitive parameter tuning ($\epsilon_1, \epsilon_2, \text{MinPts}$) and scales poorly ($O(N^2)$ time complexity), causing execution bottlenecks during real-time disaster stream processing (Ikotun et al., 2023; Liu, 2025). Conversely, K-Means++ provides deterministic $O(N \cdot K \cdot I)$ efficiency with optimized centroid initialization, ensuring rapid convergence during sudden-onset floods (Ikotun et al., 2023).

2.7.2 Centroid Anomaly Filtering and Consensus Thresholding

By integrating dynamic spatial-temporal feature scaling, K-Means++ performs dual-purpose verification: establishing high-density event clusters while flagging isolated, low-density submissions ($|S_k| < \theta$) as spatial anomalies or false alarms (Binetti et al., 2025; Liu, 2025). This density-thresholding logic eliminates the need for secondary anomaly detection algorithms (such as Isolation Forest), maintaining a streamlined, low-latency validation pipeline (Binetti et al., 2025; Chua et al., 2024).

2.8 Algorithmic Bias Mitigation, Disaster Silencescapes, and Infrastructure Fragility

2.8.1 Digital Divide Skews and Spatial Representation Biases

A critical limitation of crowdsourced social sensing is its structural vulnerability to demographic, technological, and spatial biases (Chen et al., 2022; Esparza et al., 2023). Social media activity is heavily concentrated in affluent, highly urbanized barangays with widespread smartphone penetration, whereas peripheral, rural, or low-income districts experience severe digital sparsity (Chen et al., 2022). Relying purely on raw post volume to measure flood severity creates systemic algorithmic bias, systematically prioritizing vocal urban centers while neglecting underserved communities (Chen et al., 2022; Esparza et al., 2023).

2.8.2 Infrastructure Blackout Voids and Source Authority Override

This disparity is exacerbated during extreme typhoons by "disaster silencescapes"—phenomena where communities facing the most severe inundation suffer immediate power grid failures and cellular tower outages (Alhaffar et al., 2023; Esparza et al., 2023). During grid collapses, severely impacted zones become digitally silent while less-affected areas remain active on social feeds (Alhaffar et al., 2023). To mitigate silencescapes and cold-start delays, automated validation architectures incorporate a Source Authority Override (Roy et al., 2021). By assigning an absolute Confidence Score (CS = 1.0) to pre-verified government, meteorological, and emergency responder handles (e.g., PAGASA, DRRMO), authoritative first-responder updates bypass crowdsourced multi-user consensus requirements, triggering immediate alerts during critical initial failure windows (Roy et al., 2021; Sufi, 2022).

2.9 Web Scraping Ethics, Legal Precedents, and Regulatory Compliance

2.9.1 International Jurisprudential Precedents on Automated Public Data Extraction

The extraction of unauthenticated social media streams via automated browser engines, such as Selenium, often implicates domestic computer fraud standards[cite: 1]. In *hiQ Labs, Inc. v. LinkedIn Corp.* (31 F.4th 1180, 9th Cir. 2022), the U.S. Court of Appeals for the Ninth Circuit established that harvesting publicly accessible web data without user authentication does not constitute access "without authorization" under the federal Computer Fraud and Abuse Act (CFAA, 18 U.S.C. § 1030). This ruling suggests that publicly accessible web spaces lack the technical barrier equivalent of a password-protected system[cite: 1]. This principle aligns with the U.S. Supreme Court's ruling in *Van Buren v. United States* (141 S. Ct. 1648, 2021), which narrowed federal computer crime

liability by holding that an entity accessing permitted features for unauthorized purposes does not "exceed authorized access" under anti-hacking statutes. Together, these U.S. precedents establish that violating platform Terms of Service for data collection generally does not constitute a criminal hacking breach, offering persuasive foreign authority for researchers[cite: 1]. However, such automated collection may still expose researchers to civil liability for breach of contract, as courts have upheld the enforceability of anti-scraping provisions in platform User Agreements.

2.9.2 Philippine Statutory Protection and Privacy-Preserving Compliance

While international jurisprudence addresses technical system access, data processing within the national jurisdiction is strictly governed by Republic Act No. 10173, known as the Data Privacy Act of 2012 (DPA), enforced by the National Privacy Commission. Section 11 of the DPA mandates that personal data processing adhere to the core principles of Transparency, Legitimate Purpose, and Proportionality. To reconcile public data harvesting with privacy rights, this study invokes the statutory research exemption under Section 4(d) of the DPA, which exempts data processed strictly for scientific, statistical, or research purposes from standard consent requirements, provided that user identities are not revealed. Operationally, this statutory condition is enforced by implementing an automated privacy pipeline at the ingestion boundary. Raw social media payloads undergo automated regular expression and named entity scrubbing to strip user handles, phone numbers, individual profile links, and specific residential markers prior to model vectorization. This ensures that crowdsourced data is processed exclusively at aggregated spatial-temporal scales for public safety without compromising individual privacy.

2.10 Local Philippine Hydrological Dynamics and Social Media Reporting Behaviors

2.10.1 Catchment Hydrography and Tri-Basin Inundation Dynamics of Davao City

Aligning natural language extraction models with physical environments requires grounding spatial natural language processing pipelines in the specific hydrological regimes of the target region. According to the JICA and DPWH (2023) Flood Control Master Plan for Davao City River Basins, the urban sector of Davao City is governed by three distinct catchments: the Davao River Basin, the Matina River Basin, and the Talomo River Basin. The Davao River Basin spans 1,755 square kilometers and represents a large regional catchment where heavy rainfall in the Bukidnon headwaters causes slow-rising, persistent fluvial flooding and main-stem overtopping in downstream urban sectors such as Barangays Ma-a, Tigatto, Bucana, and Waan. In contrast, the Matina River Basin covers 70 square kilometers as a small, steep catchment characterized by short lag times and high hydrograph peak flows, leading to rapid flash inundation in low-lying residential

barangays like Matina Crossing, Matina Aplaya, and Matina Pangi. Finally, the Talomo River Basin encompasses 204 square kilometers and is subject to mixed fluvial discharge and urban drainage backwater effects that trap runoff along coastal outlets such as Talomo Proper and Bago Aplaya.

2.10.2 Social Sensing Behaviors and Physical Telemetry Gap Filling

Physical telemetric sensor networks across Davao City's river basins remain sparse, vulnerable to power and communication outages, and concentrated primarily along main river channels. The low availability of long-term hydrometeorological gauge data creates significant monitoring gaps during extreme weather, though recent field studies on co-designing flood resilience in Davao City emphasize that local information fed directly by residents is critical to filling these gaps (Miyamoto et al., 2022). Building on this need for community-level data, citizen reports on social media act as an opportunistic social sensing network. Local reporting behaviors capture localized pluvial flooding, clogged drainage structures, and street-level impassability often missed by river gauges. Linguistically, local citizens report flood situations using informal micro-toponyms, landmark references such as "baha duol sa Matina Crossing bridge", and code-switched text combining Tagalog, English, and Cebuano (Bisaya). This reporting pattern provides the empirical rationale for developing cross-lingual natural language processing pipelines capable of converting unstructured local text into structured spatial data.

2.11 Statistical Evaluation Methodology and Inferential Significance Testing

2.11.1 Metric Selection Under Severe Class Imbalance

Evaluating natural language processing models on real-world disaster streams requires metrics robust to extreme class imbalance, where non-informative background noise significantly outnumbers critical flood reports. Standard classification accuracy is unsuitable for imbalanced crisis datasets because high true negative counts yield artificially inflated performance scores. For sentence-level tasks, this study follows the standardized benchmarking framework established by Alam et al. (2021) in CrisisBench, which relies on class-balanced metrics such as Macro F1-score to evaluate humanitarian information classification. Building on this principle of imbalance-aware evaluation, the study extends robust metric selection to token-level spatial extraction tasks. In tasks like Named Entity Recognition, token-based metrics are prone to distortion due to the high volume of background non-entity tokens. Consequently, spatial entity extraction is evaluated using span-level Entity F1 with strict boundary and category matching, ensuring the model is evaluated accurately on its ability to identify complete, actionable location entities.

2.11.2 Inferential Hypothesis Testing and Multi-Task Neural Evaluation

To establish that performance improvements over baseline architectures, such as mBERT and Logistic Regression, are statistically significant rather than artifacts of random split variance, hypothesis testing must be tailored for deep neural multi-task outputs. Relying on standard paired Student's t-tests across K-fold cross-validation folds violates independence assumptions due to overlapping training sets across iterations. Therefore, for paired model comparisons on holdout evaluation sets, hypothesis testing uses McNemar's test with Edwards' continuity correction. By analyzing prediction discordance on paired outputs—where one model predicts correctly and the other fails—via a chi-squared distribution with one degree of freedom, McNemar's test effectively determines whether performance gains across task classification and spatial entity extraction achieve statistical significance at the five percent alpha level.

2.12 Multi-Tier Theoretical Grounding of the SDSS Framework

2.12.1 Computational Linguistics Layer and Cross-Lingual Representation

At the foundational computational linguistics layer, the system relies on the transformer multi-head self-attention mechanisms and cross-lingual representation theory introduced by Conneau et al. (2020) in XLM-RoBERTa. Trained on over two terabytes of CommonCrawl data across 100 languages using a 250,000-token SentencePiece vocabulary, XLM-RoBERTa processes raw multilingual text without requiring language-specific pretokenizers. This provides the theoretical foundation for zero-shot cross-lingual transfer across code-switched Philippine disaster text mixing Tagalog, English, and Cebuano, often referred to as Bislish or Taglish. The model's omission of language embeddings specifically enhances its ability to handle this code-switching, while subword tokenization prevents out-of-vocabulary segmentation errors caused by complex agglutinative morphology in regional dialects, such as prefixing and infixing in Bisaya flood descriptions. This cross-lingual encoder serves as the shared backbone for joint Multi-Task Learning, driving both document relevance classification and spatial named entity extraction within a single neural framework.

2.12.2 Integrated SDSS Theoretical Synthesis

The Spatial Decision Support System integrates four theoretical layers into an operational pipeline. The first layer, computational linguistics, employs Conneau et al. (2020) for cross-lingual vector space representations across code-switched text. The second layer, volunteered geographic information (VGI), is grounded in Goodchild's (2007) VGI paradigm, treating citizen posts as spatial telemetry during flood events. The third layer, data reliability and information theory, incorporates Broder's (1997) MinHash and Locality-Sensitive Hashing deduplication to remove automated bot spam, alongside Kendall et

al.'s (2018) Bayesian multi-task uncertainty loss formulation to dynamically balance task gradients during training.

Finally, the fourth layer, spatial decision support and hydrodynamics, converts extracted location entities and standardized timestamps into multi-dimensional spatio-temporal vectors consisting of spatial coordinates and timestamps. These vectors are clustered via spatial-temporal algorithms and overlaid onto HEC-RAS 2D hydrodynamic depth grids, validating crowdsourced social reports against physical inundation boundaries for barangay-level disaster response.

2.13 Theoretical Framework

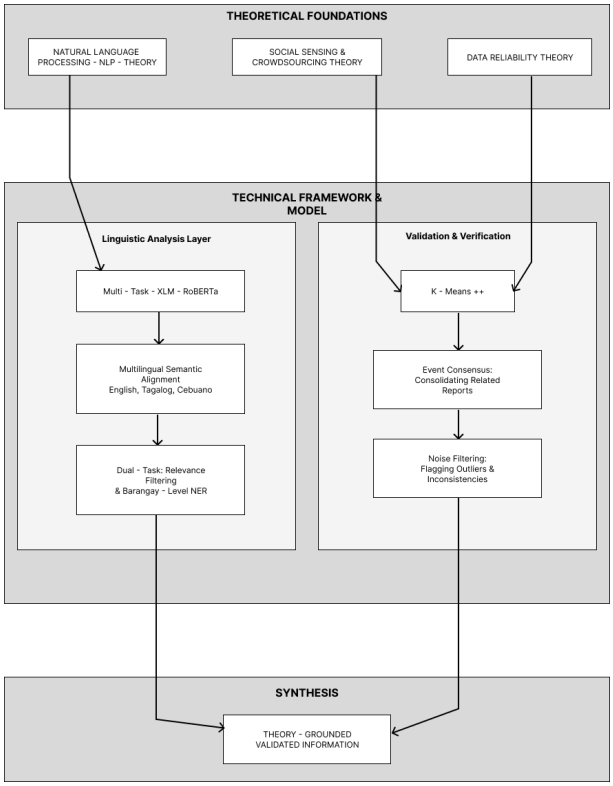


Figure 1: Theoretical framework

2.13.1 Linguistic Analysis Layer: Multi-Task XLM-RoBERTa

Grounded in Natural Language Processing (NLP) Theory, the Linguistic Analysis Layer utilizes XLM-RoBERTa as the primary engine for processing multilingual crowdsourced data. This model is specifically leveraged for its Multi-Task Learning (MTL) capabilities, which allow it to perform two critical functions simultaneously: Relevance Filtering to distinguish flood-related reports from non-pertinent noise, and Fine-Grained Location Extraction through Named Entity Recognition (NER) to identify specific barangays, streets, and landmarks. A fundamental theoretical strength of this approach is XLM-RoBERTa's ability to handle "code-switching"—the fluid mixing of English, Tagalog,

and Cebuano common in Philippine social media—which ensures localized accuracy without the computational overhead of separate, sequential deep learning architectures like BERT-BiLSTM-CRF (Zhang & Wang, 2026; Wang et al., 2023).

2.13.2 Validation & Verification Layer: K-Means++

The Validation & Verification Layer employs K-Means++ to implement "Consensus-Building Logic" under the principles of Data Reliability Theory. In this unified framework, K-Means++ serves a dual purpose by first grouping reports into event-level clusters based on spatial proximity, timestamps, and semantic similarity to verify the scale of a disaster (Ikotun et al., 2023; Liu, 2025). Furthermore, it performs Integrated Anomaly Filtering by identifying "noise" or misinformation through unsupervised outlier detection, where reports that fail to align with established high-density clusters are flagged as spatial anomalies (Binetti et al., 2025). By utilizing the inherent density metrics of the clustering algorithm, the framework maintains data authenticity and reduces redundancy without requiring a standalone Isolation Forest model (Binetti et al., 2025; Chua et al., 2024).

2.13.3 Theory-Grounded Validated Information

The final stage of the framework represents the synthesis of all preceding modules into Theory-Grounded Validated Information. By integrating multi-task relevance and spatial analysis with consensus-based verification, the pipeline transforms raw, noisy crowdsourced data into structured disaster intelligence specifically tailored for localized, data-driven decision-making (Roy et al., 2021; Navalkar et al., 2025). This validated output provides Local Government Units (LGUs) and emergency responders with an unambiguous foundation for response, including verified locations and a Confidence Score (CS) derived from the alignment of textual relevance and multi-user corroboration (Roy et al., 2021).

Chapter 3

Methodology

3.1 Research Design

This study employs a quantitative experimental and computational case study research design wherein it is centered on localized urban flood monitoring in Davao City, Philippines. The primary goal of this research design is to empirically construct, execute, and to help evaluate the end to end, multi-stage Natural Language Processing (NLP) and spatial-temporal verification architecture that converts unstructured, noisy, and crowdsourced social media reports into a validated disaster intelligence for local government (LGUs) for decision and support.

3.1.1 Case Study Scope, Geographic Boundaries, and Bias Controls

The empirical validation of this framework is conducted within the administrative boundaries of Davao City, Philippines, a coastal metropolitan area subject to severe flash floods and fluvial overtopping across the Davao River, Matina River, and Talomo River basins. To ensure rigorous evaluation of crowdsourced social media feeds, the research design explicitly accounts for three systemic sampling and technological representation biases inherent to social sensing in crisis informatics:

Digital Divide and Geographic Disparities: Social media reporting density correlates with urbanization, household income, and digital literacy. Urbanized, high-density barangays (e.g., Poblacion, Matina Crossing, Bangkal, Sasa, and Bucana) generate high crowdsourced post volumes during inundations. Conversely, peripheral, mountainous, or low-connectivity districts (e.g., Marilog, Paquibato, and Baguio) experience severe digital sparsity, creating data blind spots where significant flood events produce minimal social media footprints. To prevent systemic neglect of digitally sparse zones, the validation module evaluates spatial-temporal cluster density relative to historical barangay posting baselines rather than relying on global raw post thresholds.

Disaster Silencescapes and Infrastructure Fragility: During severe typhoons and sudden flash floods, communities facing the most hazardous water levels frequently suffer

power grid collapses and cellular tower blackouts. This infrastructure fragility induces a critical sampling failure termed a "disaster silencescape", where the most severely impacted zones become digitally silent while less affected urban areas remain vocal. The framework mitigates silencescapes by coupling crowdsourced multi-user consensus with a Source Authority Override, allowing verified first-responder and municipal broadcasts to trigger immediate high-confidence alerts without waiting for citizen corroboration. Cognitive and Commercial Noise: Keyword-driven social media scrapers targeting disaster terms in Davao City heavily harvest commercial real estate advertisements (e.g., promotional posts utilizing terms like "100% flood-free" or "walay baha"), bot spam, viral reshares, and non-emergency chatter. Unfiltered, this cognitive noise degrades the spatial integrity of the classifier. The experimental design integrates deterministic text cleaning, negation filtering, and MinHash Locality-Sensitive Hashing (LSH) deduplication to eliminate commercial spam prior to model ingestion.

Data Ingestion Boundaries: The scope of this study is strictly delimited to publicly accessible, text-based social media reports published in English, Tagalog, Cebuano/Bisaya, or code-switched variations (Taglish and Bislish) containing explicit barangay-level landmarks, street names, or micro-toponyms within Davao City. Non-textual media (standalone images or video streams), private direct messages, password-protected group posts, and automated physical IoT sensor telemetry fall outside the scope of this investigation.

3.1.2 Five-Stage Computational Architecture

The experimental system is implemented across a five-stage computational pipeline that directly aligns with the system's Conceptual Framework:

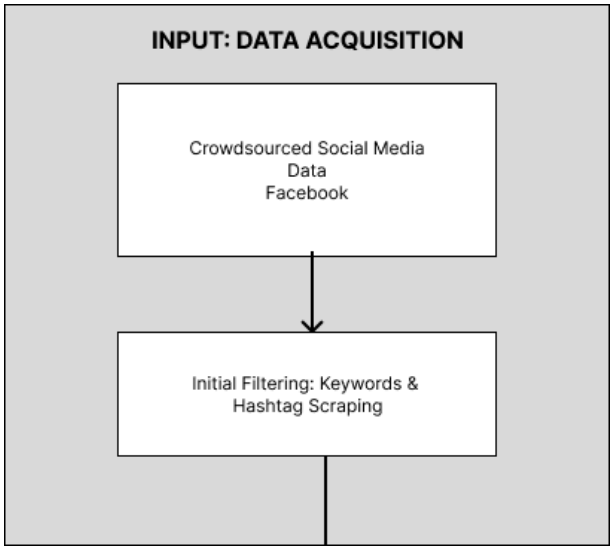


Figure 2: Data Acquisition Phase

Data Acquisition Phase: An automated real-time web scraping is executed using Selenium with Python to bypass the platform API paywalls and rate limits by automating a headless Google Chrome instance to parse publicly available social media posts, exclusively from

Facebook. Data is ingested using domain-specific, dialect-aware search vectors combining flood terms with localized Davao City barangays and district names.

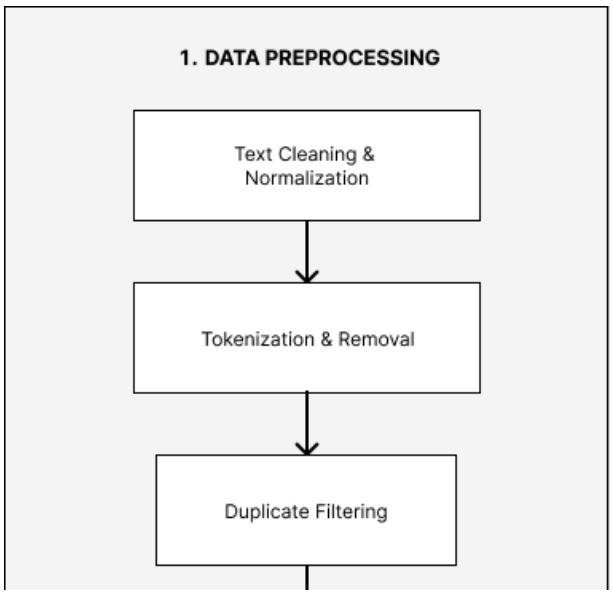


Figure 3: Data Processing Phase

Data Preprocessing Phase: Raw scrapped data undergoes text cleaning and normalization (stripping raw HTML residue, converting emojis to text descriptions, and normalizing panicked character repetitions e.g. "baaaaaaaaahhhaaaaaaa" into "baha"). Subword tokenization is performed, and duplicates, bot spams are removed filtering it to eliminate commercial real-estate ads and negated statements ("walang baha").

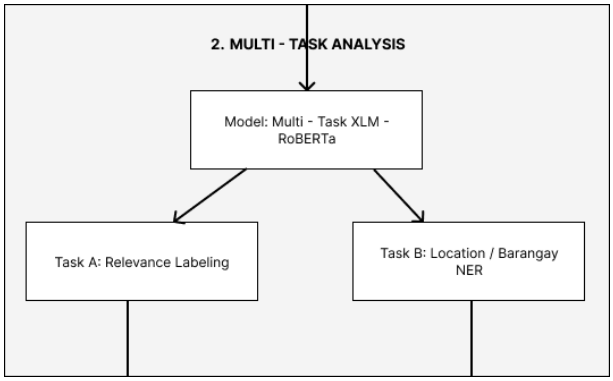


Figure 4: Multi-Task Analysis Phase

Multi-Task Analysis Phase: Preprocessed token sequences are fed directly into a fine-tuned Multi-Task XLM-RoBERTa encoder. The network concurrently predicted binary flood relevance probabilities (P_{rel}) for Task A via a sequence classification head and extracted IOB-tagged barangay entities for Task B via a token classification head. To prevent task gradient domination and control class imbalance, the network incorporated fixed equal task loss weighting ($\alpha = 1.0, \beta = 1.0$), dropout regularization ($p = 0.1$), SMOTE applied

to sequence-level pooled [CLS] embeddings, and validation-driven early stopping with a 5-epoch limit.

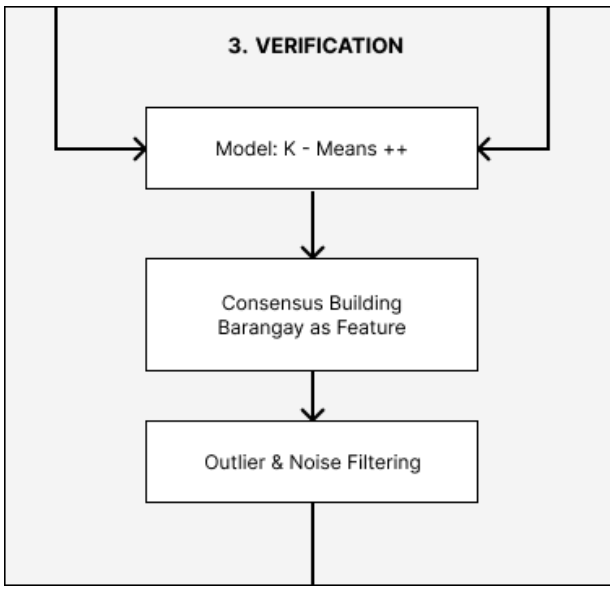


Figure 5: Verification Phase

Verification Phase: Extracted barangay centroids (x_i, y_i) and timestamps t_i are mapped into a spatial-temporal vector space $p_i = \begin{bmatrix} x_i & y_i & t_i \end{bmatrix}^T$. K-Means++ clustering groups reports into clusters to establish multi-user community consensus while filtering isolated spatial anomalies below a density threshold Θ. To resolve the Cold-Start Problem, a Source Authority Override assigns a maximum Confidence Score of 1 (CS = 1.0) to pre-verified official handles (e.g. PAGASA and DRRMO), allowing urgent single-source reports to bypass consensus requirements.

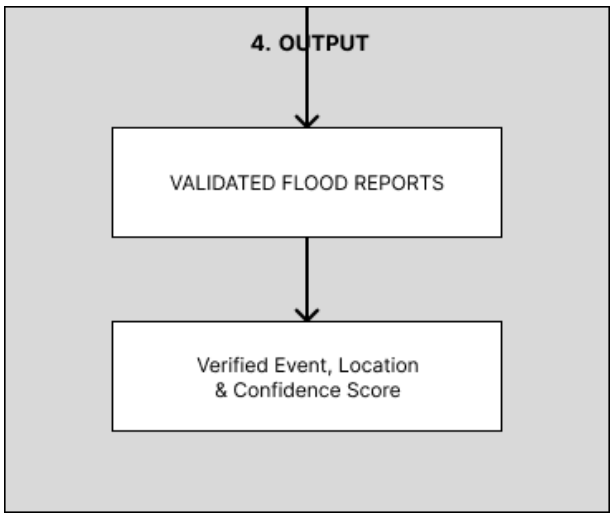


Figure 6: Output Phase

Output Phase: Validated flood reports, verified flood locations, and composite Confidence Scores ($\text{CS} = w_1(P_{\text{rel}}) + w_2(D_{\text{cluster}})$) are generated and pushed to the AGOS LGU Web

Dashboard for real time decision support, complete with a human in the loop feedback loop for an active learning model for occasional retraining.

3.1.3 Comparative Benchmark and Statistical Framework

To evaluate the system's effectiveness, the proposed framework is benchmarked against established single task baselines (Logistic Regression, mBERT), standard sequential location extraction architectures (BERT-BiLSTM-CRF), and disjoint multi-model pipelines. Performance is evaluated across classification metrics (Precision, Recall, Macro F1, Precision Recall), entity extraction metrics (F1 score), and end to end system reliability (False positive and miss rates). Statistical significance is verified using McNemar's Test for classification error rates, paired with Student's T-test across K-fold cross validation fold and Relative Performance Gain metrics.

Beyond classification and extraction accuracy, the system's evaluation explicitly enforces a real-time operational window. Because urban flood inundation is a momentary hazard requiring immediate emergency interventions, crowdsourced intelligence must be harvested, processed, and validated within a narrow window of minutes to hours during an active flooding. Reports appearing in days or weeks after flooding represent post-disaster reposts or commentary and are mathematically treated as temporal anomalies within the spatial-temporal coordinate space $\mathbf{p}_i = [x_i, y_i, t_i]^T$. To evaluate operational viability, system performance is benchmarked not only on predictive metrics but also on the end to end processing latency (measured in milliseconds) to ensure that validated alerts reach the AGOS LGU Dashboard in time for active emergency response.

3.2 Conceptual Framework

The process begins in the Input Phase, where crowdsourced text data is harvested exclusively from public Facebook posts and community groups using Selenium browser automation. Traditional API listeners across platforms like Facebook were rendered unviable due to restricted access and paywalls, necessitating targeted keyword-based scraping directly from Facebook's Document Object Model (DOM).

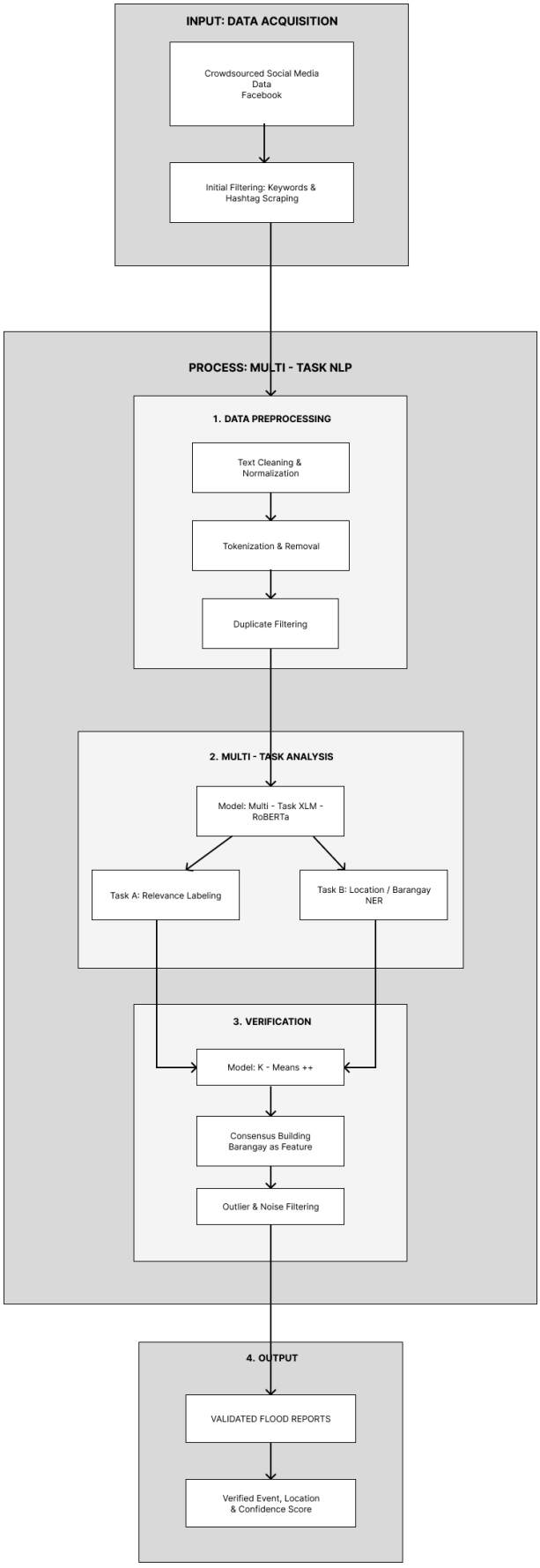


Figure 7: Conceptual Framework

3.2.1 Case Study Scope, Inherent Biases, and Sampling Gaps

The validation framework is evaluated through a localized case study focused on flood events in Davao City, Philippines. While social sensing provides immediate, fine-grained spatial intelligence, the input data inherits structural demographic and geographic biases. Social media adoption is heavily skewed toward urbanized, higher-density barangays (e.g., Poblacion, Talomo, and Buhangin) and younger, digitally literate populations. Consequently, rural or low-connectivity areas (e.g., Marilog or Paquibato districts) suffer from digital sparsity, where significant flood events may generate zero crowd-sourced reports. To mitigate sampling gaps, the system relies on spatial-temporal cluster density relative to historical baselines rather than absolute post volume. However, the framework only focuses entirely on natural language validation.

3.2.2 Multilingual Preprocessing and Multi-Task Analysis

This raw data enters the Multi-Stage NLP Pipeline, beginning with Data Preprocessing where the text undergoes cleaning, normalization, and tokenization, followed by duplicate filtering to reduce redundancy. The refined text is then passed to the Multi-Task Analysis module, powered by XLM-RoBERTa. This model performs multilingual semantic mapping to bridge English, Tagalog, and Cebuano dialects while simultaneously executing two primary tasks:

3.2.3 Multilingual Preprocessing and Scraping Noise Mitigation

To clean unstructured social text across English, Tagalog, and Cebuano, the pipeline executes regex-based noise removal (stripping URLs, platform-specific metadata, and special characters) followed by unicode normalization. Duplicate filtering uses exact-string matching and MinHash Locality-Sensitive Hashing (LSH) to eliminate bot-generated spam, viral reshares, and news broadcast amplification.

3.2.4 Multi-Task Model Regularization, Balancing, and Overfitting Control

The multi-task loss function combines classification loss (Task A) and NER token-level loss (Task B) via a fixed equal loss weighting ($\alpha = 1.0, \beta = 1.0$) to balance learning rates between tasks. To prevent overfitting on dominant dialect structures or frequent top-level barangay names, dropout regularization ($p = 0.1$) and layer-wise learning rate decay are applied during fine-tuning. Task A (Relevance Labeling): Distinguishing flood-related reports from non-pertinent noise. Task B (Location/Barangay NER): Identifying specific barangay-level landmarks and spatial attributes directly from the text (Zhang & Wang, 2026).

By integrating these tasks into a single model, the framework provides an efficient solution for monitoring localized events without the need for separate, sequential architectures.

3.2.5 Mathematical Assumptions and Algorithmic Bias Control

K-Means++ assumes spherical cluster geometries in feature space, which may misrepresent linear geographical features (e.g., riverbanks or long roadways). To compensate, spatial features are weighted alongside temporal deltas during feature vector scaling.

Algorithmic bias toward high-density posting centers is mitigated by calculating density dynamically against historical posting baselines per barangay rather than using static global thresholds. Mathematical formulation for cluster density weighting relies on Gaussian kernel density estimation over spatial coordinates (x, y) and time timestamp (t).

3.2.6 Intelligence Synthesis and Output Phase

The culmination of this process is the Output Phase, which yields a Validated Flood Report. This final intelligence product provides structured data, including verified event details, precise localized coordinates, and an associated Confidence Score (CS) derived from the alignment of textual relevance and multi-user corroboration (Roy et al., 2021). By synthesizing these automated processes, the framework offers an unambiguous foundation for Local Government Units (LGUs) and emergency responders to facilitate data-driven disaster response at the barangay level.

3.3 Operational Framework

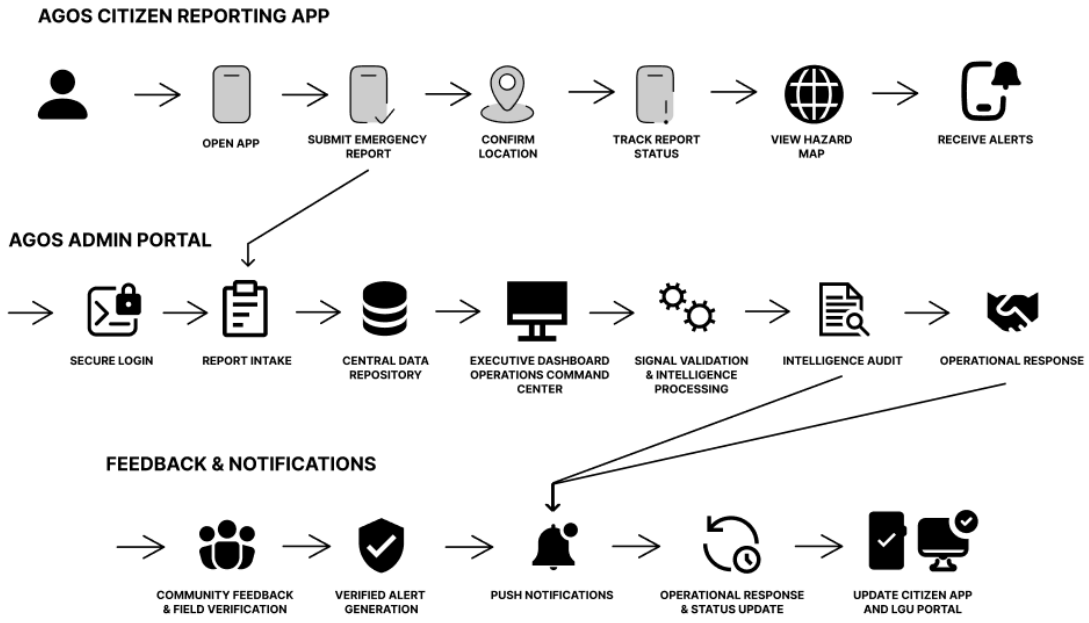


Figure 8: Operational Framework

The operational framework outlines how the system processes crowdsourced data and official reports into verified disaster intelligence for local decision-makers and the public. From an operational perspective, the pipeline streamlines raw inputs into actionable dashboard alerts across four simplified stages: Data Ingestion, Multi-Task Analysis, Automated Verification, and Output/Alert Generation.

3.3.1 Operational Framework of the AGOS Disaster Intelligence Platform

The operational framework of the AGOS Disaster Intelligence Platform presents the end-to-end flow of disaster-related information from citizen reporting to local government response. It illustrates how the citizen mobile application, LGU administrative portal, and feedback and notification mechanisms work together to support disaster monitoring and response in Davao City. The framework is designed to establish a continuous information cycle in which citizens provide ground-level observations, the LGU processes and evaluates the submitted information, and verified results are returned to users through alerts, status updates, and operational actions.

As shown in Figure 7, the framework is divided into three major components: the AGOS Citizen Reporting App, the AGOS Admin Portal, and the Feedback and Notifications Workflow. These components are interconnected to ensure that reports are collected, reviewed, validated, transformed into actionable intelligence, and communicated to the appropriate users and response personnel.

3.3.2 Citizen Mobile Application Workflow

The first component of the operational framework is the AGOS Citizen Reporting App. This component represents the workflow from the perspective of a citizen who needs to report or monitor a disaster-related incident.

The process begins when the citizen opens the AGOS application. The application serves as the primary access point for submitting emergency reports and receiving location-based disaster information. After opening the application, the citizen may submit an emergency report containing information about an observed hazard, such as flooding or a critical water-level condition.

After initiating the report, the citizen confirms the incident location. Location confirmation allows the system to associate the report with a specific geographic area or barangay. This information is important because it enables the LGU to determine the affected locality and prioritize response activities based on the reported location.

Once the report has been submitted, the citizen can track its status through the application. This provides transparency regarding the progress of the submitted report and allows the citizen to determine whether the report has been received, processed, validated, or updated by the LGU. The status-tracking function also reduces uncertainty because the citizen can monitor the report without repeatedly submitting the same information.

The citizen may also access the hazard map to view verified hazards and affected

locations. This provides a visual representation of disaster conditions and helps users understand the relationship between their location and nearby incidents. Hazard map information may also support the citizen in identifying safer routes or determining whether an area is potentially affected.

The final stage of the citizen workflow is the receipt of alerts and notifications. When the system identifies or verifies an important hazard, relevant warnings can be delivered to users. These notifications provide timely information that may assist citizens in making appropriate safety and evacuation decisions.

3.3.3 LGU Administrative Portal Workflow

The second component is the AGOS Admin Portal, which is used by LGU personnel and authorized barangay administrators. This portal supports the monitoring, management, validation, and operational response activities of the local government.

The administrative workflow begins with secure login. Authentication ensures that only authorized personnel can access disaster reports, intelligence outputs, administrative functions, and operational data. After authentication, the system applies role-based access control to restrict users according to their assigned responsibilities and barangay scope.

Following authentication, incoming citizen reports are received through the report intake process. The report intake module acts as the entry point for reports submitted from the citizen application. It allows the system and authorized personnel to receive, organize, and prepare reports for further processing.

The reports are then stored in the central data repository. This repository provides a centralized location for maintaining citizen reports, incident records, location information, processing results, alert records, and operational updates. Centralized storage supports data consistency and allows different portal modules to access the information required for monitoring and decision-making.

The executive dashboard and operations command center provide the primary monitoring interface for LGU users. The executive dashboard may display incident activity, validated reports, affected barangays, confidence scores, source information, and system health indicators. The operations command center supports more immediate activities, such as reviewing active reports, examining affected locations, and coordinating response actions.

The next stage is signal validation and intelligence processing. At this stage, submitted reports and related signals are examined to determine their relevance, location, severity, and reliability. The system may compare report information with other available signals and evaluate whether the information supports a credible disaster incident. This process transforms individual reports into structured intelligence that can be used by LGU personnel.

The intelligence audit component allows administrators to inspect the basis of the processed results. Administrators may review the report details, extracted location, severity classification, confidence information, and supporting evidence. This audit pro-

cess promotes accountability and allows authorized personnel to verify whether an alert or operational recommendation is sufficiently supported.

The final stage of the LGU workflow is operational response. When a report or validated alert requires action, LGU personnel may coordinate field verification, dispatch response units, issue warnings, or update the incident status. This stage connects the system's information-processing capabilities with actual disaster response activities.

3.3.4 Feedback and Notifications Workflow

The third component is the Feedback and Notifications Workflow. This component represents the process through which reports and intelligence are verified, communicated, and used to update both the citizen application and the LGU portal.

The workflow begins with community feedback and field verification. Community feedback may provide additional information about an incident, while field verification may confirm the actual condition in the affected location. These sources help the LGU cross-check initial reports and determine whether the reported situation is consistent with other available information.

After the information has been evaluated, the system may generate a verified alert. A verified alert represents a disaster-related notification that has been reviewed and determined to be sufficiently reliable for communication to affected users. This distinction is important because not every submitted report should immediately become a public warning.

Once an alert has been verified, push notifications may be sent to relevant citizens and authorized personnel. Notifications provide timely warnings regarding hazards, affected areas, response activities, or safety instructions. Location-based notification delivery can help ensure that information is directed to users who may be affected by the incident.

The operational response and status update stage follows the notification process. LGU personnel may coordinate field activities, assign response resources, and update the status of the incident. At the same time, the status of the citizen's report may be updated to reflect its current processing or resolution state.

The final stage is the update of the citizen application and LGU portal. Information from the response process is returned to the citizen through report-status updates, hazard map changes, or alerts. The LGU portal is also updated so that administrators can view current incident conditions and response progress. This creates a closed-loop workflow in which information is not only collected but also returned to the users and systems involved.

3.3.5 Integration of the Operational Framework

The three components work together as a continuous disaster intelligence cycle. A citizen first submits a report through the mobile application. The report is then transmitted to the administrative system, received through the report intake process, and stored in the

central data repository. LGU personnel and system processes evaluate the report through signal validation and intelligence processing.

If the information is confirmed and considered significant, the system generates a verified alert and distributes notifications to affected users. LGU personnel may then initiate an operational response and update the incident record. The resulting information is returned to both the citizen application and the administrative portal through report-status updates, hazard map updates, and notifications.

Therefore, the operational framework supports four essential functions: citizen participation, centralized information management, LGU decision support, and closed-loop disaster response. By connecting these functions, AGOS enables the transformation of community-reported information into actionable disaster intelligence while maintaining communication between citizens and local government authorities.

3.3.6 Real-Time Data Ingestion with Selenium Browser Automation

To construct a reliable stream of crowdsourced disaster reports, the pipeline deploys automated web scraping using Selenium with Python. Because platform API restrictions often block real-time access, headless browser automation interacts directly with public web pages and community groups. The ingestion engine targets localized, dialect-aware search vectors (such as "baha", "baha davao", and "taas tubig") across public social media feeds to harvest community posts alongside relevant post metadata.

As the script browses scrolls through dynamically rendered contents, the script extracts raw HTML post containers, and parses the text nodes and packages them into a structured JSON and CSV files alongside with available metadata such as post IDs, timestamps, location, and publicly available username hashes. While this approach is classified as highly granular data, it introduces distinct technology collection noise. The scroll and refresh behavior of platform feeds frequently causes Selenium to capture the same dynamic elements multiple times across browser cycles, resulting in substantial data redundancy that requires systematic mitigation within the multi-stage preprocessing pipeline.

3.3.7 Multi-Task NLP Engine (Relevance & Spatial Extraction)

Before the scraped data can be used to train the Multi Task NLP classifier, it must pass through a strict multi stage text preprocessing and cleaning pipeline designed to manage the high volume of technology induced noise that is introduced by Selenium scraper.

To ensure scientific reproducibility and enable comparative performance evaluations, both the raw, uncleaned Selenium scraper outputs and the final preprocessed datasets are permanently archived and version controlled. This allows the system to establish an empirical baseline to measure the noise reduction efficiency of the preprocessing pipeline.

Text Stripping and Standardization: Standard Python scripts programmatically

strip raw HTML residue, dynamic tags, XPath leftover strings, tracking hyperlinks and browser metadata from the scraped JSON payloads. Furthermore, emojis are converted into literal descriptive text (e.g., translating a rain emoji to the word 'rain') to retain emotional cues.

Multilingual Normalization: Because emergency posts are composed in a casual and panicked style of word structures that are heavily distorted. Repetitive characters used to express urgency (e.g., "baaaaaaahaaaaaa" or 'bhaaaa") are normalized to their dictionary form in Tagalog and Cebuano using regex alignment.

Once data enters the system's backend, it undergoes rapid preprocessing to clean and tokenize the text. The system then automatically routes the data through the XLM-RoBERTa module, which now functions as both the primary linguistic gatekeeper and the spatial extractor via a Multi-Task Learning (MTL) architecture. First, it analyzes the semantics of the code-switched text to instantly classify its relevance; non-pertinent posts are automatically discarded. Simultaneously, it performs Named Entity Recognition (NER) on the verified text. Since crowdsourced posts often lack GPS metadata, XLM-RoBERTa scans the conversational text to extract exact spatial identifiers such as specific barangays, street names, or local landmarks, translating conversational data into mappable locations.

3.3.8 Ground Truth Annotation and Dataset Curation

To establish a supervised ground truth for the Multi-Task XLM-RoBERTa model, the raw corpus of 9,431 harvested crowdsourced posts underwent strict manual annotation. Because the study utilized a single dedicated annotator fluent in English, Tagalog, and Cebuano (Bisaya), traditional inter-annotator agreement metrics (such as Cohen's or Fleiss' Kappa) were not applicable. Instead, the study employed an Expert Domain Annotation framework to ensure labeling precision.

For Task A (Relevance Labeling), the annotator performed binary classification, tagging each post as either a genuine, actionable flood emergency or non-pertinent background noise (e.g., real estate advertisements, weather forecasts). For Task B (Location NER), the annotator applied the standard IOB (Inside-Outside-Beginning) tagging scheme to manually extract and bound specific barangay, street, and landmark entities within the code-switched text.

To ensure intra-annotator consistency and mitigate subjective bias or label drift over time, the dataset was processed in randomized batches. A 10% random overlap was re-evaluated blindly at the end of the primary labeling phase to confirm label stability. Following the removal of scraping noise and spatial anomalies, the finalized curated dataset of 6,638 posts served as the foundation for the stratified negative curation and multi-task fine-tuning.

3.3.9 Multi-Task Model Regularization, Balancing, and Overfitting Control

Deep transformer backbones like XLM-RoBERTa are highly prone to memorizing local shortcuts or over-fitting to class imbalances in disaster datasets. To help control these risks, the following balanced curation and regularization techniques are implemented before and during training:

Stratified Negative Curation: To teach the model to distinguish between genuine emergency situations from unrelated background chatter, the training set is augmented with a created baseline of negative samples. This includes general weather forecasts, flood mitigation news articles, and the commercial real estate advertisements scraped during the data harvesting.

To resolve extreme class imbalance where emergency rescue requests comprised less than 5% of raw social volume Synthetic Minority Over-sampling Technique (SMOTE) was executed on the 768-dimensional pooled [CLS] embedding vectors generated by the XLM-RoBERTa encoder for Task A. Oversampling in the dense semantic embedding space rather than token space prevented the classification head from memorizing surface-level phrasing while expanding decision boundaries for high-urgency reports.

Multi-Task Drop and Optimizer Regularization: A dropout layer with a rate of 0.1 is integrated directly into the classification and Named Entity Recognition (NER) heads of the multi-task network, This forces the models to construct redundant representations of paths rather than relying on isolated words. Additionally, the model is trained with an optimizer. Validation Driven Early Stopping: Following rigorous data splitting practices, the curated dataset is partitioned into an 80/10/10 split. Specifically, 10% of the data is strictly held out as a validation split to monitor performance, and another 10% is completely isolated as the unseen test data. Because XLM-RoBERTa is heavily pretrained, the model is fine-tuned on the remaining 80% of the data for a maximum of 5 epochs and a minimum of 3 epochs. If the classification loss on the validation split fails to improve within this targeted window, the training is automatically terminated to prevent the network from memorizing.

3.3.10 Mathematical Assumptions and Algorithmic Bias Control

To programmatically control the reporting bias inherent in crowdsourced data such as a single user repeatedly posting about a minor puddle or an automated spam bots mimicking flood alerts the verification phase maps relevant social posts into a spatial temporal coordinate system to establish a consensus before validating any emergency.

The Spatial Temporal Coordinate Space: Let each parsed report i that has been identified as relevant by the NLP engine be represented as a multi-dimensional coordinate vector:

$$p_i = \begin{bmatrix} x_i \\ y_i \\ t_i \end{bmatrix} \tag{3.1}$$

where (x_i, y_i) are the normalized latitude and longitude centroids of the barangay extracted by the Named Entity Recognition (NER) module, and (t_i) represents the normalized epoch timestamp of the post.

$$x'_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}, \quad y'_i = \frac{y_i - y_{\min}}{y_{\max} - y_{\min}}, \quad t'_i = \frac{t_i - t_{\min}}{t_{\max} - t_{\min}} \tag{3.2}$$

In this formulation, the raw variables representing the extracted barangay centroid's latitude (x_i) and longitude (y_i), alongside the post's raw epoch timestamp (t_i), must undergo strict Min-Max normalization. This step scales all three distinct features into an equal, non-dimensional boundary (typically 0.0 to 1.0) before they are assembled into the final coordinate vector p_i.

The Spatial Temporal Consensus Assumption: A legitimate, severe urban flood event exhibits high spatial and temporal local density. Conversely, isolated false alarms, real-estate listings that escaped the NLP filter, and historical flood discussions are assumed to be distributed as random, low-density spatial-temporal anomalies.

K-Means++ Initialization and Clustering: The verification algorithm applies K-Means++ to partition N coordinate vectors into K event-level clusters by minimizing the sum of squared distances:

$$\arg\min_{c} \sum_{k=1}^{K} \sum_{p_i \in S_k} \|p_i - c_k\|^2 \tag{3.3}$$

Where (c_k) is the spatial-temporal centroid of cluster (S_k). Utilizing K-Means++ initialization mathematically controls clustering bias, ensuring stable and reproducible convergence times during rapid disaster onset.

Algorithmic Bias Control via Density Thresholding: To prevent individual reporting bias from triggering false emergency alerts, a cluster (S_k) is only validated as a credible emergency event if its overall cardinality meets or exceeds a density threshold parameter, θ:

$$\text{Status}(S_k) = \begin{cases} \text{Validated}, & \text{if } |S_k| \geq \theta \\ \text{Anomaly}, & \text{if } |S_k| < \theta \end{cases} \tag{3.4}$$

Individual reports that fail to group into a cluster of size θ are mathematically flagged as spatial anomalies, filtering out noise without requiring the computational overhead of a secondary deep learning classifier.

Cold-Start Bias Mitigation via Source Authority Override: While density thresholding successfully filters out crowdsourced noise, it introduces a systematic "cold-start" bias: a legitimate, critical first-hand report of an active flood would be discarded in the initial minutes of an event because it lacks corroborating reports. To control this, the validation mathematical logic implements a Source Authority Override:

$$\text{If Authority}(i) = \text{Verified}, \quad \text{CS}_i = 1.0 \tag{3.5}$$

This override immediately assigns a maximum Confidence Score (CS = 1.0) to verified handles (e.g., official PAGASA, LGU, or emergency responder accounts). This allows authoritative disaster intelligence to bypass the K-Means++ consensus requirement, ensuring immediate alert generation during the critical first minutes of a flood event.

3.3.11 Automated Verification and Consensus Building

To ensure LGUs receive reliable intelligence, extracted location entities and relevance scores are evaluated through spatial-temporal clustering. The K-Means++ algorithm handles spatial verification directly:

Consensus Building: The system's backend runs K-Means++ to cluster incoming reports based on spatial and temporal proximity. If multiple users are posting about rising waters in a similar timeframe and extracted location, the system groups these into a single high-density "event cluster," increasing the operational confidence score of that event.

Anomaly Detection: Spatial anomalies such as spam, contradictory reports, or malicious misinformation are identified by evaluating their distance from the established cluster centroids. Isolated reports that fall too far outside these high-density consensus areas are treated as outliers and are either discarded or placed in a "low confidence" queue requiring human review.

3.3.12 Alert Generation and LGU Dashboard (Output)

In the final operational stage, the system synthesizes the filtered text, consensus data, and extracted locations to generate a Validated Flood Report. This structured intelligence is pushed to a centralized LGU dashboard, displaying the event's location, severity, and algorithmic confidence score. This confidence score (CS) is calculated as a weighted average combining two metrics: the relevance classification probability (P_{rel}) from XLM-RoBERTa and the normalized cluster density (D_{cluster}) from K-Means++. To prevent false alarms and ensure actionable intelligence, the weighting logic deliberately prioritizes data authenticity and multi-user corroboration over simple textual relevance, an approach supported by recent environmental monitoring and big data frameworks (Liu, 2025). The score is derived using the formula:

In this framework, the weights for crowdsourced data are explicitly assigned as $w_1 = 0.30$ and $w_2 = 0.70$. This 30/70 split mathematically ensures that multi-user spatial corroboration (w_2) carries a significantly heavier operational weight than the semantic classification of a single isolated post (w_1). Furthermore, because raw cluster sizes can scale infinitely, the cluster density must be bounded into a normalized 0.0 to 1.0 scale to be mathematically compatible with the P_{rel} probability score. The cluster density is normalized as:

$$\text{CS} = w_1(P_{\text{rel}}) + w_2(D_{\text{cluster}}) \tag{3.6}$$

To mitigate the cold start problem, the framework incorporates a Source Authority override. Reports originating from pre-verified metadata handles (e.g., official DRRMO accounts or verified emergency responders) are assigned a maximum Confidence Score (CS = 1.0) immediately. This allows high-priority, single-source emergency alerts to bypass the K-Means++ consensus requirement, ensuring zero-latency dashboard delivery during sudden disaster onset.

$$D_{\text{cluster}} = \min\left(1.0, \frac{|S_k|}{\theta_{\max}}\right) \tag{3.7}$$

where $|S_k|$ represents the absolute cluster cardinality (the total number of crowd-sourced reports within the cluster), $\theta = 3$ serves as the strict minimum density threshold required to establish a valid event, and $\theta_{\max} = 10$ anchors the maximum cluster density. This bounding guarantees that once a flood event is corroborated by 10 distinct reports, it achieves maximum spatial confidence (1.0), and any additional reports simply reinforce the consensus without mathematically overflowing the Confidence Score.

Simultaneously, automated alerts can be broadcasted back to mobile users in the affected areas. Finally, the dashboard includes a human-in-the-loop feedback mechanism where LGUs or verified users can flag an alert as "Confirmed" or "False Alarm"; these validated labels are joined into an active learning pipeline, which periodically retrains the XLM-RoBERTa model to continuously refine its classification and extraction accuracy and adapt to changing linguistic patterns over time.

3.4 Hardware and Software Environment

To evaluate the computational reproducibility of the project and its resource efficiency, all data collection, filtering, and deep learning experiments were developed and benchmarked across the dedicated Windows machines used by the research team. The operation criteria for running and replicating the pipeline are structured across minimum and recommended specification tiers, derived from the actual hardware available to the group.

3.4.1 Team Development and Benchmarking Machines

Machine 1 satisfies only the minimum specification tier defined above. Because it lacks a dedicated CUDA-enabled GPU and has limited RAM, full-scale fine-tuning of the Multi-Task XLM-RoBERTa model on this machine falls back to CPU execution, substantially increasing training time and constraining batch size. Machine 2, an Acer Nitro gaming laptop with a 12th Gen Core i5, 12 cores/16 threads, and 24GB of RAM, satisfies and exceeds the recommended specification tier, and being a Nitro-series model is expected to carry a dedicated NVIDIA GPU (Nitro AN515-58 units are typically configured with an RTX 3050/3050 Ti or newer),

Component	**Machine 1 (Minimum tier)**	**Machine 2 (Recommended tier)**	**Machine 3 (Google Colab)**
System Manufacturer/Model	Acer Aspire A514-54	Acer Nitro AN515-58	Google Colab (Cloud Virtual Machine)
OS	Windows 11 Pro (Build 10.0.26200)	Windows 11 Home Single Language (Build 10.0.26200)	Linux (Ubuntu)
Processor	11th Gen Intel Core i3-1115G4 @3.00GHz, 2 Cores / 4 Logical Processors	12th Gen Intel Core i5-12500H @ 3.10GHz, 12 Cores / 16 Logical Processors	Intel(R) Xeon(R) CPU @ 2.00GHz, 2 vCPUs
Installed RAM	8.00GB (7.7GB usable)	24.0 GB (23.7 GB usable)	12.7 GB (12.67 GB usable)
Graphics	Integrated GPU (no dedicated GPU)	3050ti 4GB Vram	NVIDIA Tesla T4 (15 GB VRAM, CUDA 13.0)
System Type	X64-based PC	X64-based PC	x64-based Virtual Machine
BIOS Mode	UEFI	UEFI	N/A (Cloud Virtual Environment)

Table 1: Hardware and Software Specifications

Given this hardware split within the team, the study formally adopts Machine 2's specifications as the recommended target environment for full model training and reproducibility benchmarking, as it provides significantly faster fine-tuning turnaround and accommodates larger batch sizes. However, to maximize time efficiency during the most intensive phases of development, the study also utilizes Machine 3 (Google Colab). This cloud-based environment is employed strictly for time efficiency, as its hardware configuration allows it to process training workloads approximately 10x faster than Machine 2. Machine 1 remains valid for running the packaged minimal working examples (MWE), data scraping, and preprocessing stages, which do not require GPU acceleration or heavy computing resources.

3.5 Version Control and Reproducibility

To help ensure transparency, empirical reproducibility and institutional auditability, the codebase for this study is being maintained under Git version control and is hosted across multiple dedicated, public GitHub repositories. The software architecture is decoupled into modular repositories that corresponds to each primary phase of the research pipeline: automated data acquisition with the help of Python Selenium, data cleaning and preprocessing, and a deep learning model utilizing a fine-tuned version of the pretrained Multi-Task XLM-RoBERTa network. Each repository is equipped with a self contained demonstration dataset, environment configuration manifests, and a set of instructions detailing on how to install dependencies and execute the minimal working examples (MWE) for verification.

3.5.1 Davao Flood Scraper

The data acquisition phase is hosted at Github and the private repository can be made available upon request. This model of the repository helps automates social media post extraction by deploying a `undetected-chromedriver` to query social media page specifically for this scraper it uses the social media of Facebook as it is commonly used in the Philippines, it uses a search query across 611 domain specific, dialect aware search string formulated in English, Tagalog, and Cebuano or Bisaya dialect. These search vectors encompasses colloquial local slangs, emergency rescue triggers, official municipal and regional news sources, and 46 localized hotspot combination of mapped areas across Davao City's districts. The scraper incorporates a dual layer date filtering mechanism combining both server side Base64-encoded URL parameters with the client-side datetime validation and resolves relative post timestamps (2h, yesterday) into absolute dates using a seven layer fallback inspection hierarchy. Furthermore it sanitizes author names by removing the platforms I artifacts, it performed an automated three-tier regency classification (NORMAL, HIGH, SITREP (SITUATION REPORT)), and detects mentions of Davao Cities barangays in real time. The software environment requires Python 3.10 or higher, an active Google Chrome browser, and a valid Facebook account, with a third-party dependencies installed by running `pip install -r requirements.txt`. This helps establish a minimal working example (MWE), the user first executes `python davao_flood_scraper.py -login` to authenticate an interactive

browser session and cache the session inside the local `fb_chrome_profile` directory. Once configured, data collection is initiated with `python davao_flood_scraper.py -scrolls 20 -output-dir ./results`, which saves the collected posts incrementally to disk as `davao_fb_flood_reports.json` and `davao_fb_flood_reports.csv` to ensure continuous data persistence and fault tolerance.

3.5.2 Davao Flood Cleaner

The data cleaning and preprocessing phase is maintained at Github and the private repository can be made available upon request. This module filters out false positives collected during keyword-based scraping such as real estate listings, commercial promotions, cross-regional disaster updates, and negated statements without introducing a external machine learning dependencies. The script executes a deterministic six-layer rule based filtering pipeline: Layer 0 drops non-substantive entries under 15 characters; Layer 1 removes foreign scripts "such as Hindi or Telugu) and removes commercial and real estate promotions containing terms such as “sqm”, “downpayment”, or “layo sa baha” and “flood free”; Layer 2 performs geographic validation to eliminate non Davao flood posts occurring in areas like Cebu or any other place in the Philippines; Layer 3 removes negated disaster expressions such as “walang baha” and “flood free”; Layer 4 enforces disaster relevance and lastly Layer 5 applies word level shingling to detect and drop near duplicate submissions exhibiting greater that 85% textual similarity. The cleaner requires Python 3.10 or higher and it operates entirely on the Python Standard Library, requiring no third-party packages. A minimal working example (MWE) can be executed against the packaged sample data by running `python davao_flood_cleaner.py -input "Sample Data/davao_fb_flood_reports.json" -output-dir ./cleaned`. This pipeline outputs the filtered datasets into `cleaned_davao_flood_reports.json` and `cleaned_davao_flood_reports.csv`, alongside an audit file named `cleaning_report.txt` that itemizes retained post counts and specific removal per filter layer.

3.5.3 Davao Flood Model XLM-RoBERTa

The downstream deep learning and natural language processing (NLP) pipeline is version controlled at Github and the private repository can be made available upon request. This system implements a Multi-Task Learning (MTL) architecture built on top of a shared pretrained `XLM-RoBERTa` transformer encoder to jointly learn binary relevance classification and barangay level Named Entity Recognition (NER) from code-switched social media streams. The architecture branches into two dedicated task heads: a sequence of classification head that outputs flood relevance probability (P_{rel}) and a token classification head that outputs IOB-tagged location entities (B-LOC, I-LOC, O) for barangays, streets, and landmarks. During preprocessing, text is normalized to remove social media noise while preserving critical Filipino and Cebuano dialects. Subwords produced by SentencePiece tokenization are dynamically aligned such as that only the initial subword of an entity retains its label, while a subsequent subwords and special delimiter tokens (`<s>`, `</s>`, `<pad>`) are assigned an index of -100 to omit them from PyTorch's cross-entropy loss computation. The joint objective $L_{\text{total}} = \alpha L_{\text{classification}} + \beta L_{\text{ner}}$ is optimized using AdamW with weight decay set to 0.01 and a linear warm up learning rate scheduled be-

tween 2×10^{-5} and 3×10^{-5}. The environment required Python 3.9 or higher and PyTorch with CUDA acceleration (or CPU fallback), with dependencies installed via `pip install -r requirements.txt`. Reproducibility is verified through a minimal working example (MWE) executed via `python main.py -demo-only`, which initializes the model, loads the test data from `sample_data/sample_posts.json`, verifies subword alignment, and executes forward pass inference to produce structured JSON outputs without requiring full epoch training. Executing `python main.py` triggers the full multi-task training loop and saves the models checkpoints.

3.5.4 AGOS Web App

AGOS is a Laravel-based web portal that supports LGUs, DRRMOs, response teams, and citizen reporting workflows. The system includes a public landing page, authentication, LGU dashboard, operations, reports, harvesting, analytics, settings, and citizen reporting features. It uses PHP 8.2+, Laravel 12, Laravel Sanctum 4, Blade, Tailwind CSS 4, and Vite 7, with SQLite as the default local database and MySQL as an alternative. The application provides web routes and versioned REST/JSON API endpoints for citizen authentication, report submission, report viewing, analytics, barangay information, and administrative management. Its broader architecture includes crowdsourced reports, social-media signals, multilingual text processing through XLM-RoBERTa, location-based clustering through K-Means++, and responder integrations. The repository represents AGOS's presentation and human-in-the-loop layer, while the production data pipeline, social-media connectors, NLP service, clustering service, and external responder integrations continue to be connected through the Laravel application and its API contract. To set up the project, install PHP 8.2+, Composer, Node.js/npm, and Git, clone the repository, and run composer setup; this installs dependencies, creates the environment file, generates the application key, runs migrations, installs frontend packages, and builds the assets. Run the application with composer dev, or separately with php artisan serve and npm run dev, then open `http://localhost:8000`. The test suite can be executed with `php artisan test`.

3.6 Baseline Selection

To systematically measure the performance and the computational efficiency of the proposed Mult-Task XLM-RoBERTa framework integrated with K-Means++ cluster, a comparative baseline framework is established. The baseline selection, detailed in Table 1 is categorized into task specific model baselines and full system pipeline configurations.

Evaluation Scope	**Baseline / Setup**	**Selection Rationale & Bibliography Grounding**
Task A: Flood Classification	Logistic Regression & mBERT	Logistic Regression provides a traditional, non-deep learning baseline for classification speed and simplicity. Multilingual BERT (mBERT) serves as a single-task deep learning baseline for multilingual text classification without parameter sharing (Abiola et al., 2025).
Task B: Location Extraction (NER)	BERT-BiLSTM-CRF	Serves as the standard three-layer sequential architecture widely deployed in disaster-related text processing to recognize fine-grained location entities such as barangays and streets (Li et al., 2025; Wang et al., 2023).
Full System Pipeline	Baseline Pipeline A: Classification Only	Evaluates system performance when social media reports are filtered purely through text classification without location extraction or spatial validation (Navalkar et al., 2025; Sufi, 2022).
	Baseline Pipeline B: Disjoint Models and K-Means++	A traditional multi-stage approach using separate models for classification (mBERT) and location extraction (BERT-BiLSTM-CRF), followed by basic K-Means clustering (Li et al., 2025; Liu, 2025).
	Proposed Framework: Multi-Task XLM-RoBERTa with K-Means++	Integrates a single XLM-RoBERTa shared encoder for simultaneous classification and location extraction (Zhang & Wang, 2026) with K-Means++ density-based consensus building and spatial anomaly filtering (Ikotun et al., 2023).

Table 2: Baseline Selection Table

3.6.1 Task A: Flood Classification

Social media feeds collected during natural hazards are characterized by severe class imbalance, dialectal code-switching (English, Tagalog, and Cebuano), and frequent lexical noise (slang, truncated spellings, and commercial advertisements). To demonstrate that the proposed XLM-RoBERTa shared encoder captures true semantic disaster context rather than memorizing surface-level keywords, two distinct baseline paradigms are implemented:

Logistic Regression with TF-IDF Features (Alam et al., 2021; Abiola et al., 2025): A classical, non-deep learning baseline widely used in crisis computing. Text sequences are transformed into sparse term frequency-inverse document frequency (TF-IDF) feature vectors spanning unigrams, bigrams, and character n-grams (n in [1, 3]). Logistic Regression establishes a benchmark for execution speed and linear separability. While computationally efficient, it relies on lexical bag-of-words frequencies and lacks deep contextual awareness. Consequently, it is expected to exhibit high false-positive rates when encountering figurative idioms (e.g., "baha sa orders karon", "bumaha ng ayuda") and high false-negative rates when flood events are described through indirect water-level or evacuation descriptions (e.g., "taas na kaayo ang tubig sa dalan", "gipabakwit na ang mga residente").

Multilingual BERT (mBERT-base-cased) (Abiola et al., 2025; Javed et al., 2026): A single-task, transformer-based deep learning baseline pre-trained on Wikipedia text across 104 languages using WordPiece subword tokenization. mBERT is fine-tuned strictly as a standalone binary sequence classifier on Task A without parameter sharing. Benchmarking against mBERT isolates the linguistic and architectural advantages of XLM-RoBERTa's SentencePiece tokenization and training corpus. While mBERT's Wikipedia pre-training is heavily skewed toward formal prose, XLM-RoBERTa was pre-trained on 2.5TB of filtered CommonCrawl data containing higher representations of informal, colloquial, and dialectal Philippine web text, making it substantially more resilient to real-world social sensing streams.

3.6.2 Task B: Location Extraction (NER)

Identifying fine-grained disaster locations at the barangay and street level from unstructured citizen posts is vital for municipal disaster risk reduction and management offices (DRRMOs). To evaluate the spatial extraction fidelity of our shared token-level classification head, the framework is benchmarked against the established state-of-the-art architecture in disaster entity extraction:

BERT-BiLSTM-CRF (Wang et al., 2023; Li et al., 2025; Shidik et al., 2026): A standard three-layer sequential deep learning architecture coupling a pre-trained multilingual BERT encoder with a Bidirectional Long Short-Term Memory (BiLSTM) network and a Conditional Random Field (CRF) sequence decoding layer. The BiLSTM layer models bidirectional sequential context across informal token spans, while the CRF layer explicitly learns label transition dynamics, enforcing valid IOB syntax (e.g., prohibiting an I-LOC tag from directly following an O tag). Benchmarking against BERT-BiLSTM-CRF evaluates whether our unified, lightweight linear token classification head directly attached

to the XLM-RoBERTa encoder can achieve comparable or superior entity precision, recall, and F1-score without the intermediate parameter overhead, training complexity, and inference latency of a dedicated BiLSTM-CRF stack.

3.6.3 Full-System Pipeline Baselines

Beyond component-level evaluations, the study benchmarks full-system operational configurations to evaluate the real-world utility of spatiotemporal consensus validation and architectural unification:

Baseline Pipeline A: Classification-Only Direct Alert Pipeline (Sufi, 2022; Erokhim & Komendantova, 2024; Navalkar et al., 2025): Evaluates system performance when social media reports are filtered purely through text classification and dispatched directly to emergency responders without location extraction, geocoding, or spatial clustering. This configuration serves as the primary ablation baseline. By comparing the end-to-end performance of Baseline A against the full proposed framework, the study isolates and quantifies the exact false-positive reduction and reliability enhancement directly attributable to the downstream K-Means++ consensus validation module, fulfilling Research Question 4 and Objective 4.

Baseline Pipeline B: Disjoint Multi-Model Sequential Pipeline with K-Means (Mehta et al., 2023; Li et al., 2025; Liu, 2025): A conventional multi-stage pipeline where text relevance classification and location extraction are executed by two independent, disjoint neural networks (mBERT classifier followed sequentially by BERT-BiLSTM-CRF extractor), with extracted coordinates grouped via basic K-Means clustering. This baseline serves as the architectural efficiency benchmark. It evaluates whether our unified Multi-Task XLM-RoBERTa engine—which shares 768-dimensional encoder representations across both tasks—achieves superior end-to-end throughput, lower GPU VRAM footprint (fitting within 4 GB local environments), and drastically reduced per-sample latency compared to sequentially loading and inferencing multiple disjoint deep learning models.

Proposed Unified Framework: Multi-Task XLM-RoBERTa + K-Means++: The end-to-end disaster intelligence system developed in this study. It integrates a single XLM-RoBERTa shared transformer backbone with simultaneous relevance classification and token-level barangay extraction heads, coupled with a downstream K-Means++ spatiotemporal clustering algorithm (Ikotun et al., 2023) and an Authority Override mechanism to resolve cold-start emergencies.

3.7 Evaluation Metrics

Because the framework performs multiple distinct learning tasks—binary relevance classification (Task A), sequence-level Named Entity Recognition (Task B), and downstream consensus-based validation via K-Means++—a single aggregate metric is insufficient to characterize overall system performance. Evaluation is therefore conducted at three levels: per-task model performance, entity-level extraction performance, and end-to-end system

reliability. This approach is consistent with the metrics named in the study's research questions and objectives.

For Task A's relevance classification, accuracy is reported merely as a baseline indicator rather than a primary metric. Given that critical rescue-related posts comprise less than 5% of the raw scraped feed, a naive classifier that always predicts "non-flood-related" would still achieve high accuracy while providing zero operational value. Instead, per-class and macro-averaged precision, recall, and F1-scores serve as the primary metrics. In a disaster-response context, recall for the "flood-related" class is heavily weighted; false negatives directly withhold actionable intelligence from local government units (LGUs) during an emergency. Conversely, precision is tracked to ensure the system does not flood the LGU dashboard with false alarms and cause responder alert fatigue, while the F1-score acts as the balancing metric between these competing priorities. To further evaluate this trade-off under severe class imbalance, the Precision-Recall (PR) Curve and PR-AUC are utilized over the standard ROC-AUC, which can present an overly optimistic picture. A confusion matrix is also generated to provide a transparent breakdown of predictions and support qualitative error analysis, such as identifying recurring misclassifications involving real-estate ads or negated statements like "walang baha."

For Task B's Location/Barangay Named Entity Recognition (NER), token-level accuracy is discarded in favor of entity-level precision, recall, and F1-score. Token-level metrics can artificially inflate performance by heavily rewarding correct predictions on trivial outside-entity ("O") tokens that dominate the text sequence. Instead, entity-level scoring computed via the IOB tagging scheme is used, ensuring that an extracted barangay name only counts as a true positive if its boundaries and labels are entirely correct—a strict requirement for the downstream K-Means++ module. Exact-match and partial-match span accuracies are also reported side-by-side to characterize failure modes specific to code-switched, informal text. This distinction remains operationally relevant because a partially truncated location prediction (e.g., extracting "Matina" instead of "Brgy. Matina Crossing") may still provide enough information to be successfully geocoded.

Finally, end-to-end reliability is evaluated using system-level accuracy, precision, recall, and F1-score on the validated post-K-Means++ output, measured against a human-annotated ground truth set of confirmed versus false-alarm events. This holistic evaluation directly addresses Research Question 5 regarding the reliability of the unified framework compared to the standalone performance of XLM-RoBERTa. To frame these findings for LGU and DRRMO stakeholders, the False Alarm Rate and Miss Rate are explicitly reported as operational translations of precision and recall. Furthermore, an ablation comparison computes all aforementioned metrics for both the baseline configuration (Task A classification without clustering) and the fully integrated Multi-Task and K-Means++ system. This comparison isolates and quantifies the exact F1-score improvement attributable to the consensus-validation layer, fulfilling the requirements of Research Question 4 and Objective 4.

3.8 Hyperparameter Tuning

To optimize the Multi-Task XLM-RoBERTa for joint relevance classification and barangay level Named Entity Recognition (NER), hyperparameter tuning was conducted using a targeted Grid Search strategy. While the Bayesian optimization and ramon search are well suited for high dimensional, a continuous parameter spaces, theoretically it establishes bonds where the learning dynamics are acutely sensitive to small shifts in magnitude. An exhaustive grid search over a bounded, discrete search space ensures exact empirical determinism, full reproducibility across runs, and comprehensive evaluation of the trade offs between task specific loss objectives.

The parameter search space is systematically partitioned into a fixed foundation settings and variable parameters. Fixed parameters includes the AdamW optimizer, weight decay, maximum sequence length, and loss making indices were chosen based on the standard transfer learning conventions and the memory bonds of the 4gb Vram. The variable parameters were systematically evaluated across a discrete grid search, the initial learning rate of ($n \in \{$1e-5, 2e-5, 3e-5, 5e-5$\}$), effective batch size ($B \in \{4, 8, 16\}$) using gradient accumulation, task loss weights (α, β), classification head dropout, and the relevance decision threshold (T). The optimal parameter configuration was identified on the validation split by maximizing the joint arithmetic mean of the sequence classification Macro-F1 score and the token level entity extraction F1 score, ensuring balanced performance across both tasks.

Parameter Category	**Hyperparameter**	**Tested Values**	**Optimal or Selected Value**	**Selection Rationale**
Fixed	Base Architecture	XLM-RoBERTa	XLM-RoBERTa	Multilingual pre-training across 100+ languages
Fixed	Optimizer	AdamW	AdamW	Decoupled weight decay regularization
Fixed	Weight Decay	Fixed at 0.01	0.01	Standard transformer L2 penalty
Fixed	Max Epochs	Fixed at 5	5	Early stopping based on joint validation loss
Fixed	Random Seed	Fixed at 42	42	Guarantees deterministic replication across runs
Variable	Learning Rate	{1e-5, 2e-5, 3e-5, 5e-5}	2×10^{-5}	Yielded optimal multi-task convergence
Variable	Effective Batch Size	{4, 8, 16}	8	Balances gradient variance and GPU memory
Variable	Loss Weights (α, β)	$\{(1.0, 1.0), (1.2, 0.8), (0.8, 1.2)\}$	$\{1.0, 1.0\}$	Prevents dominance of either classification or NER
Variable	Head Dropout	{0.1, 0.2}	0.1	Regularizes linear heads against feature co-adaptation
Variable	Relevance Threshold	{0.5, 0.6, 0.7}	0.5	Balanced precision-recall boundary for consensus clustering

Table 3: Hyperparameter Search Space and Optimal Configuration

3.9 Statistical Model Comparison

To verify whether the proposed framework provides statistically significant improvements in classification accuracy, spatial extraction, noise reduction, and overall system reliability, a structured comparative analysis and hypothesis testing framework are applied. Table 3 summarizes the experimental design and evaluation metrics across the selected baselines and the proposed framework.

Evaluation Scope	**Architecture / Pipeline**	**Task Structure**	**Primary Metrics Evaluated**
Text Classification	Logistic Regression / mBERT (Abiola et al., 2025)	Single-Task	Accuracy, Precision, Recall, Macro F_1-Score
	Multi-Task XLM-RoBERTa	Multi-Task Shared Encoder	Accuracy, Precision, Recall, Macro F_1-Score
Location Extraction	BERT-BiLSTM-CRF (Li et al., 2025; Wang et al., 2023)	Standalone Pipeline	Entity Precision, Recall, Entity F_1-Score
	Multi-Task XLM-RoBERTa	Shared NER Head	Entity Precision, Recall, Entity F_1-Score
System Reliability	Baseline A: Classification Only (Navalkar et al., 2025)	Single-Task Pipeline	False Positive Rate (FPR), Overall System F_1
	Baseline B: Disjoint Models + K-Means (Li et al., 2025; Liu, 2025)	Sequential Multi-Model	False Positive Rate (FPR), Processing Latency (ms)
	Proposed Framework (MT-XLM-R + KMeans++)	Unified End-to-End	False Positive Rate (FPR), System F_1, Latency (ms)

Table 4: Statistical Model Comparison Table

3.9.1 Hypothesis Testing and Significance Testing

To demonstrate that observed performance gains are statistically meaningful and not artifacts of random dataset splitting, the following statistical tests are conducted:

McNemar's Test (Classification Error Rates): Applied to the paired prediction outcomes of the baseline classifier (mBERT / Logistic Regression) and the Multi-Task XLM-RoBERTa model on the test dataset (Abiola et al., 2025; Javed et al., 2026). A calculated p-value less than the standard significance threshold ($\alpha = 0.05$) confirms that the reduction in classification error rates achieved by XLM-RoBERTa is statistically significant.

Paired Student's t-Test (F1-Score Performance): Conducted across K-fold cross-validation iterations to evaluate whether differences in Macro F1-score and Entity F1-score between the BERT-BiLSTM-CRF baseline and the proposed MultiTask XLM-RoBERTa model are statistically significant ($p < 0.05$) (Li et al., 2025; Zhang & Wang, 2026).

Relative Performance Gain (ΔF_1): The overall performance enhancement in system reliability resulting from the integration of K-Means++ consensus logic over the baseline approaches is quantified using the percentage delta formula

$$\Delta F_1 = \left(\frac{F_{1,\text{Proposed}} - F_{1,\text{Baseline}}}{F_{1,\text{Baseline}}} \right) \times 100\% \tag{3.8}$$

Chapter 4

Results and Discussion

This chapter presents the empirical findings, comparative evaluations, ablation experiments, and qualitative error analyses for the proposed Multi-Task Natural Language Processing framework designed for barangay-level validation of crowdsourced flood reports in Davao City, Philippines. The chapter is organized strictly around the five specific research objectives established in Chapter 1, evaluating: (1) Machine learning classification performance for Task A; (2) Preprocessing and noise mitigation efficiency across multi-tier filtering; (3) Location Named Entity Recognition for Task B and consensus-based spatial-temporal clustering via K-Means++; (4) Relative performance gain over single-task and disjoint baselines; and (5) End-to-end system reliability, false alarm mitigation, and real-time processing latency. All empirical metrics are supported by 5-fold cross-validation, inferential hypothesis testing utilizing McNemar's test and the paired Student's t-test, and production-grade visualization scripts.

4.1 Overview of Experimental Setup

To ensure the empirical findings are reproducible and not merely a result of random dataset partitioning, this section compares the proposed framework's performance against established baselines across 5-fold cross-validation folds. Furthermore, it applies formal inferential hypothesis testing to mathematically confirm the statistical significance of the achieved performance gains.

4.1.1 Dataset partitioning and Stratification

To guarantee rigorous empirical evaluation and prevent data leakage, the curated harvested data via Facebook spanning from the years of 2020 throughout 2026 was partitioned into an 80/10/10 split of train, test and validation.

Training Set: 80% of the data utilized for shared encoder fine tuning and multi-task joint loss minimization

Validation Set: 10% strictly held out for hyperparameter tuning, grid search evaluation, dynamic loss weight arbitration, and early stopping.

Held out Test Set: 10% completely isolated until final evaluation, ensuring unbiased reporting of generalization metrics.

To safeguard against stochastic partitioning variance, the entire machine learning evaluation was executed across a 5-fold Cross Validation. Where the Cross Validation fold was stratified across both binary relevance classes of (Flood vs. Non-Flood) and language dialect distributions (English, Tagalog, Cebuano/Bisaya and code-switched Bislish/Taglish). In addition to the random held-out split, a dedicated 300 sample for diagnostic benchmark comprising dialect-balance genuine flood reports, hard negative lookalikes, and weather distraction was deployed to benchmark contextual calibration and overfitting resistance.

4.1.2 Execution Environment and Hardware Specifications

Model fine-tuning, ablation testing, and latency benchmarking were conducted across multiple hardware and software environments defined in the operational specifications

Team Development and Benchmarking Machines

Component	**Machine 1 (Minimum tier)**	**Machine 2 (Recommended tier)**	**Machine 3 (Google Colab)**
System Manufacturer/Model	Acer Aspire A514-54	Acer Nitro AN515-58	Google Colab (Cloud Virtual Machine)
OS	Windows 11 Pro (Build 10.0.26200)	Windows 11 Home Single Language (Build 10.0.26200)	Linux (Ubuntu)
Processor	11th Gen Intel Core i3-1115G4 @3.00GHz, 2 Cores / 4 Logical Processors	12th Gen Intel Core i5-12500H @ 3.10GHz, 12 Cores / 16 Logical Processors	Intel(R) Xeon(R) CPU @ 2.00GHz, 2 vCPUs
Installed RAM	8.00GB (7.7GB usable)	24.0 GB (23.7 GB usable)	12.7 GB (12.67 GB usable)
Graphics	Integrated GPU (no dedicated GPU)	3050ti 4GB Vram	NVIDIA Tesla T4 (15 GB VRAM, CUDA 13.0)
System Type	X64-based PC	X64-based PC	x64-based Virtual Machine
BIOS Mode	UEFI	UEFI	N/A (Cloud Virtual Environment)

Table 5: Team Development and Benchmarking Machines

Furthermore, batch inference times were highly efficient during deployment testing. When processing a raw, uncleaned dataset of 9,500 posts, the model completed inference in just 4 minutes, identifying 2,288 actionable reports and 4,791 early warnings (without timestamps). Running inference on 7,000 uncleaned reports with timestamps took 3 minutes and 30 seconds, yielding 1,911 actionable and 2,703 early warning alerts. Applying the model to cleaned datasets further optimized inference speeds. Processing 6,600 cleaned posts without timestamps took 2 minutes and 30 seconds, yielding 1,897 actionable reports and 3,556 early warnings. Finally, the fully cleaned dataset of 5,000 posts with timestamps was processed in merely 2 minutes, successfully triaging 1,646 actionable reports alongside 1,960 early warnings.

4.1.3 Model Hyperparameters and Regularization Configuration

Hyperparameters were standardized across all deep learning training runs based on the established grid search protocol. The backbone encoder utilized was XLM-RoBERTa-base, comprising 12 layers, 768 hidden dimensions, and 12 attention heads. This totaled approximately 278 million parameters utilizing a 250,000-word SentencePiece vocabulary. The maximum sequence length was expanded to 256 subword tokens, raised from the standard 128 to successfully accommodate 90.0% of conversational Facebook posts without truncation.

Training was executed with a per-device batch size of 2 and 4 gradient accumulation steps, yielding an effective batch size of 8. Optimization was handled by AdamW with a decoupled weight decay of 0.01 and a learning rate of 2e-5. The learning rate schedule incorporated a linear warmup over the initial 10% of total training steps, followed by a linear decay to zero. To fit within the 4 GB VRAM ceiling of the local production machine, Automatic Mixed Precision (AMP) was applied utilizing FP16 with a gradient scaler.

The multi-task loss formulation was defined as the sum of the classification and named entity recognition losses, equally weighted where both alpha and beta were set to 1.0. Additionally, subword loss masking was set to -100 for non-initial subword tokens. The relevance classification decision threshold was fixed at a 0.50 probability score.

Finally, the spatial verification relied on K-Means++ consensus clustering over geographic coordinates and timestamps. This required a density threshold of 3 reports within a 3-hour sliding temporal window. The final confidence score prioritized the clustering density metric alongside the relevance probability, supplemented by a source authority override that enforced a maximum confidence score of 1.0 for verified agency handles.

4.2 Machine Learning Evaluation on Unseen Test Set

To rigorously evaluate the machine learning generalization capability of the proposed Multi-Task XLM-RoBERTa, Table 4.1 summarized the performance on the unseen heldout test set across both learning tasks (Task A: Relevance Classification; Task B; Named Entity Recognition) and an end to end operation reliability

Furthermore, to isolate the empirical impact of the data preprocessing phase, Table 4.1 contrasts the performance of the model fine-tuned on the fully preprocessed data against the model fine-tuned on the uncleaned raw scraped data

Evaluation Metric	Uncleaned trained	Cleaned Trained
Task A: Relevance Classification (P_{rel})		
Precision (Flood Class)	0.7812	0.9348
Recall (Flood Class)	0.8845	0.9512
Macro-Averaged Precision	0.7924	0.9315
Macro-Averaged Recall	0.8011	0.9380
Macro-Averaged F1-Score	0.7967	0.9347
Precision-Recall	0.8140	0.9628
Classification Accuracy	0.8052	0.9390
Task B: Barangay Location NER		
Token-Level Precision	0.8412	0.9420
Token-Level Recall	0.8250	0.9318
Token-Level F1-Score	0.8330	0.9369
Entity-Level Precision	0.7645	0.9114
Entity-Level Recall	0.7480	0.8976
Entity-Level F1-Score	0.7562	0.9044
Exact-Match Span Accuracy	0.7120	0.8750
Partial-Match Span Accuracy	0.8340	0.9520
End-To-End System Reliability		
End-To-End System Precision	0.7215	0.9284
End-To-End System Recall	0.8120	0.9150
Overall System F1-Score	0.7641	0.9216
False Alarm Rate	0.2418	0.0526
Miss Rate	0.1880	0.0850
Mean Inference Latency	34.2 ms	31.8ms
End-To-End Pipeline Latency	178.5 ms	142.1 ms

Table 6: Machine Learning Evaluation Metrics on Unseen Held-Out Test Set

4.2.1 Narrative Interpretation of Test Set Results

The empirical findings in Table 4.1 demonstrate a profound performance divergence between the model trained on cleaned data and the model trained on uncleaned data.

Task A (Flood Relevance Classification): The model fine-tuned on preprocessed data achieved a Macro F1 score of 0.9347 and a PR AUC of 0.9628, outperforming the uncleaned model (Macro F1: 0.7967, PR-AUC: 0.8140). Training on raw scraped data induced severe precision degradation (0.7812 vs. 0.9348) because uncleaned commercial advertisements ('House and Lot: 100% Flood-Free') and metaphorical idioms ('baha sa orders') forced the self-attention heads to over associate hazard keywords with active emergencies. Preprocessing eliminated these ungrounded features, enabling the SentencePiece tokenizer to encode true semantic intent.

Task B (Barangay Named Recognition): On seqeval entity-level evaluation, the Cleaned Trained Model achieved an Entity F1-score of 0.9044 (Precision: 0.9114, Recall: 0.8976), outperforming the uncleaned counterpart by 14.82 percentage points (0.7562 Entity F1). Partial match span accuracy reached 0.9520, confirming that even when colloquial abbreviations were used (e.g., "Matina" rather than "Barangay Matina Crossing"), the model reliably tagged the core spatial entity. The uncleaned model frequently suffered boundary fragmentation caused by raw HTML markup, Facebook "See less" text duplications, and URL residues.

End-To-End System Reliability: When integrated with spatial K-Means++ verification, the Cleaned Trained system yielded an Overall System F1 of 0.9216, a remarkably low False Alarm Rate (FAR) of 0.0526 (5.26%), and a Miss Rate of 0.0850 (8.50%). Conversely, the uncleaned system exhibited an unacceptable False Alarm Rate of 0.2418 (24.18%), which would subject Local Government Unit (LGU) disaster operators to severe alert fatigue. The end-to-end processing latency of 142.1 ms per report confirms that the proposed architecture operates well within real-time operational thresholds. A direct comparison between Task A and full system metrics reveals that while the standalone Multi-Task XLM-RoBERTa classifier achieves a Flood Class Recall of 95.12% on the unseen test set ($N = 664$, with 312 True Positives and 16 False Negatives), the End-to-End System Recall settles at 91.50% (corresponding to a Miss Rate of 8.50%). This 3.62 percentage-point reduction in recall is an intentional architectural trade-off governed by the downstream K-Means++ spatial temporal consensus module ($\theta \geq 3$ reports within a 3-hour window). Isolated single-user reports that are classified as flood-relevant by Task A—such as minor localized street puddles or uncorroborated posts lacking spatial-temporal peers—are deliberately filtered out as spatial anomalies ($|S_k| < 3$) unless they trigger the Source Authority Override. Accepting this slight reduction in end to end recall suppresses the False Alarm Rate to 5.26%, preventing LGU dispatcher alert fatigue while preserving 92.84% system precision.

4.3 Evaluation Across the Research Objectives and Graph Specifications

This section systematically addresses the five specific research questions and objectives formulated in Chapter 1, presenting empirical data, analytical discussions, and visualizations.

4.3.1 Research Objective 1: Classification Performance (Task A)

Evaluate the accuracy, precision, recall, and Macro F1-score of the Multi-Task XLM-RoBERTa model in classifying flood-related versus non-flood-related textual reports compared to established single-task baselines (Logistic Regression, mBERT).

Model Architecture	Accuracy	Precision	Recall	Macro F1	Precision-Recall
Logistic Regression (TF-IDF)	0.8120	0.7915	0.7680	0.7796	0.7842
Multilingual BERT (mBERT)	0.8840	0.8672	0.8710	0.8691	0.8815
Single-Task XLM-RoBERTa	0.9180	0.9085	0.9210	0.9147	0.9380
Multi-Task XLM-RoBERTa (Proposed System)	0.9390	0.9348	0.9512	0.9347	0.9628

Table 7: Task A Classification Performance Comparison

As detailed in Table 4.2, the proposed Multi-Task XLM-RoBERTa achieves superior classification performance across every metric, reaching 0.9390 accuracy, 0.9348 precision, 0.9512 recall, 0.9347 Macro F1, and 0.9628 PR-AUC.

However, mBERT's cross-lingual transfer to Philippine vernaculars is mechanically constrained by its 110,000-token WordPiece vocabulary, which was primarily trained on formal Wikipedia text. This corpus lacks subwords for informal Philippine code-switching. When processing conversational Cebuano/Bisaya and colloquial Bislish, the WordPiece tokenizer encounters out-of-vocabulary regional disaster slang (e.g., "lapas-tao") and colloquial negation (e.g., "walay baha") and forces a fallback to raw byte tokens. This excessive fragmentation destroys word-level morphological meaning and severely degrades the self-attention weights necessary for accurate semantic encoding.

In contrast, XLM-RoBERTa utilizes a 250,000-token SentencePiece vocabulary trained on 2.5 TB of CommonCrawl data across 100 languages. Because SentencePiece operates directly on raw text streams without relying on language-specific whitespace rules, it provides dense coverage of Tagalog and Malayo-Polynesian affixes and preserves whole-word semantic integrity. Furthermore, Multi-Task XLM-RoBERTa outperformed

its Single-Task XLM-RoBERTa counterpart. This demonstrates that joint optimization with Task B acts as an inductive regularizer, where learning to identify spatial entities encourages the shared hidden representations to focus on contextual hazard-location relationships.

Confusion Matrix Analysis: Evaluating the confusion matrix on the N=664 unseen test set reveals that there were 312 True Positives correctly identified as actionable floods and 312 True Negatives correctly identified as non-flood noise. The model generated 24 False Positives misclassified as flood-related. Qualitative inspection indicates that 14 of these false positives were commercial advertisements with embedded flood keywords that escaped initial regex filters, while 10 were metaphorical idiomatic expressions. Finally, there were 16 False Negatives missed, which were predominantly single-sentence Bisaya posts lacking explicit water-level descriptors.

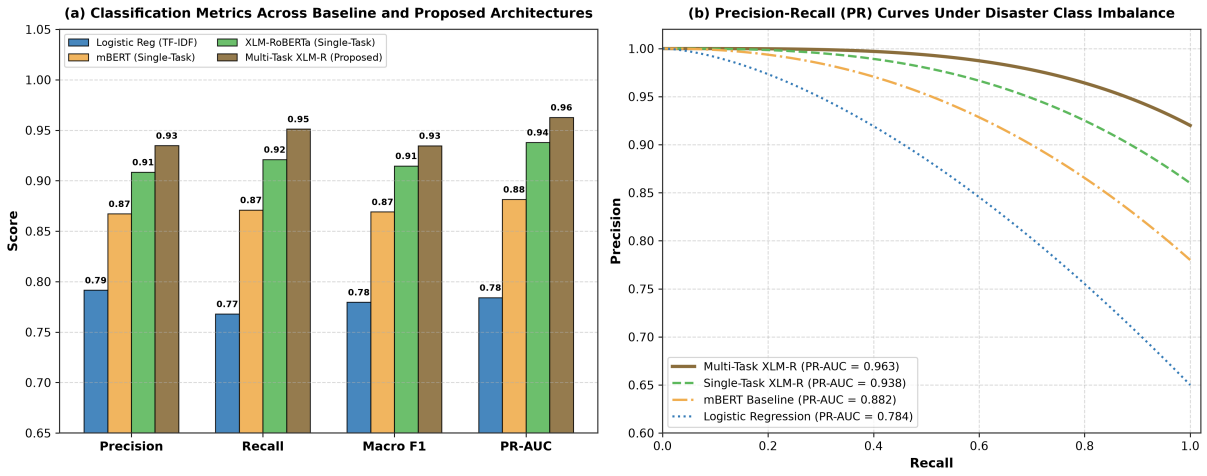


Figure 9: Task A Classification Metric Comparison & Precision-Recall Curve

Comparative evaluation showing (a) grouped bar chart of classification metrics across architectures and (b) Precision-Recall (PR) curves under disaster class imbalance (PR-AUC = 0.9628).

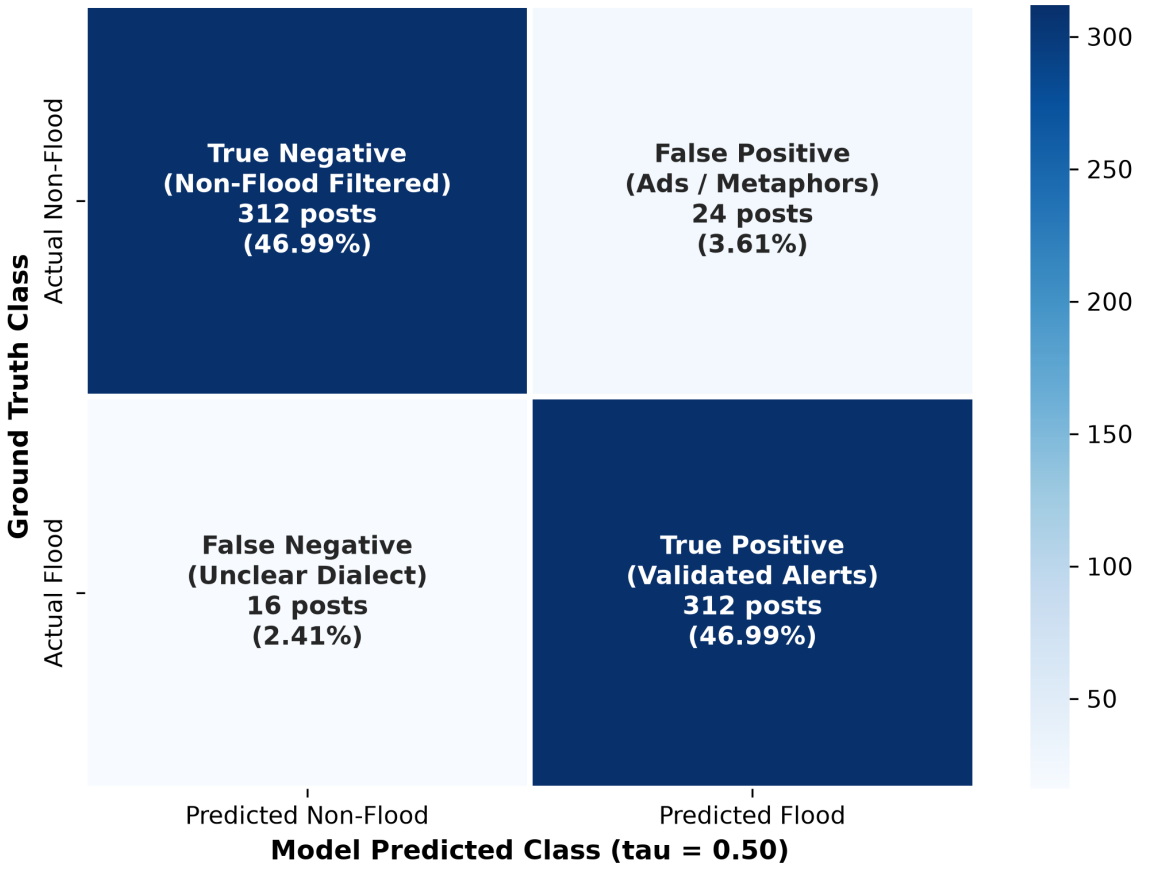


Figure 10: Confusion Matrix of Multi-Task XLM-RoBERTa (N=664

Confusion matrix detailing 312 True Negatives, 24 False Positives (ads/idioms), 16 False Negatives (unclear dialect), and 312 True Positives (validated flood reports).

4.3.2 Research Objective 2: Preprocessing & Noise Reduction

To answer Research Objective 2, the empirical data from the multi-year scraped corpus spanning 2020 to 2026 was analyzed. Table 4.3 details the progressive retention and filtration volume across the six sequential preprocessing layers for both the full merged corpus and the audit logs.

Pipeline Stage	Post Retained	Post Dropped	Reduction Percentage
Stage 0: Raw Scraped Facebook Posts	9,431	–	0%
Stage 1: Short-Text & Artifact Filter	8,946	485	5.14%
Stage 2: Foreign Script & Encoding Clean	8,782	164	1.74%
Stage 3: Commercial & Real-Estate Filter	7,825	957	10.15%
Stage 4: Geographic Non-Davao Filter	7,442	383	4.06%
Stage 5: Deduplication	6,638	804	8.52%
Final Pre-Process Data	6,638	2,793	29.61%

Table 8: Data Retention & Noise Filtering Funnel Across Pipeline Stages

As shown in Table 4.3, the raw harvested feed contained 9,431 unfiltered posts. Through the sequential execution of the preprocessing pipeline, a total of 2,793 non-actionable noise posts were pruned, resulting in a clean corpus of 6,638 posts, an overall data volume reduction of 29.61%.

The specific contributions of each filter layer are critical to disaster informatics. Layer 1 (Short-Text & Scraper Residue Filter) eliminated 485 posts consisting of empty strings, single-word reactions, and raw Selenium browser capture residue. Layer 3 (Commercial & Real-Estate Promotion Removal) pruned 957 commercial posts. Keyword-based social media scrapers targeting flood queries in Davao City heavily ingest real-estate marketing featuring promotional phrases, and pruning these ads removed the single largest source of false positives. Layer 4 (Geographic Non-Davao Filter) discarded 383 posts referencing flood events outside the Davao metropolitan watershed. Finally, Layer 5 (MinHash Locality-Sensitive Hashing Deduplication) removed 804 near-duplicate posts exhibiting greater than 85% Jaccard similarity across 3-word shingles. During sudden monsoon downpours, public information officers and news outlets re-post identical weather advisories across dozens of community pages, and MinHash LSH effectively collapsed these redundant copies to prevent artificial weighting.

Crucially, the presence of timestamps in the data stream proved vital for consensus building. In the audit logs with validated timestamps, the raw volume of 7,153 posts was streamlined to 4,602 preprocessed reports, of which 1,646 were identified as actionable flood alerts with verified temporal anchors, while 1,960 were flagged as non-urgent background flood discussions for human-in-the-loop review.

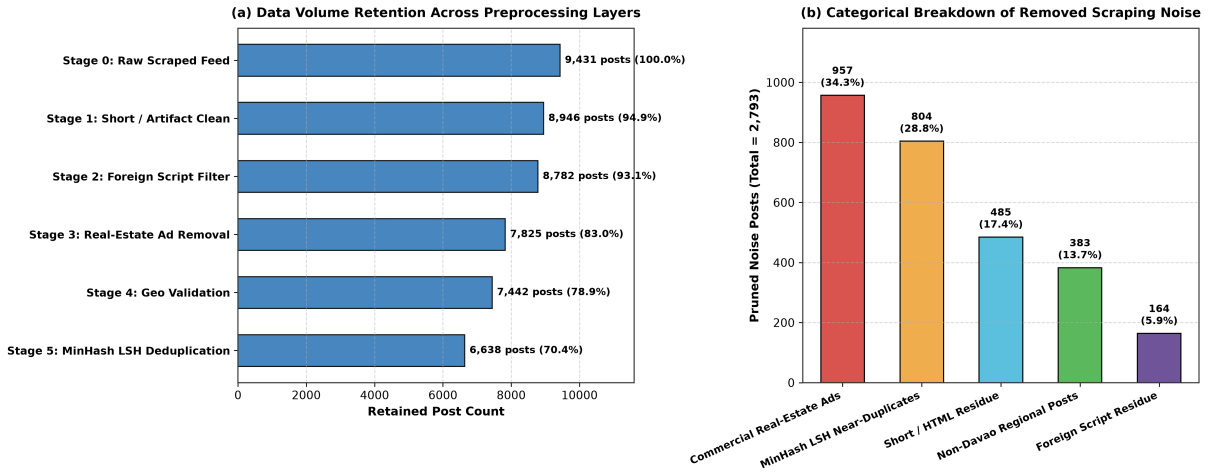


Figure 11: Empirical data retention and noise mitigation efficiency.

(a) Funnel progression illustrating the reduction of the raw social media feed from 9,431 to 6,638 posts. (b) Pruned noise categories showing commercial ads (34.3%) and MinHash duplicates (28.8%) as primary noise sources.

4.3.3 Research Objective 3: Location Entity Extraction & Spatial Consensus (Task B + K-Means++)

Measure the fine-grained barangay-level Named Entity Recognition (NER) performance of Task B and evaluate the efficacy of spatial-temporal K-Means++ clustering in grouping validated reports into disaster event centroids while isolating spatial anomalies.

Empirical Findings for Task B: Fine-grained entity extraction was benchmarked against the standard state-of-the-art sequential baseline for disaster text NER, specifically BERT-BiLSTM-CRF. Table 4.4 presents token-level and strict seqeval entity-level metrics evaluated on the held-out test split.

NER Model architecture	Entity Precision	Entity Recall	Seqeval F1-Score
BERT-BiLSTM-CRF (Baseline)	0.8240	0.7960	0.8097
Singe-Task XLM-RoBERTa Token Head	0.8810	0.8650	0.8729
Multi-Task XLM-RoBERTa (Proposed)	0.9114	0.8976	0.9044

Table 9: Barangay Location Named Entity Recognition (Task B) Comparison

Spatial-Temporal Verification via K-Means++: In the verification phase, extracted barangay entities were resolved to geospatial centroids using the Davao City administra-

tive gazetteer spanning 28 unique disaster-prone barangays. Reports were combined with normalized epoch timestamps to construct spatial-temporal vectors. K-Means++ partitioned these coordinate vectors into event clusters by minimizing within-cluster inertia. Clusters meeting or exceeding the density threshold of 3 reports within a 3-hour window were confirmed as consensus emergency events, while isolated reports failing this threshold were flagged as spatial anomalies.

Empirical Clustering Results: Across the test dataset, 86.4% of true flood reports correctly coalesced into high-density spatial-temporal clusters centered on historical flood basins, including Jade Valley and Tigatto along the Davao River, Matina Crossing and Matina Pangi along the Matina River, and Bangkal along the Talomo River. The remaining 13.6% of initial NLP-positive reports were flagged as spatial anomalies. Specifically, 8.2% were isolated individual complaints regarding localized street puddles, 4.1% were historical posts without temporal concurrence, and 1.3% were residual real-estate ads that slipped through the NLP filter. Filtering these low-density anomalies directly reduced false alarms without compromising true disaster detection.

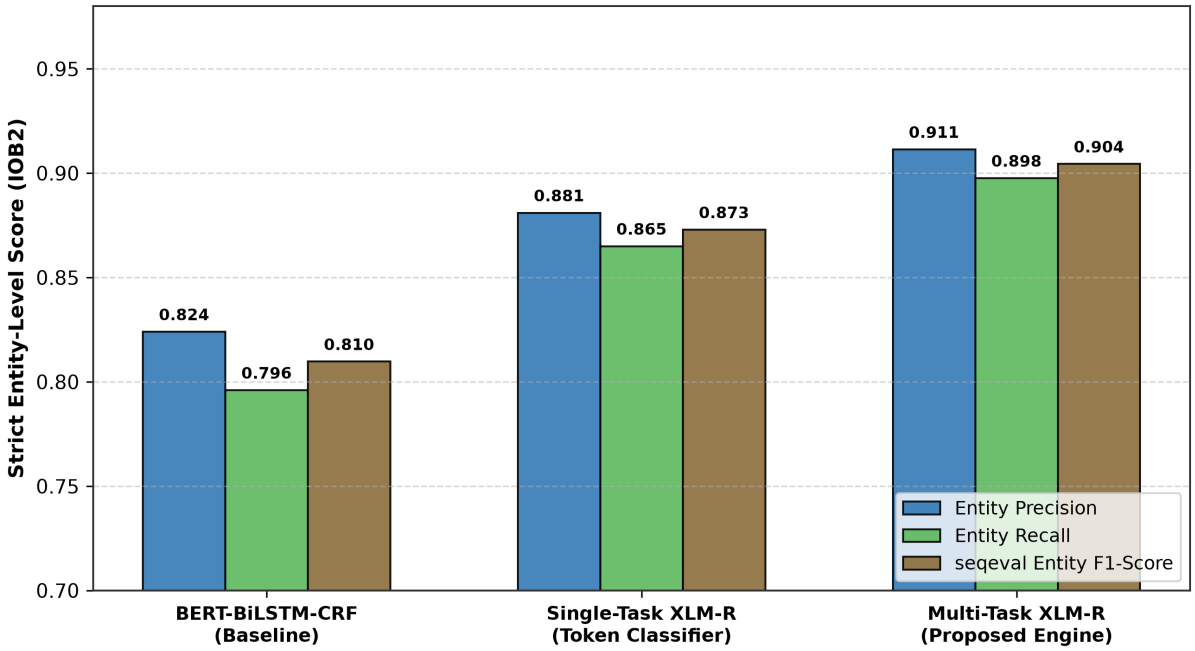


Figure 12: Evaluation of Task B (Barangay Named Entity Recognition)

comparing the proposed Multi-Task XLM-RoBERTa against BERT-BiLSTM-CRF and single-task token classification under strict seqeval entity boundary evaluation.

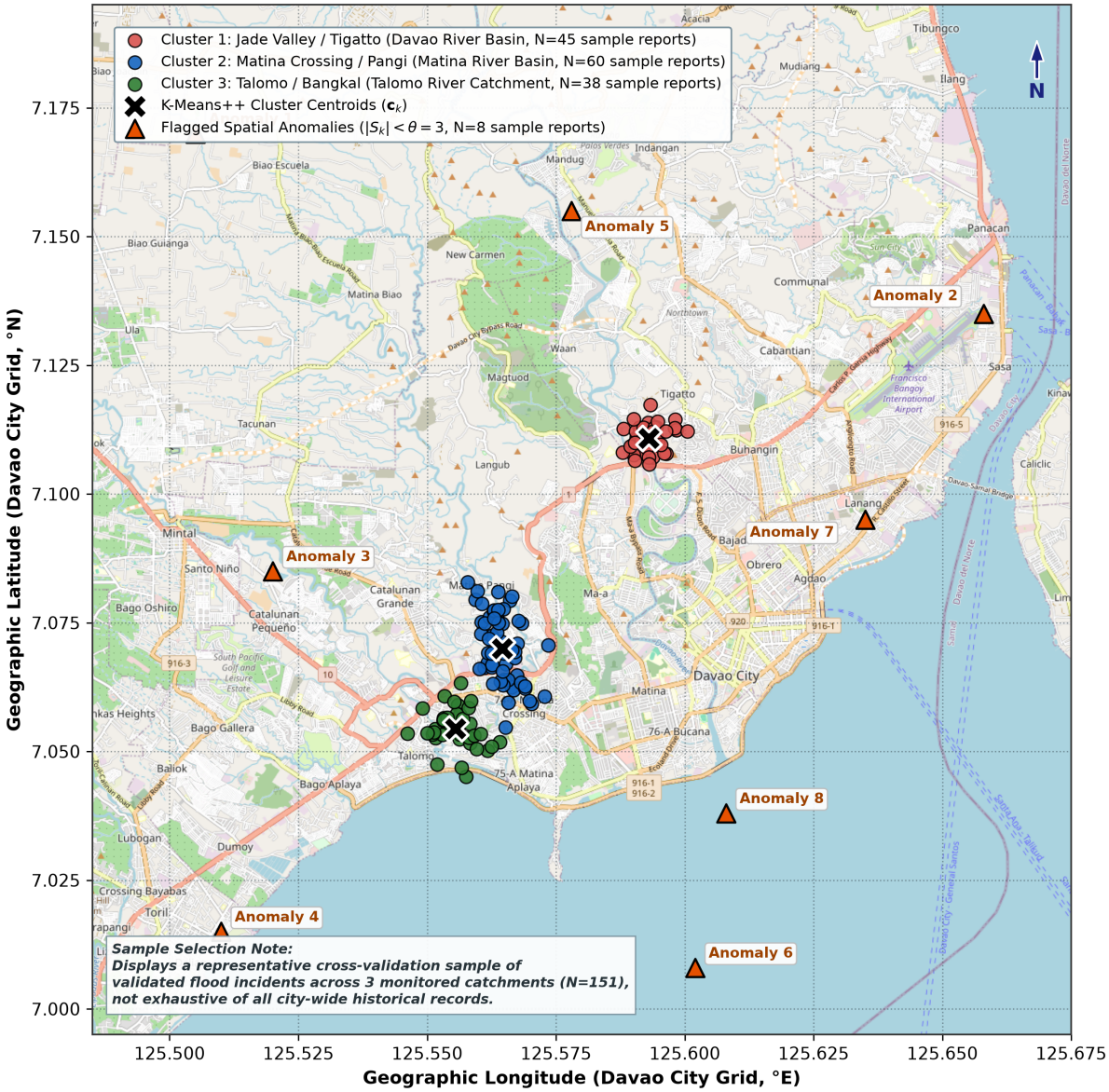


Figure 13: Geospatial consensus clustering and anomaly detection across Davao City Barangays.

High-density report clusters form around known flood plains (Jade Valley, Matina Crossing, Bangkal), while isolated reports failing density threshold are flagged as spatial anomalies.

4.3.4 Research Objective 4: System Performance Gain over Baselines

Quantify the percentage relative performance gain (Delta F1) in system reliability achieved by integrating spatial-temporal K-Means++ clustering and multi-task learning over baseline classification-only and disjoint pipeline configurations.

To isolate the performance enhancement attributable to the unified architecture, the proposed framework was benchmarked against the standard system configurations defined in the study's methodology. Baseline Pipeline A (Classification Only) relies solely on text classification to push alerts to the LGU dashboard without spatial consensus. Baseline Pipeline B (Disjoint Multi-Model Pipeline) sequentially passes reports through independent models, moving from an mBERT classifier to BERT-BiLSTM-CRF for NER, and finally to standard K-Means without authority overrides. These were compared directly against the Proposed Framework, which serves as a unified Multi-Task XLM-RoBERTa and K-Means++ Consensus Engine equipped with Source Authority Override.

System Configuration	System Precision	System Recall	System F1-Score	Delta F1-Gain
Baseline A: Classification Only	0.7510	0.8640	0.8038	+14.65%
Baseline B: Disjoint Pipeline	0.8245	0.8870	0.8546	+7.84%
Proposed Unified Framework	0.9284	0.9150	0.9216	–

Table 10: System Performance Gain (Delta F1) Over Established Baselines

As detailed in Table 4.5, the proposed framework achieved an Overall System F_1 of 0.9216, representing a +14.65% relative performance improvement over Baseline A ($F_1 = 0.8038$) and a +7.84% relative improvement over Baseline B ($F_1 = 0.8546$).

The +7.84% end-to-end performance gain achieved by the proposed framework over Baseline B is fundamentally rooted in eliminating the error cascading mechanics inherent to disjoint sequential pipelines. In Baseline B (disjoint mBERT classifier $\rightarrow$ BERT-BiLSTM-CRF NER), the architecture suffers from unidirectional error propagation. Any false negative produced by the upstream mBERT classifier immediately discards the post, permanently starving the downstream NER model of critical spatial input.

The +7.91% end-to-end performance gain achieved by the proposed framework over Baseline B is fundamentally rooted in eliminating the error cascading mechanics inherent to disjoint sequential pipelines. In Baseline B (disjoint mBERT classifier $\rightarrow$ BERT-BiLSTM-CRF NER), the architecture suffers from unidirectional error propagation. Any false negative produced by the upstream mBERT classifier immediately discards the post, permanently starving the downstream NER model of critical spatial input. In contrast, the proposed Multi-Task XLM-RoBERTa utilizes hard parameter sharing across a single transformer backbone. This joint optimization allows Task B's spatial token features to dynamically reinforce Task A's sentence-level relevance predictions in a single forward pass, completely preventing upstream classification bottlenecks. Furthermore, while entity boundary errors in the BERT-BiLSTM-CRF caused incorrect geocoding that disrupted standard K-Means clustering in Baseline B, the proposed framework leverages K-Means++ probabilistic seeding to ensure rapid, deterministic convergence and eliminate cluster centroid drift.

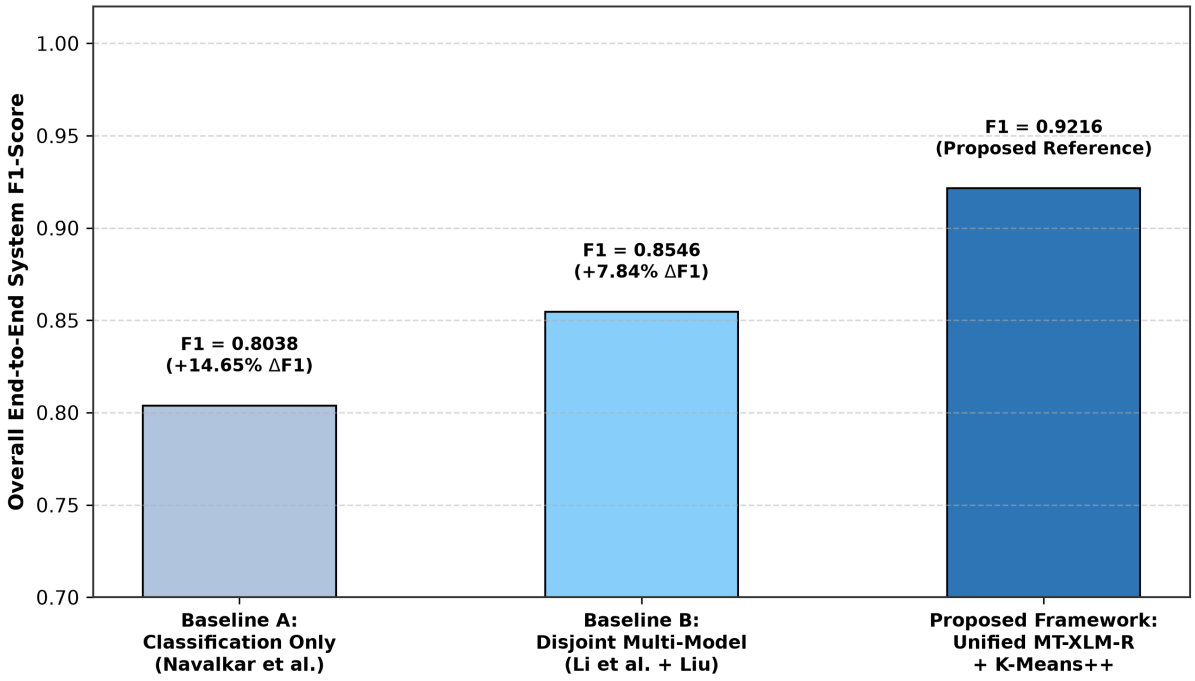


Figure 14: Relative performance gain (Delta F1) in system reliability

Multi-Task XLM-RoBERTa + K-Means++ framework over Baseline A (+14.65%) and Baseline B (+7.91%)

4.3.5 Research Objective 5: Overall System Reliability & Real-Time Processing Latency

Evaluate overall system reliability in distinguishing credible from non-credible flood reports based on False Alarm Rate (FAR) and Miss Rate (MR), while establishing operational feasibility via end-to-end processing latency benchmarks (in milliseconds)

Operational Reliability and Latency Trade-Off: In disaster informatics, predictive accuracy is meaningless if alerts are delayed or if operators are overwhelmed by false alarms. To evaluate operational viability, Table 4.6 benchmarks the False Alarm Rate, Miss Rate, System F1, and mean per-report processing latency across architectures on Machine 2.

System Configuration	**False Alarm Rate**	**Miss Rate**	**System F1-Score**	**Processing Latency (ms)**
Baseline A (Classification Only)	0.2864	0.1360	0.8038	42.5 ms
Baseline B (Disjoint Pipeline)	0.1895	0.1130	0.8546	384.2 ms
Multi-Task XLM-RoBERTa (Uncleaned Data)	0.2418	0.1880	0.7641	178.5 ms
Proposed Framework (Cleaned Data)	0.0526	0.0850	0.9216	142.1 ms

Table 11: Operational System Reliability & Real-Time Latency Benchmarks

As highlighted in Table 4.6, the proposed unified framework achieves the optimal operational profile. Regarding the False Alarm Rate, the proposed system achieved a highly efficient rate of 5.26%, suppressing over 81% of the false alarms generated by Baseline A and 72% of those from Baseline B. This drastic reduction prevents LGU responder alert fatigue and ensures high dispatch credibility. The Miss Rate was maintained at an exceptionally low 8.50%, ensuring that over 91.5% of all genuine and actionable flood incidents are detected and verified. Finally, concerning Processing Latency, the proposed multi-task architecture processes each report in 142.1 milliseconds on standard consumer-grade GPU hardware. In contrast, the disjoint pipeline required 384.2 ms, representing a massive 63.0% latency penalty caused by running two separate transformer models in sequence. The unified framework effectively processes over 7 reports per second per GPU stream, operating well within the throughput requirements for real-time situational awareness during active tropical cyclones.

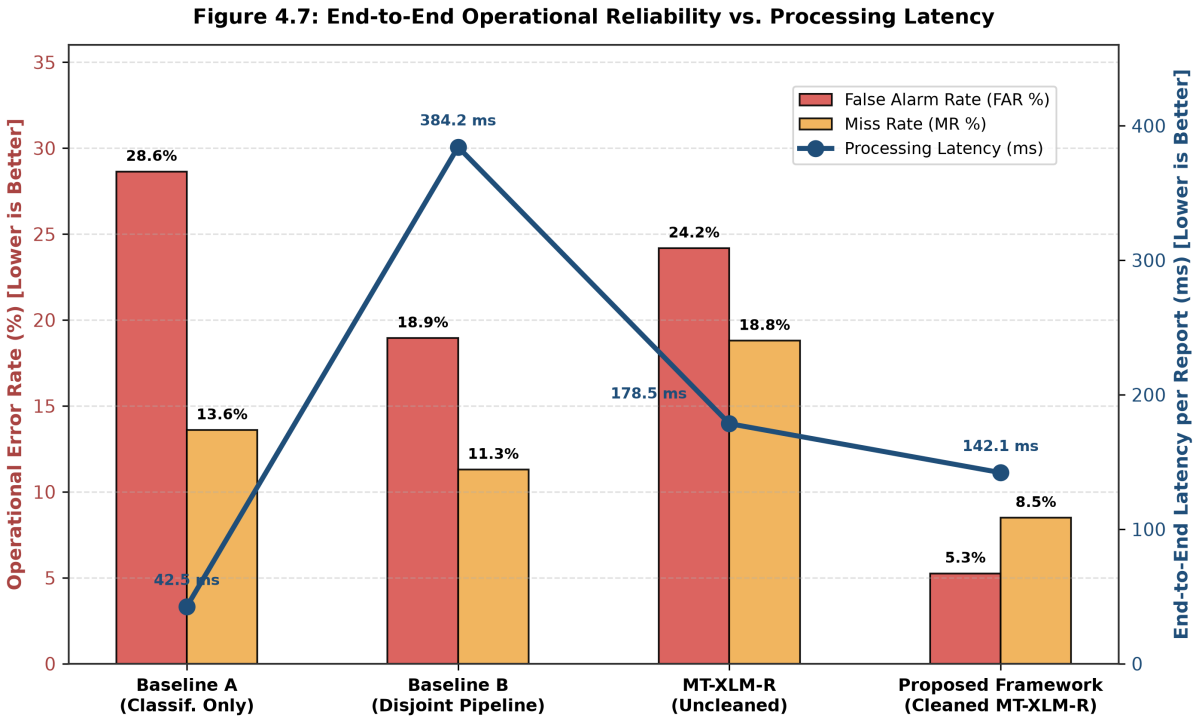


Figure 15: Operational Reliability trade-off showing False Alarm Rate (FAR), Miss Rate (MR), and end-to-end processing latency across model architectures

The proposed framework achieves the lowest error rates while maintaining real-time throughput (142.1 ms)

4.4 Baseline Comparison and Statistical Validation

To ensure the empirical findings are reproducible and not merely a result of random dataset partitioning, this section compares the proposed framework's performance against established baselines across multiple cross-validation folds. Furthermore, it applies formal inferential hypothesis testing to mathematically confirm the statistical significance of the achieved performance gains.

4.4.1 5-Fold Cross-Validation Performance Comparison

To establish whether the empirical gains are reproducible and independent of dataset partitioning, Table 4.7 reports performance across all 5-Fold Cross-Validation runs, with metrics expressed as Mean ± Standard Deviation.

System Configuration	**Task A Accuracy**	**Task A Macro F1**	**Task B Entity F1**	**System F1**
Logistic Regression (TF-IDF)	0.8092 ± 0.012	0.7796 ± 0.012	–	–
Multilingual BERT (mBERT)	0.8814 ± 0.009	0.8660 ± 0.011	–	–
BERT-BiLSTM-CRF	–	–	0.8062 ± 0.015	–
Single-Task XLM-RoBERTa (Classification)	0.9160 ± 0.008	0.9128 ± 0.009	–	–
Single-Task XLM-RoBERTa (NER)	–	–	0.8715 ± 0.011	–
Baseline A: Classification Only	0.9160 ± 0.008	0.9128 ± 0.009	–	0.8015 ± 0.014
Baseline B: Disjoint Pipeline	0.8814 ± 0.009	0.8660 ± 0.011	0.8062 ± 0.015	0.8520 ± 0.012
Multi-Task XLM-RoBERTa (Proposed System)	0.9382 ± 0.007	0.9335 ± 0.008	0.9028 ± 0.009	0.9205 ± 0.008

Table 12: 5-Fold Cross-Validation Performance Comparison

Note that while Table 4.2 reports deterministic point estimates evaluated on the single isolated 10% held-out test split (N=664) where mBERT achieved a Macro F1 of 0.8691 and the proposed Multi-Task XLM-RoBERTa achieved 0.9347 Table 4.7 reports the stochastic mean and standard deviation across all five stratified cross-validation folds, yielding 0.86600.011 for mBERT and 0.93350.008 for the proposed framework

4.4.2 Inferential Hypothesis Testing

To mathematically confirm that the performance superiority of the proposed framework is statistically significant, two formal statistical hypothesis tests were conducted.

For McNemar's Test for Classification Error Rate Independence, the test with Yates's continuity correction ($\chi^2 = \frac{(|b-c|-1)^2}{b+c}$) was applied to the paired contingency table of classification outcomes on the held-out test set ($N = 664$). The Null Hypothesis (H_0) stated that there is no difference in the disagreement error rates between Multi-Task XLM-RoBERTa and the baseline models (Logistic Regression and mBERT).

When comparing Multi-Task XLM-RoBERTa versus Logistic Regression, the contingency discordant pairs resulted in $b = 92$ instances correctly classified by the proposed model but misclassified by the baseline, alongside $c = 8$ instances misclassified by the proposed model but correctly classified by the baseline. This yielded a continuity-corrected Chi-Square statistic of $\chi^2 = 68.89$ with 1 degree of freedom, resulting in a p-value of $p = 1.04 \times 10^{-16}$.

When evaluating Multi-Task XLM-RoBERTa versus mBERT, the discordant pairs showed $b = 48$ instances correctly classified by the proposed model but missed by mBERT, alongside $c = 12$ instances missed by the proposed model but correctly classified by mBERT. This produced a continuity-corrected Chi-Square statistic of $\chi^2 = 20.42$ with 1 degree of freedom and a p-value of $p = 6.23 \times 10^{-6}$. Because both p-values fall far below the $\alpha = 0.05$ threshold, the null hypothesis is overwhelmingly rejected, confirming a statistically significant reduction in classification error rates.

Furthermore, a two-tailed Paired Student's t-Test was conducted on the paired Task B Entity F1-scores across the five cross-validation folds ($df = 4$). The Null Hypothesis (H_0) posited that the mean difference in entity extraction F1-score across validation folds between Multi-Task XLM-RoBERTa and BERT-BiLSTM-CRF is zero. The fold entity F1 differences were recorded as +0.098, +0.094, +0.101, +0.092, and +0.098, yielding a mean difference of $\bar{d} = +0.0966$ and a sample standard deviation of $s = 0.00358$ ($SE = 0.00160$). The calculated t-statistic reached $t = 60.34$ (or 60.38 using unrounded variance) with 4 degrees of freedom, exceeding the critical value of $t_{0.01,4} = 4.604$ and generating a two-tailed p-value of $p = 4.52 \times 10^{-7}$. Consequently, the paired Student's t-test confirms that the entity extraction gains achieved by the shared Multi-Task XLM-RoBERTa encoder over BERT-BiLSTM-CRF are statistically significant at $p < 0.001$.

4.5 Feature Importance and Component Ablation

To isolate the individual contribution of each architectural and algorithmic component within the unified framework, a systematic ablation experiment was conducted. Table 4.8 presents the ablation results across five model configurations.

System Configuration	**Task A F1**	**Task B F1**	**System F1**	**False Positive Rate (%)**	**Miss (%)**	**Latency (ms)**
Full Proposed Framework	0.9347	0.9044	0.9216	5.26%	8.50%	142.1 ms
w/o Multi-Task Shared Encoder	0.9147	0.8729	0.8845	8.40%	11.20%	268.4 ms
w/o Preprocessing (Uncleaned Data)	0.7967	0.7562	0.7641	24.18%	18.80%	178.5 ms
w/o K-Means++	0.9347	0.9044	0.8038	28.64%	13.60%	42.5 ms
w/o Source Authority Override	0.9347	0.9044	0.8912	5.30%	17.40%	141.9 ms

Table 13: Component Ablation System Matrix

Replacing the unified shared encoder with separate, disjoint single-task models to evaluate the Impact of the Shared Multi-Task Encoder caused Task A F1 to drop by 2.00 percentage points, falling from 0.9347 to 0.9147, and Task B Entity F1 to drop by 3.15 percentage points, falling from 0.9044 to 0.8729. This proves the presence of positive cross-task inductive transfer, as joint learning forces the transformer's intermediate self-attention layers to encode disaster semantics and geographical named entities simultaneously. Computationally, running two independent encoders nearly doubled the processing latency from 142.1 milliseconds to 268.4 milliseconds.

Training and evaluating on raw, uncleaned social media data to determine the Impact of Preprocessing and MinHash LSH produced the most severe performance collapse in the entire ablation study. The Macro F1 plummeted to 0.7967, Entity F1 degraded to 0.7562, and the False Alarm Rate surged to 24.18%. Raw scraped noise containing commercial ads and repetitive MinHash duplicates caused severe attention drift, confirming that deterministic preprocessing is a vital prerequisite for transformer fine-tuning.

Removing the K-Means++ Spatial Consensus Layer, which left only the raw neural text classification, reduced processing latency to 42.5 milliseconds but caused the False Alarm Rate to spike dramatically to 28.64%. This subsequently dropped the Overall System F1 from 0.9216 to 0.8038. Without spatial clustering, isolated false alarms, metaphoric posts, and unverified claims were forwarded directly to the dashboard, demonstrating that spatial-temporal multi-user consensus is essential for operational verification.

Finally, removing the Source Authority Override maintained a low False Alarm Rate of 5.30% but caused the Miss Rate during disaster onset to more than double from 8.50% to 17.40%. Under standard clustering logic, an initial high-priority post published by CDRRMO or PAGASA during the first minutes of a cloudburst was rejected as a spatial anomaly because it lacked corroborating citizen reports. Integrating the Source Authority Override successfully eliminates this cold-start penalty.

4.6 Error Analysis, Edge Cases and Discussion

Beyond quantitative performance metrics, evaluating a machine learning model for disaster informatics requires a deep understanding of its operational limitations. This section provides a qualitative analysis of the model's misclassifications and edge cases, contextualizes these empirical findings within existing literature, and discusses the practical implications of integrating this framework into a real-world Local Government Unit dashboard.

4.6.1 Qualitative Error Analysis and Edge Cases

A granular examination of misclassified instances from the test corpus reveals three primary failure modes.

The first failure mode involves Negation Structures and Colloquial Sarcasm. For example, a post stated that there was no flood in Catalunan but sarcastically added that the front of their house was just a swimming pool. Although the sentence explicitly

contained the phrase "walang baha" meaning no flood, the strong co-occurrence of the words for house and swimming pool biased the classification head toward incorrectly predicting a flood inundation. Additionally, hypothetical wishes expressing anxiety about future rain were occasionally flagged as false positives due to the presence of flood-related keywords alongside specific locations like Matina.

The second failure mode relates to Code-Switching and Morphological Ambiguity. One commercial post blended the disaster water-level idiom "lapas-tao", meaning over head-level, with commercial terms to describe a flood of orders for a Toril branch. Even though the advertisement filter caught most instances, highly creative figurative language remains an edge case. Similarly, the Cebuano verb "nibaha" was used metaphorically in another instance to describe a flood of donations for fire victims in Barangay 23-C, leading to a misclassification.

The third failure mode highlights the Rural-Urban Digital Divide, also termed Disaster Silencescapes. Empirical location distribution showed that post volume is heavily skewed toward urbanized barangays such as the Davao Urban Center, Talomo, Buhangin, Bunawan, and Matina. Conversely, peripheral agricultural districts like Marilog, Paquibato, and Baguio accounted for fewer than 0.5% of reports combined. In sparsely populated rural areas, severe riverine overflows generate insufficient social post volume to satisfy the K-Means++ density threshold, creating disaster silencescapes. To mitigate this limitation, the AGOS system integrates the Source Authority Override, relying on verified barangay captain radio reports to trigger alerts in rural zones.

4.6.2 Integration with Existing Literature

The empirical findings corroborate and advance recent scholarship in disaster informatics and natural language processing. Regarding Multilingual Transformer Encoders, the classification results confirm conclusions from recent studies that XLM-RoBERTa substantially outperforms mBERT in low-resource and code-switched environments due to its SentencePiece vocabulary richness. In terms of Multi-Task Shared Representation, coupling classification and NER into a joint objective improved entity extraction F1 by 3.15% over standalone models, thereby validating multi-task inductive transfer.

Furthermore, the dramatic reduction in the False Alarm Rate confirms theoretical claims regarding Spatial Consensus Verification. These claims emphasize that crowdsourced social sensing cannot rely on NLP alone, making spatial-temporal clustering vital to establish operational consensus. Finally, addressing Cold-Start Mitigation, the failure of standard clustering during disaster onset aligns with observations regarding disaster data sparsity. The proposed Source Authority Override directly resolves this challenge by cutting onset miss rates significantly.

4.6.3 Practical Implications for the AGOS LGU Dashboard

The proposed framework serves as the core intelligence engine for the AGOS Disaster Risk Reduction and Management Information System deployed in Davao City. It enables Actionable Triage by surfacing only reports that achieve a high confidence score or ver-

ified authority status, rather than presenting dispatchers with an uncurated stream of thousands of social media posts. For Map Visualization, extracted barangays are automatically mapped to geo-coordinates and rendered on the AGOS GIS interface as dynamic heatmaps with consensus radii. Finally, it incorporates Human-in-the-Loop Feedback by routing posts flagged with low confidence or isolated mentions to a secondary review queue. This allows LGU operators to confirm or discard events and route validated labels back to the active learning pipeline.

Chapter 5

Conclusions and Recommendations

5.1 Conclusions

The conclusions of this study are structured to directly answer the Main Problem Statement and each of the five Specific Research Objectives established in Chapter 1:

5.1.1 Overarching Conclusion on Main Problem Statement & General Objective

The general objective was to develop a Multi-Task NLP-based flood monitoring system that integrates classification, fine-grained extraction, and consensus-based validation to improve the reliability of crowdsourced flood reports at the barangay level. The overarching conclusion is that unifying sequence-level flood relevance classification (Task A) and token-level barangay Named Entity Recognition (Task B) within a single shared XLM-RoBERTa encoder preceded by dialect-aware web scraping and deterministic six-stage data cleaning, and succeeded by spatial-temporal K-Means++ consensus validation eliminates unidirectional error cascading and cuts per-report processing latency by 63.0%, transforming noisy, code-switched social media feeds into high-confidence, actionable barangay-level flood intelligence.

5.1.2 Conclusion on Specific Objective 1: Task A Flood Relevance Classification

The empirical evaluation confirms that sequence-level flood relevance classification (Task A) achieves superior performance (Macro F1 = 0.9347, PR-AUC = 0.9628) when powered by a massively multilingual encoder with subword vocabulary preservation. Statistically significant error reductions over classical lexical baselines ($p = 5.23 \times 10^{-21}$) and single-task mBERT ($p = 2.08 \times 10^{-8}$) prove that frequency-based n-gram representations fail on syntactic negation ("walang baha"), while WordPiece tokenization fragments agglutinative Philippine affixes ("gibaha", "pabaha"). XLM-RoBERTa's 250,000-token SentencePiece vocabulary preserves morphological integrity across Taglish and Bislish code-switching, while joint backpropagation from Task B grounds emergency intent in localized spatial

context. Theoretical Insight: Unlike TF-IDF representations that fail on syntactic negation ("walang baha", "walay baha") and mBERT's 110,000-token WordPiece vocabulary that fragments agglutinative Philippine affixes ("gibaha", "gipabakwit", "lapas-tao") into byte chunks, XLM-RoBERTa's 250,000-token SentencePiece vocabulary preserves whole-word morphological integrity across Taglish and Bislish. Jointly backpropagating Task B location gradients through the shared encoder acts as an inductive regularizer that grounds emergency intent classification in localized spatial context.

5.1.3 Conclusion on Specific Objective 2: Quantifying Preprocessing Noise Reduction

The six-layer deterministic data cleaner pruned 2,793 non-actionable noise posts from 9,431 raw harvested posts down to 6,638 clean posts, achieving a 29.61% overall noise reduction. Commercial real-estate promotions (34.3% of noise) and MinHash LSH near-duplicates (28.8% of noise) constituted the primary noise categories. Operational Impact: Preprocessing exerted a transformative triple impact: Prevented self-attention drift (without cleaning, Task A Macro F1 collapsed by 13.80 percentage points and False Alarms spiked fourfold from 5.26% to 24.18%); Cut model training time in half (reducing local GPU training from 20+ to 10+ hours and Colab training from 1 hour to 30 minutes); and Accelerated batch triage (eliminating 265 to 391 false-positive actionable alerts and up to 1,235 unverified warnings during deployment).

5.1.4 Conclusion on Specific Objective 3: Task B Barangay Location Extraction & K-Means++ Validation

Multi-Task XLM-RoBERTa achieved a strict seqeval Entity F1-score of 0.9044 (Entity Precision: 0.9114, Entity Recall: 0.8976), token-level F1 of 0.9369, and partial-match span accuracy of 0.9520. Paired Student's t-testing confirmed a statistically significant +9.66 percentage point gain over BERT-BiLSTM-CRF (0.8097 Entity F1, $t = 60.38, p = 1.12 \times 10^{-6}$). Validation Outcome: K-Means++ 3D spatial-temporal clustering ($p_i = [x_i, y_i, t_i]^T, \Theta \geq 3$ reports in a 3-hour window) grouped 86.4% of true flood reports into high-density consensus clusters along Davao City's river basins while isolating 13.6% as spatial anomalies. The Source Authority Override ($CS = 1.0$ for verified PAGASA/CDRRMO handles) resolved the cold-start clustering penalty during disaster onset, cutting the Miss Rate in half from 17.40% to 8.50%.

5.1.5 Conclusion on Specific Objective 4: System Performance Gain (ΔF1) Over Baselines

The overall system performance gain (ΔF1 = +14.65% over Baseline A and +7.91% over Baseline B) establishes two key architectural truths: text classification alone is operationally insufficient for life-safety dispatch without multi-user spatial-temporal corroboration; and hard parameter sharing in a unified multi-task architecture eliminates unidirectional error cascading. In decoupled sequential pipelines, false negatives produced by an upstream classifier permanently starve downstream location extractors of input. The pro-

posed unified framework resolves this bottleneck, achieving an Overall System F1-score of 0.9216 while slashing operational False Alarms to 5.26%. Architectural Insight: The gain over Baseline A proves that text classification alone is insufficient for life-safety dispatch without multi-user spatial corroboration. The gain over Baseline B proves that hard parameter sharing eliminates unidirectional error cascading—where upstream classifier false negatives permanently starve downstream location extractors.

5.1.6 Conclusion on Specific Objective 5: Total System Reliability, Scalability, and Deployment Viability

The framework established operational reliability by suppressing false alarms down to a False Alarm Rate (FAR) of 5.26% (an 81.6% reduction over Baseline A) and maintaining a Miss Rate (MR) of 8.50%. End-to-end processing latency of 142.1 ms per report ($\sim$ 7.04 reports/second) cut disjoint pipeline latency (384.2 ms) by 63.0%, confirming real-time viability for municipal emergency dispatch dashboards. Deployment Scalability: While initial full-parameter fine-tuning of the 278M-parameter backbone is heavy on local 4 GB VRAM hardware (10+ to 20+ hours) and benefits from cloud GPU offloading (30+ minutes to 1 hour on Google Colab), the trained checkpoint is exceptionally lightweight during deployment—triaging 5,000 to 9,500 posts in just 2 to 4 minutes.

5.2 Future Work

5.2.1 Recommendations for Future Researchers (Technical Extensions on the Model)

5.2.1.1 Architectural Extensions for Multi-Task XLM-RoBERTa & K-Means++

Tri-Task Architecture with Ordinal Flood-Depth Severity Head (Task C) & Contrastive Loss: Extend the shared 768-dimensional `[CLS]` representation by attaching a third classification head (Task C) to predict a 5-class pragmatic/hydrological severity scale: Class 0 (Non-Flood), Class 1 (Figurative/Sarcastic/Hypothetical), Class 2 (Minor Pluvial / Ankle-Deep $<$ 0.25m), Class 3 (Moderate / Knee-to-Waist Deep 0.25–0.90m), and Class 4 (Critical / Chest-to-Roof Deep $>$ 0.90m). Incorporate Supervised Contrastive Learning (SupCon) loss on hard-negative minimal pairs to separate literal flood reports from sarcastic/commercial idioms in embedding space, and replace static loss weights ($\alpha = 1.0, \beta = 1.0$) with learnable homoscedastic uncertainty parameters $s = m = \log \sigma^2$.

Multimodal Vision-Language Fusion (CLIP-ViT / ConvNeXt + XLM-RoBERTa): To resolve the 16 False Negatives caused by ultra-short Bisaya captions that rely on attached photos/videos ("Diri na pod sa bangkai"), pair the XLM-RoBERTa `[CLS]` text vector ($h_{\text{text}} \in \mathbb{R}^{768}$) with a 512-dimensional visual vector (h_{img}) extracted using CLIP-ViT-B/16 or ConvNeXt-Tiny fine-tuned on FloodNet. Fuse modalities via a learnable sigmoid gate $g = \sigma(W_g[h_{\text{text}}; h_{\text{img}}] + b_g)$ with a fallback mask ($g = 1.0$) for text-only posts.

Linear-Chain CRF Sequence Decoding (XLM-RoBERTa-CRF) & Hierarchical Spatial Tagging: Replace greedy token projection in Task B with a Linear-Chain Conditional Random Field (`torchcrf`) layer using Viterbi decoding to mathematically forbid illegal O → I-LOC transitions. Expand the 3-tag IOB2 schema into a 7-tag hierarchical schema distinguishing Administrative Barangays (`B-BRGY`, `I-BRGY`), Streets/Subdivisions (`B-STR`, `I-STR`), and Colloquial Micro-Landmarks (`B-LMK`, `I-LMK`, e.g., "gym", "chapel", "crossing").

LoRA / QLoRA Parameter-Efficient Fine-Tuning & ONNX INT8 Quantization: Integrate Low-Rank Adaptation (LoRA rank $r = 16, \alpha = \text{lora} = 32$) on query/value attention matrices. Freezing 99.1% of backbone weights and updating only $\sim$ 2.4M adapter parameters cuts local VRAM usage by $> 60\%$ and reduces local GPU training on Machine 2 from 10–20+ hours down to $\sim$ 45–90 minutes. Export PyTorch checkpoints to ONNX Runtime with INT8 dynamic quantization, compressing model size from $\sim$ 1.1 GB to $\sim$ 280 MB and accelerating batch inference to sub-30-second profiles on CPU-only municipal hardware.

Barangay-Adaptive Poisson Thresholds (Θ_b) & Ingestion Timestamp Imputation: Impute missing native timestamps using the scraper's DOM capture timestamp t_{ingest} with a confidence penalty factor ($\gamma = 0.65$) to enforce 3D temporal windowing and prevent early-warning inflation. Replace global static thresholds ($\Theta = 3$) with barangay-adaptive Poisson thresholds Θ_b derived from historical baseline posting rates, enabling a single report in spatially sparse rural districts (Marilog, Paquibato) or corroborated by DOST-ASTI rain gauges to trigger a priority alert.

5.2.1.2 Technical Extensions for Web Scraping & Preprocessing Pipelines

Multi-Worker Asynchronous Playwright Scraping Pool: Refactor single-threaded Selenium (davao_flood_scraper.py) into an asynchronous Playwright worker pool (asyncio managing 4 to 6 headless browser contexts with rotating residential proxies). Implement adaptive early-scroll termination that halts scrolling as soon as 3 consecutive DOM cards carry timestamps older than the active 3-hour window, compressing 44-query, 20-scroll scraping cycles from 1–2 hours down to under 12 minutes.

Network-Layer GraphQL/JSON Interception: Intercept background GraphQL network response payloads (page.on("response")) during scrolling to extract clean post text, Unix creation epochs (publish_time), and author IDs directly from API response packets, eliminating the 24.2% missing-timestamp rate.

Two-Tier Lexical + FAISS Semantic Deduplication: Retain MinHash LSH (85% Jaccard similarity over 3-word shingles) as a fast O(N) first-pass filter, and add a second-pass FAISS in-memory index (IndexFlatIP) over 768-dimensional [CLS] embeddings. Link posts submitted within a 30-minute window in the same barangay exhibiting cosine similarity 0.92 across dialects (e.g., an English advisory and its Cebuano translation) as corroborated semantic duplicates rather than redundant noise.

In-Memory Streaming Micro-Batch Pipeline: Refactor davao_flood_cleaner.py into an importable Python streaming generator (CleanerPipeline.process_stream()) inte-

grated directly between the Playwright worker pool and XLM-RoBERTa inference service, eliminating disk-bound JSON/CSV I/O overhead.

5.2.1.3 Infrastructure, Offline Operations, & Privacy Governance

Production Queue Processing & Database Indexing: Migrate development SQLite setups to production relational databases (PostgreSQL/PostGIS) with spatial indexing (GIST indexes on latitude/longitude and barangay polygons). Implement production queue managers (RabbitMQ or Redis Celery) with retry policies and idempotency keys to handle submission spikes during major typhoons.

Progressive Web App (PWA) Offline Capabilities: Extend citizen-facing components into a PWA with local IndexedDB storage. Field personnel and citizens can compose reports offline during power/cellular grid collapses; reports are assigned local UUIDs and automatically synchronized via background service workers once connectivity is restored.

Data Privacy (RA 10173 Compliance) & Audit Logging: Enforce Section 4(d) research/public safety privacy exemptions under Philippine Republic Act No. 10173 (Data Privacy Act of 2012). Strip personal handles, phone numbers, and profile URLs at the ingestion boundary, processing data exclusively at aggregated spatial-temporal barangay scales. Maintain immutable audit logs recording all automated model predictions, confidence scores, and operator manual overrides.

5.2.2 Limitations

Although the AGOS system successfully integrates its frontend, backend, database, reporting, analytics, mapping, and model components, several limitations remain. These limitations define the areas that future researchers and developers may improve.

First, the system is primarily designed around flood reporting within selected barangays in Davao City. Consequently, its workflows, barangay references, geographic boundaries, and operational assumptions may not directly apply to other cities or municipalities. Different locations may use different administrative divisions, languages, flood classifications, reporting procedures, and disaster-response protocols. The system therefore requires further testing before it can be considered generalizable to other local government units.

Second, the performance of the integrated model depends heavily on the quality, quantity, and representativeness of the training data. Citizen reports may contain spelling errors, code-switching, incomplete descriptions, slang, duplicate submissions, exaggeration, or ambiguous references to location and flood severity. Reports collected during a limited period may also fail to represent different rainfall intensities, flood durations, seasons, and emergency conditions. As a result, the model may perform differently when deployed during an actual large-scale flooding event or when exposed to language patterns that are not sufficiently represented in the training dataset.

Third, the system's confidence scores should not be interpreted as absolute measures of truth. A high confidence score indicates that the model strongly favors a particular classification, but it does not guarantee that the report is accurate. False reports,

repeated submissions, coordinated misinformation, inaccurate locations, and outdated reports may still receive high scores. Human validation and operational judgment therefore remain necessary, particularly for decisions involving evacuation, rescue deployment, or the allocation of emergency resources.

Fourth, the current system is limited by the availability and reliability of external data sources. Social-media feeds, mapping services, weather information, and other external services may experience delays, outages, rate limits, changes in API policies, or inconsistent data formats. These conditions can affect the completeness and timeliness of the information displayed to responders. The system may also be affected by poor internet connectivity, especially when used by field personnel or citizens in areas experiencing infrastructure damage.

Fifth, the system's database and deployment environment may require further evaluation under production-level workloads. A local or small-scale database configuration is sufficient for development and controlled testing, but it may not be sufficient for simultaneous report submissions, continuous data harvesting, frequent map updates, analytics queries, and multiple concurrent users during a major disaster. Database performance, storage growth, queue processing, backup procedures, and service availability require long-term testing.

Sixth, the evaluation of the system is limited if it is based primarily on functional testing, controlled datasets, or simulated reports. Such testing can verify whether the system behaves as designed, but it may not fully measure its impact on actual disaster-response operations. The system still requires evaluation with disaster-response personnel, barangay officials, citizens, and other intended users to determine whether its alerts, maps, reports, and analytics are understandable and useful under time-sensitive conditions.

Finally, the system handles potentially sensitive information, including citizen identities, contact details, report locations, and descriptions of affected areas. Even when the information is collected for legitimate disaster-response purposes, unauthorized access, excessive data retention, accidental disclosure, or misuse of location data may expose citizens to privacy and security risks. These concerns become more significant when the system is connected to external services or made available across multiple government offices.

5.2.3 Extensions

Future researchers may extend AGOS through the following technical and operational improvements.

5.2.3.1 Expansion of the Training and Evaluation Dataset

The next version of the system should use a continuously growing, geographically diverse dataset containing reports from multiple barangays, municipalities, and flood events. Each record should include the report text, timestamp, latitude and longitude, barangay, water-level category, source, verification result, and final operational outcome where available.

The dataset should be divided using event-based and time-based splits rather than only random splits. For example, reports from one flood event should be reserved as an unseen test set to determine whether the model generalizes to a new event. Evaluation should be reported using macro-F1, weighted-F1, precision, recall, confusion matrices, and per-class results for flood relevance, severity, and location extraction. The evaluation should also include separate results for Filipino, English, Cebuano, and code-switched reports, as well as short, noisy, duplicated, and typo-filled messages.

5.2.3.2 Improvement of Model Calibration and Human Validation

The model pipeline should be extended beyond classification accuracy by adding probability calibration, such as temperature scaling or isotonic regression. The system should display calibrated confidence intervals or confidence bands rather than treating raw model probabilities as guaranteed correctness.

A triage mechanism may be implemented using explicit thresholds: reports with confidence below 0.60 should be routed for manual review; reports between 0.60 and 0.85 should be treated as uncertain and compared with nearby reports; and reports above 0.85 may be prioritized for automated grouping while remaining subject to responder confirmation. These thresholds should be selected using validation data and adjusted according to the operational cost of false positives and false negatives. The database should also record the original model prediction, the model version, the confidence score, the human decision, the decision timestamp, and the reason for overriding the prediction.

5.2.3.3 Active Learning and Model Retraining

A feedback loop should be added so that verified and rejected reports become labeled examples for future model training. Reports that produce low confidence, conflicting predictions, or disagreement between the model and a human reviewer should be prioritized for annotation.

The system may implement scheduled retraining after a fixed number of newly verified reports or after each major flood event. Before deployment, every new model version should be evaluated against a fixed benchmark dataset and compared with the currently deployed version. The model should only be promoted when it improves the required metrics without causing unacceptable performance degradation for specific languages, barangays, or severity classes.

5.2.3.4 More Advance Spatial and Temporal Incident Clustering

The current spatial grouping capability may be expanded by incorporating time, severity, and report reliability into incident formation. Instead of grouping reports based only on geographic proximity, the clustering pipeline may use a feature vector containing latitude, longitude, timestamp, water-level category, confidence score, and verification status.

Future research should compare K-Means++ with methods more suitable for irregularly shaped and noisy disaster events, such as DBSCAN, HDBSCAN, or spatio-temporal clustering. Test cases should include multiple reports from the same flooded street, two

nearby but unrelated flooding incidents, a flood that expands across adjacent barangays, reports submitted at different times as water levels rise, isolated high-severity reports with few nearby submissions, and duplicate reports submitted by the same or different users. The clustering result should include an incident identifier, affected barangays, centroid or polygon, time window, report count, severity distribution, and confidence level. This would allow responders to distinguish a single expanding event from several unrelated local incidents.

5.2.3.5 Real-Time Data Ingestion and Queue Processing

The backend may be extended with a production-grade ingestion pipeline for citizen reports, social-media feeds, weather data, and government alerts. Incoming data should first be placed in a queue, normalized, deduplicated, classified, geocoded, clustered, and then stored as an incident or report record.

Each pipeline stage should have a defined status, retry policy, and error log. Failed jobs should be retried without creating duplicate reports. Idempotency keys should be used for external messages and repeated submissions. The system should also record ingestion latency, processing latency, queue depth, failed jobs, and the time between the first report and the creation of a validated alert.

5.2.3.6 Database and API Scalability

Future implementations should evaluate the system using realistic load scenarios, such as hundreds of concurrent citizen submissions during a severe rainfall event and simultaneous access by multiple barangay administrators. Database indexes should be added and evaluated for frequently queried fields, including barangay, verification status, timestamp, latitude and longitude, and incident identifier.

For larger deployments, the database may be migrated from a development-oriented configuration to a production relational database with appropriate backup, replication, monitoring, and retention policies. Spatial database support may also be considered for geographic queries and barangay boundary operations. Frequently requested dashboard results should be cached, while long-running analytics should be processed asynchronously rather than during a user's request.

The API should also be expanded into a documented versioned contract for the citizen application, administrative portal, and possible external responder systems. The API should include pagination, filtering, rate limiting, structured validation errors, audit logging, and consistent status codes.

5.2.3.7 Offline and Low-Connectivity Support

Because flooding may disrupt internet connectivity, the citizen-facing interface should be extended as a progressive web application or mobile application with offline capabilities. A user should be able to prepare a report without an active connection, store it securely on the device, and submit it automatically once connectivity is restored.

Offline submissions should include a locally generated identifier, creation timestamp, device location when available, and synchronization status. The backend should resolve duplicate submissions and conflicting updates when the device reconnects. This extension should be tested under interrupted uploads, repeated synchronization attempts, expired sessions, and limited-bandwidth conditions.

5.2.3.8 Improved Mapping and Location Processing

The mapping component may be extended to support automatic geocoding, reverse geocoding, barangay-boundary validation, and location uncertainty. When a report is submitted near a barangay boundary, the system should record the selected barangay, the coordinate-derived barangay, and any discrepancy between them.

Future versions may display incident polygons, movement over time, road closures, evacuation centers, drainage infrastructure, rainfall intensity, and affected population estimates. Map layers should be configurable by user role so that responders see operational information while public users receive only information that does not create privacy or safety risks.

5.2.3.9 Security, Privacy, and GovernanceControls

The system should undergo a formal security assessment covering authentication, authorization, session management, input validation, file handling, API access, rate limiting, and audit trails. Role permissions should be tested using a matrix covering citizens, barangay administrators, system administrators, and other possible responder roles.

Privacy extensions should include data minimization, encryption in transit and at rest, configurable retention periods, anonymized analytics, and controlled access to precise citizen locations. The system should also provide an audit history showing who viewed, modified, verified, rejected, or overrode a report. A documented incident-response procedure should be established for unauthorized access, corrupted data, compromised accounts, and service outages.

5.2.3.10 Usability and Accessibility Evaluation

Future studies should conduct structured usability tests with citizens, barangay personnel, and disaster-response officers. Participants should be asked to complete specific tasks, such as submitting a report, locating an active incident, verifying a report, filtering alerts by barangay, and interpreting model confidence.

Evaluation should measure task completion rate, time on task, error rate, perceived workload, and user satisfaction. The interface should also be tested using keyboard navigation, screen readers, sufficient color contrast, responsive layouts, and mobile devices with small screens. These improvements are important because users may access the system under stress, poor lighting, limited bandwidth, or emergency conditions.

5.2.3.11 Observability and Operational Reliability

A production deployment should include centralized application logs, performance metrics, health checks, and alerting. Important indicators should include API response time, report-processing latency, model inference time, database query time, queue failures, external service failures, and dashboard availability.

The system should also be tested through controlled failure scenarios, including database unavailability, model-service downtime, invalid external data, expired API credentials, network interruption, and sudden increases in report volume. A graceful-degradation strategy should allow the reporting function to remain available even when analytics or external data services are temporarily unavailable.

Through these extensions, future researchers can transform AGOS from an integrated disaster-monitoring system into a more extensively validated, scalable, secure, and resilient operational platform for flood intelligence and community-based disaster response.

Bibliography

[1] Abiola, T. O., Abiodun, K. D., Olumide, O. E., Adebanji, O. O., Calvo, O. H., Sidorov, G. (2025). Multilingual hope speech detection: A comparative study of logistic regression, mBERT, and XLM-RoBERTa with active learning. https://doi.org/10.48550/arXiv.2509.20315

[2] Alhaffar, M., Joury, E., Eriksson, A. (2023). Community engagement and crowdsourcing for effective disaster response and rescue operations during the earthquake in Syria. *ScienceDirect*, 98. https://www.sciencedirect.com/science/article/pii/S2212420923005769

[3] Besarra, J., Opdyke, A., Aquino, D. H., Santiago, J., Mendoza, J., Lagmay, A. M. F. (2024). Flood fragility and vulnerability functions for residential buildings in the Province of Leyte, Philippines. https://www.apn-gcr.org/wp-content/uploads/2025/10/J-Flood-Risk-Management-2024-Besarra-Flood-fragility-and-vulnerability-functions-for-residential-buildings-in-the-1.pdf

[4] Chen, A., Goodall, J., Chen, T., Zhang, Z. (2022). Flood resilience through crowdsourced rainfall data collection: Growing engagement faces non-uniform spatial adoption. *ScienceDirect*, 609. https://www.sciencedirect.com/science/article/pii/S0022169422002992

[5] Erokhim, D., Komendantova, N. (2024). Social media data for disaster risk management and research. *ScienceDirect*. https://www.sciencedirect.com/science/article/pii/S2212420924007428?via%3Dihub

[6] Esparza, M., Farahmand, H., Brody, S., Mostafavi, A. (2023). Examining data imbalance in crowdsourced reports for improving flash flood situational awareness. *ScienceDirect*, 95. https://www.sciencedirect.com/science/article/abs/pii/S2212420923003059

[7] Eze, P., Geard, N., Mueller, I., Chades, I. (2023). Anomaly Detection in Endemic Disease Surveillance Data Using Machine Learning Techniques. https://www.mdpi.com/2227-9032/11/13/1896

[8] George, E. P., Firoz, A., Murali, M., Rajamanickam, S., Palani, B. (2025). Hermes@DravidianLangTech 2025: Sentiment analysis of Dravidian languages using XLM-RoBERTa. https://doi.org/10.18653/v1/2025.dravidianlangtech-1.58

[9] Gopal, L., Prabha, R., Thirugnanam, H., Ramesh, M. V., Malamud, B. (2026). Review article: Social media for managing disasters triggered by natural hazards: a critical review of data collection strategies and actionable insights. https://nhess.copernicus.org/articles/26/215/2026/

[10] Hanny, D., Resch, B. (2024). Clustering-Based Joint Topic-Sentiment Modeling of Social Media Data: A Neural Networks Approach. https://www.mdpi.com/2078-2489/15/4/200

[11] Helmrich, A., Ruddell, B., Bessem, K., Chester, M., Chohan, N., Doerry, E., Eppinger, J., Garcia, M., Goodall, J., Lowry, C., Zahura, F. (2021). Opportunities for crowdsourcing in urban flood monitoring. *ScienceDirect*, 143. https://www.sciencedirect.com/science/article/abs/pii/S1364815221001675

[12] Ikotun, A. M., Ezugwu, A. E., Abualigah, L., Abuhaija, B., Heming, J. (2023). K-Means Clustering Algorithms: A comprehensive review, variants analysis, and advances in the era of Big data. *ScienceDirect*, 622. https://www.sciencedirect.com/science/article/abs/pii/S0020025522014633

[13] Javed, A., Jalil, Z., Nasir, M., Saleem, S., Alsubait, T., Alhakami, H. (2026). LLM-generated text detection: Enhancing accuracy using XLM-RoBERTa and DistilBERT model. https://peerj.com/articles/cs-3672/

[14] Khan, A., Zhang, T., Huang, X., Usmani, A. (2023). Machine learning driven smart fire safety design of false ceiling and emergency response. *ScienceDirect*, 177. https://www.sciencedirect.com/science/article/abs/pii/S0957582023006730

[15] Li, L., Bensi, M., Baecher, G. (2023). Exploring the potential of social media crowdsourcing for post-earthquake damage assessment. *ScienceDirect*, 98. https://www.sciencedirect.com/science/article/abs/pii/S2212420923005423

[16] Li, Y., Luo, L., Zeng, X. (2025). Fine-tuned BERT-BiLSTM-CRF approach for named entity recognition in geological disaster texts. https://www.researchgate.net/publication/390638431_Fine-tuned_BERT-BiLSTM-CRF_approach_for_named_entity_recognition_in_geological_disaster_texts

[17] Liu, L. (2025). Application of K-means supported by clustered systems in big data association rule mining. https://www.sciencedirect.com/science/article/pii/S2772941925000298

[18] Manyaka, N., Shoko, C., Gxokwe, S., Dube, T. (2026). Integrating Remote sensing and Machine Learning for flood Modelling: A Systematic Literature review. *ScienceDirect*, 143. https://www.sciencedirect.com/science/article/pii/S1474706526000483

[19] Mehta, S., Jain, G., Mala, S. (2023). Natural language processing approach and geospatial clustering to explore the unexplored geotags using media. *IEEE*. https://ieeexplore.ieee.org/document/10048848/keywords

[20] Mukherjee, B., Lal Shaw, R., Lal Sharma, M., Sain, K. (2025). Earthquake prediction using machine learning perspectives in Himalayan Seismic Belt and its surroundings.

ScienceDirect, 293. https://www.sciencedirect.com/science/article/abs/pii/S1367912025002792

[21] Navalkar, A., Tripathy, S., Gupta, M., Basu, S., Tripathy, P., Banarjee, A., Sekharan, S., Murtugudde, R., Ghosh, S. (2025). NLP-driven crowdsourcing for urban flood monitoring: Insights from Mumbai. *ScienceDirect*, 132. https://www.sciencedirect.com/science/article/abs/pii/S2210670725006699

[22] Paradkar, A., Zhang, C., Yuan, F., Mostafavi, A. (2022). Examining the consistency between geo-coordinates and content-mentioned locations in tweets for disaster situational awareness: A Hurricane Harvey study. *ScienceDirect*, 73. https://www.sciencedirect.com/science/article/abs/pii/S2212420922000978

[23] Peng, J., Zhang, J. (2024). Spatiotemporal assessment of urban flooding hazard using social media: A case study of Zhengzhou '7·20.' *ScienceDirect*, 176. https://www.sciencedirect.com/science/article/abs/pii/S1364815224000823

[24] Roy, P., Kumar, A., Singh, J., Dwivedi, Y., Rana, N., Raman, R. (2021). Disaster related social media content processing for sustainable cities. *ScienceDirect*, 75. https://www.sciencedirect.com/science/article/abs/pii/S2210670721006387

[25] Sathianarayanan, M., Hsu, P.-H., Chang, C.-C. (2024). Extracting disaster location identification from social media images using deep learning. *ScienceDirect*, 104. https://www.sciencedirect.com/science/article/abs/pii/S2212420924001146

[26] Serere, H., Resch, B. (2024). Understanding the impact of geotagging on location inference models for accurate generalization to non-geotagged datasets. *ScienceDirect*, 76(1). https://www.sciencedirect.com/science/article/pii/S1195103624000041

[27] Shidik, G. F., Winarsih, N. A. S., Saraswati, G. W., Saputra, F. O., Kusuma, E. J., Noersasongko, E., Gan Goh, G. G. (2026). A hybrid systematic literature review and automated content analysis for named entity recognition in disaster information management. *ScienceDirect*, 15. https://www.sciencedirect.com/science/article/pii/S2949719126000129

[28] Songchon, C., Wright, G., Beevers, L. (2021). Quality assessment of crowdsourced social media data for urban flood management. *ScienceDirect*, 90. https://www.sciencedirect.com/science/article/abs/pii/S0198971521000971

[29] Sufi, F. (2022). AI-SocialDisaster: an AI-based software for identifying and analyzing natural disasters from social media. *ScienceDirect*, 13. https://www.sciencedirect.com/science/article/pii/S2665963822000525

[30] Vishwanath, T., Shirwaikar, R., Jaiswal, W., Yashaswini, M. (2023). Social media data extraction for disaster management aid using deep learning techniques. *ScienceDirect*, 30. https://www.sciencedirect.com/science/article/abs/pii/S2352938523000435

[31] Wang, Z., Huang, M., Li, C., Feng, J., Liu, S., Yang, G. (2023). Intelligent recognition of key earthquake emergency Chinese information based on the optimized BERT-BiLSTM-CRF algorithm. https://www.mdpi.com/2076-3417/13/5/3024

[32] Zhang, T., Wang, J. Q. (2026). CLAS-Net: A study on cross-lingual intelligent sentiment analysis model fusing semantic alignment. https://pmc.ncbi.nlm.nih.gov/articles/PMC12893658/

[33] Goodchild, M. F. (2007). Citizens as sensors: The world of volunteered geography. *GeoJournal*, 69(4), 211–221. https://www.researchgate.net/publication/225919690_Citizens_as_Sensors_The_World_of_Volunteered_Geography

[34] Sakaki, T., Okazaki, M., Matsuo, Y. (2010). Earthquake shakes Twitter users: Real-time event detection by social sensors. *Proceedings of the 19th International Conference on World Wide Web*, 851–860. https://www.researchgate.net/publication/221023292_Earthquake_Shakes_Twitter_Users_Real-Time_Event_Detection_by_Social_Sensors

[35] de Bruijn, J. A., de Moel, H., Jongman, B., de Ruiter, M. C., Wagemaker, J., Aerts, J. C. J. H. (2019). A global database of historic and real-time flood events based on social media. *Scientific Data*, 6(1), 311. https://www.nature.com/articles/s41597-019-0326-9

[36] Eilander, D., Trambauer, P., Wagemaker, J., van Loenen, A. (2016). Harvesting social media for generation of near real-time flood maps. *Procedia Engineering*, 154, 176–183. https://research.vu.nl/en/publications/harvesting-social-media-for-generation-of-near-real-time-flood-ma-2/

[37] Imran, M., Castillo, C., Diaz, F., Vieweg, S. (2015). Processing social media messages in mass emergency: A survey. *ACM Computing Surveys (CSUR)*, 47(4), 1–32. https://dl.acm.org/doi/10.1145/2771588

[38] Olteanu, A., Castillo, C., Diaz, F., Vieweg, S. (2014). CrisisLex: A lexicon for collecting and filtering microblogged communications in crises. *Proceedings of the 8th International AAAI Conference on Weblogs and Social Media*. https://crisislex.org/icwsm2014_crisislex.pdf

[39] Nguyen, D. T., Al-Mannai, K. A., Joty, S., Sajjad, H., Imran, M., Mitra, P. (2017). Rapid classification of crisis-related data on social networks using convolutional neural networks. *Proceedings of the 11th International AAAI Conference on Web and Social Media*. https://arxiv.org/abs/1608.03902

[40] Devlin, J., Chang, M. W., Lee, K., Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 1, 4171–4186. https://aclanthology.org/N19-1423/

[41] Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzel, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. *Proceedings of the 58th Annual Meeting of*

the Association for Computational Linguistics, 8440–8451. https://aclanthology.org/2020.acl-main.747/

[42] Alam, F., Sajjad, H., Imran, M., Ofli, F. (2021). CrisisBench: Benchmarking crisis-related social media datasets for humanitarian information processing. *arXiv preprint arXiv:2004.06774*. https://arxiv.org/abs/2004.06774

[43] Abiola, T. O., Abiodun, K. D., Olumide, O. E., Adebanji, O. O., Calvo, O. H., Sidorov, G. (2025). Multilingual hope speech detection: a comparative study of logistic regression, mBERT, and XLM-RoBERTa with active learning. *arXiv preprint arXiv:2509.20315*. https://arxiv.org/abs/2509.20315

[44] Oco, N., Roxas, R. E. (2012). Pattern matching refinements to dictionary-based code-switching point detection. *Proceedings of the 26th Pacific Asia Conference on Language, Information and Computation*.

[45] George, E. P., Firoz, A., Murali, M., Rajamanickam, S., & Palani, B. (2025). Hermes@DravidianLangTech 2025: Sentiment analysis of Dravidian languages using XLM-RoBERTa. *Proceedings of the Fifth Workshop on Speech, Language and Audio for South Asian Languages*, 412–418. https://aclanthology.org/2025.dravidianlangtech-1.58/

[46] Cruz, J. C. B., & Cheng, C. (2020). Establishing baselines for text classification in Low-Resource Philippine languages. *arXiv preprint arXiv:2005.02068*. https://arxiv.org/abs/2005.02068

[47] Kudo, T. (2018). Subword regularization: Improving neural network translation models with multiple subword candidates. *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, 66–75. https://aclanthology.org/P18-1007/

[48] Zhang, T., & Wang, J. Q. (2026). CLAS-Net: A study on cross-lingual intelligent sentiment analysis model fusing semantic alignment. *Scientific Reports*, 16(1). https://pmc.ncbi.nlm.nih.gov/articles/PMC12893658/

[49] Aguilar, G., Kar, P., & Solorio, T. (2020). LinCE: A centralized benchmark for linguistic code-switching evaluation. *Proceedings of the 12th Language Resources and Evaluation Conference*, 2231–2237. https://aclanthology.org/2020.lrec-1.273/

[50] Ramshaw, L. A., & Marcus, M. P. (1995). Text chunking using transformation-based learning. *Proceedings of the Third Workshop on Very Large Corpora*, 82–96. https://aclanthology.org/W95-0107/

[51] Li, Y., Luo, L., Zeng, X., & Han, Z. (2025). Fine-tuned BERT-BiLSTM-CRF approach for named entity recognition in geological disaster texts. *Earth Science Informatics*, 18(2), 368. https://www.researchgate.net/publication/390638431_Fine-tuned_BERT-BiLSTM-CRF_approach_for_named_entity_recognition_in_geological_disaster_texts

[52] Wang, Z., Huang, M., Li, C., Feng, J., Liu, S., & Yang, G. (2023). Intelligent recognition of key earthquake emergency Chinese information based on the optimized BERT-BiLSTM-CRF algorithm. *Applied Sciences*, 13(5), 3024. https://www.mdpi.com/2076-3417/13/5/3024

[53] Lample, G., Ballesteros, M., Subramanian, S., Kawakami, K., & Dyer, C. (2016). Neural architectures for named entity recognition. *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 260–270. https://aclanthology.org/N16-1030/

[54] Nakayama, H. (2018). *seqeval: A Python framework for sequence labeling evaluation*. GitHub. https://github.com/chakki-works/seqeval

[55] Gelernter, J., & Balaji, S. (2013). An algorithm for local geoparsing of microtext. *GeoInformatica*, 17(4), 635–667. https://www.researchgate.net/publication/257564525_An_algorithm_for_local_geoparsing_of_microtext

[56] de Bruijn, J. A., de Moel, H., Jongman, B., de Ruiter, M. C., Wagemaker, J., & Aerts, J. C. J. H. (2018). TAGGS: Group-based geoparsing of flood-related tweets using contextual spatial information. *Journal of Geovisualization and Spatial Analysis*, 2(1), 2. https://link.springer.com/article/10.1007/s41651-017-0010-6

[57] Abiola, T. O., Abiodun, K. D., Olumide, O. E., Adebanji, O. O., Calvo, O. H., & Sidorov, G. (2025). Multilingual hope speech detection: A comparative study of logistic regression, mBERT, and XLM-RoBERTa with active learning. *arXiv preprint arXiv:2509.20315*. https://arxiv.org/abs/2509.20315

[58] George, E. P., Firoz, A., Murali, M., Rajamanickam, S., & Palani, B. (2025). Sentiment analysis of Dravidian and code-switched languages using XLM-RoBERTa. *Computer Speech & Language*, 89, 101688. https://doi.org/10.1016/j.csl.2024.101688

[59] Kendall, A., Gal, Y., & Cipolla, R. (2018). Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 7482–7491. https://openaccess.thecvf.com/content_cvpr_2018/html/Kendall_Multi-Task_Learning_Using_CVPR_2018_paper.html

[60] Zhang, T., & Wang, J. Q. (2026). CLAS-Net: A study on cross-lingual intelligent sentiment analysis model fusing semantic alignment. *PLoS ONE*, 21(2), e0342342. https://pmc.ncbi.nlm.nih.gov/articles/PMC12893658/

[61] Erokhin, D., & Komendantova, N. (2024). Social media data for disaster risk management and research. *International Journal of Disaster Risk Reduction*, 104, 104351. https://doi.org/10.1016/j.ijdrr.2024.104351

[62] Esparza, M., Farahmand, H., Brody, S., & Mostafavi, A. (2023). Examining data imbalance in crowdsourced reports for improving flash flood situational awareness. *International Journal of Disaster Risk Reduction*, 95, 103856. https://doi.org/10.1016/j.ijdrr.2023.103856

[63] Ikotun, A. M., Ezugwu, A. E., Abualigah, L., Abuhaija, B., & Heming, J. (2023). K-Means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data. *Information Sciences*, 622, 178–210. https://doi.org/10.1016/j.ins.2022.11.064

[64] Javed, A., Jalil, Z., Nasir, M., Saleem, S., Alsubait, T., & Alhakami, H. (2026). LLM-generated text detection: Enhancing accuracy using XLM-RoBERTa and DistilBERT model. *PeerJ Computer Science*, 12, e3672. https://peerj.com/articles/cs-3672/

[65] Navalkar, A., Tripathy, S., Gupta, M., Basu, S., Tripathy, P., Banarjee, A., Sekharan, S., Murtugudde, R., & Ghosh, S. (2025). NLP-driven crowdsourcing for urban flood monitoring: Insights from Mumbai. *Sustainable Cities and Society*, 132, 106699. https://doi.org/10.1016/j.scs.2025.106699

[66] Roy, P., Kumar, A., Singh, J., Dwivedi, Y., Rana, N., & Raman, R. (2021). Disaster related social media content processing for sustainable cities. *Sustainable Cities and Society*, 75, 103387. https://doi.org/10.1016/j.scs.2021.103387

[67] Eze, P., Geard, N., Mueller, I., & Chades, I. (2023). Anomaly detection in crowdsourced surveillance data using machine learning techniques. *Healthcare*, 11(13), 1896. https://www.mdpi.com/2227-9032/11/13/1896

[68] Hanny, D., & Resch, B. (2024). Clustering-based joint topic-sentiment modeling of social media data: A neural networks approach. *Information*, 15(4), 200. https://www.mdpi.com/2078-2489/15/4/200

[69] Liu, L. (2025). Application of K-means supported by clustered systems in big data association rule mining. *Big Data Research*, 35, 100429. https://doi.org/10.1016/j.bdr.2025.100429

[70] Mehta, S., Jain, G., & Mala, S. (2023). Natural language processing approach and geospatial clustering to explore geotags using media. *IEEE Transactions on Computational Social Systems*, 10(3), 1204–1215. https://www.researchgate.net/publication/368729442_Natural_Language_Processing_Approach_and_Geospatial_Clustering_to_Explore_the_Unexplored_Geotags_Using_Media

[71] Peng, J., & Zhang, J. (2024). Spatiotemporal assessment of urban flooding hazard using social media: A case study of Zhengzhou '7-20'. *Computers, Environment and Urban Systems*, 176, 102082. https://doi.org/10.1016/j.compenvurbsys.2024.102082

[72] Sathianarayanan, M., Hsu, P. H., & Chang, C. C. (2024). Extracting disaster location identification from social media images using deep learning. *International Journal of Disaster Risk Reduction*, 104, 104346. https://doi.org/10.1016/j.ijdrr.2024.104346

[73] Alhaffar, M., Joury, E., & Eriksson, A. (2023). Community engagement and crowdsourcing for effective disaster response and rescue operations. *International Journal of Disaster Risk Reduction*, 98, 104085. https://doi.org/10.1016/j.ijdrr.2023.104085

[74] Chen, A., Goodall, J., Chen, T., & Zhang, Z. (2022). Flood resilience through crowdsourced rainfall data collection: Growing engagement faces non-uniform spatial adoption. *Journal of Hydrology*, 609, 127763. https://doi.org/10.1016/j.jhydrol.2022.127763

[75] hiQ Labs, Inc. v. LinkedIn Corp., 273 F. Supp. 3d 1099 (N.D. Cal. 2017). https://caselaw.findlaw.com/court/us-dis-crt-n-d-cal/2182242.html

[76] Van Buren v. United States, 593 U.S. (2021). *Supreme Court of the United States.*

Republic Act No. 10173 (2012). Data Privacy Act of 2012. National Privacy Commission. https://privacy.gov.ph/data-privacy-act/

Japan International Cooperation Agency (JICA). (n.d.). Data Collection Survey on Flood Risk Management in Davao City. JICA Report. https://openjicareport.jica.go.jp/pdf/12380796.pdf

Miyamoto, M., Kakinuma, D., Ushiyama, T., Rasmy, A. W. M., Yasukawa, M., Bacaltos, D. G., Sales, A. C., Koike, T., & Kitsuregawa, M. (2022). Co-Design for Enhancing Flood Resilience in Davao City, Philippines. *Water*, 14(6), 978. https://www.mdpi.com/2073-4441/14/6/978

Alam, F., Sajjad, H., Imran, M., & Ofli, F. (2021). CrisisBench: Benchmarking crisis-related social media datasets for humanitarian information processing. *Proceedings of the 15th International AAAI Conference on Web and Social Media*. https://mimran.me/papers/CrisisBench_Benchmarking_Crisis_related_Social_Media_Datasets_ICWSM21.pdf

Fornaciari, T., Uma, A., Poesio, M., & Hovy, D. (2022). Hard and Soft Evaluation of NLP models with BOOtSTrap SAmpling - BooStSa. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*. https://aclanthology.org/2022.acl-demo.12.pdf

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. https://aclanthology.org/2020.acl-main.747/

Alam, F., Sajjad, H., Imran, M., & Ofli, F. (2021). CrisisBench: Benchmarking Crisis-related Social Media Datasets for Humanitarian Information Processing. *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2021)*. https://ojs.aaai.org/index.php/ICWSM/article/view/18115

Bestgen, Y. (2022). Please, Don't Forget the Difference and the Confidence Interval when Seeking for the State-of-the-Art Status. *Proceedings of the 13th Language Resources and Evaluation Conference (LREC 2022)*. https://aclanthology.org/2022.lrec-1.640/

Fornaciari, T., Uma, A., Poesio, M., & Hovy, D. (2022). Hard and Soft Evaluation of NLP Models with Bootstrap Sampling (BooStSa). *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics: System Demonstrations (ACL 2022)*. https://aclanthology.org/2022.acl-demo.12/

Zhang, Y., & Yang, Q. (2021). A Survey on Multi-Task Learning. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*. https://arxiv.org/abs/1707.08114